

JBX-6300 series

CALIBRATION PROGRAM OPERATION

For the proper use of the instrument, be sure to read this instruction manual. Even after you read it, please keep the manual on hand so that you can consult it whenever necessary.

JBX-6300 series

CALIBRATION PROGRAM OPERATION

**Please be sure to read this instruction manual carefully,
and fully understand its contents prior to the operation
or maintenance for the proper use of the instrument.**

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Note: For servicing and inquiries, please contact your JEOL service office.

NOTATIONAL CONVENTIONS AND GLOSSARY

■ Examples for general notations

– CAUTION – :	Important precautions for use, which, if not followed, may result in damage to or problems with the device itself.
 :	Additional points to remember regarding the operation.
 :	A reference to another section, chapter or manual.
1, 2, 3 :	Numbers indicate a series of operations that achieve a task.
 :	A diamond indicates a single operation that achieves a task.
File :	The names of menus, commands, or parameters displayed on the screen are denoted with bold letters.
File–Exit :	Selecting a menu item from a pulldown menu is denoted by linking the menu and the item with a dash (–). For example, File–Exit means selecting Exit from the File menu.
 :	Keys on the keyboard are denoted by enclosing their names in a box.

■ Examples for mouse terminology

Mouse pointer:	A mark, displayed on the screen, which moves following the movement of the mouse. It is used to specify a menu item, command, parameter value, and other items. Its shape changes according to the situation.
Click:	To press and release the left mouse button.
Right-click:	To press and release the right mouse button.
Double-click:	To press and release the left mouse button twice quickly.
Drag:	To hold down the left mouse button while moving the mouse.

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SAFETY PRECAUTIONS

Although this instrument is protected with safety device which prevents the occurrence of accident that could result in an injury, harm, and damage to the users or instrument itself, the safety feature may not work properly if you use the instrument for the purpose of use not intended or in an improper usage. For the proper use of the instrument, please be sure to read all of the instructions, descriptions, notices, and precautions contained in this manual carefully to understand them fully prior to the operation or maintenance. This section, "Safety Precautions," contains important information related to safety for using of the instrument.

The safety indications and their meanings are as follows:

- | | |
|---|---|
|  | DANGER: An imminently hazardous situation which, if not avoided, will result in death or serious injury. |
|  | WARNING: A potentially hazardous situation which, if not avoided, could result in death or serious injury. |
|  | CAUTION: A potentially hazardous situation which, if not avoided, may result in minor or moderate injury, or a situation that could result in serious damage to facilities or acquired data. |

Labels bearing the following symbols are attached to dangerous locations on the instrument. Do not touch any of these locations with your hands or anything else.



Examples of symbols

-  • Use the instrument properly within the scope of the purpose and usage described in its brochures and manuals.
-  • Never open/remove protective parts (exterior panels) and parts that can't be opened/removed without use of tool (including key), or disconnect/connect the cables/connectors that are not described in this manual.
-  • Never attempt to do any works of disassembling/assembling the instrument other than those described in this manual.
-  • Never make modifications that include installing substitute parts and disabling safety devices or other safety features.
-  • Never disconnect the grounding wire or move it from the prescribed position. Failure to follow this instruction could result in electric shock.
-  • To avoid falling, do not climb onto the operation table and console during daily operation or during maintenance or inspection.
-  • When you dispose of the instrument or liquid or other waste, follow all applicable laws and regulations, and dispose of it in a proper manner without polluting the environment.
-  • Be sure to read the "Safety Precautions" section of the manuals for the accessories attached to or built into the instrument.
-  • If anything is unclear, please contact your JEOL service office.



WARNING for Installation



- **Do not attempt to install the instruments by yourself.**

Installation work requires professional expertise and JEOL is responsible for the installation of the instruments and related attachments purchased from JEOL.

Consult your JEOL service office.

1 GENERAL

This manual explains the Calibration Program that runs in your JEOL Electron Beam Lithography System. This starts from the EBX menu and offers the user an interface for calibration of the e-beam lithography system, which is used to adjust the writing conditions of the hardware.

2 WINDOW GENERAL

The operations of the Calibration Program fall into two categories: operations from the Calibration (...) window (hereinafter referred to as the Calibration window) and operations from parameter-setting windows.

Clicking the **calibration** icon on the EBX main menu window opens the Calibration window of this program. The subprograms can be executed from the Calibration window and the result of the execution is displayed on the window.

The parameter-setting window can also be opened from the Calibration window. Parameter editing of the subprograms and subprogram executions can be performed on the parameter-setting window.

2.1 Calibration window

Function	Outline	Remarks
Changing, deleting, and copying the condition file	You can change the condition file to be used, and also create, delete, and copy a new condition file.	See Section 2.3.
Subprogram selection	You can select the subprogram to be used.	
Opening parameter-setting window	You can open the parameter-change window of the subprogram of which the parameters are to be revised.	See Section 2.2.
Subprogram execution	You can execute the subprogram.	
Displaying results	You can display the results of subprogram execution.	

2.2 Parameter-Setting Window

Function	Outline	Remarks
Revising and saving parameters	You can edit and save the subprogram parameters.	
Subprogram execution	You can execute the subprogram.	

2.3 Condition File

One condition file contains the individual parameters for each subprogram of all the subprograms.

- More than one condition file can be prepared.
- The condition file can be created, edited, copied and deleted.
- The name of the condition file is called “condition name.”
- The condition name can consist of up to 63 characters.

In a condition name, you can use upper- and lower-case letters, numbers, and printable symbols that can be used in filenames. However, you cannot use slashes, backslashes, single quotes, or non-printing characters including spaces.

An error occurs if an inappropriate condition name is designated.

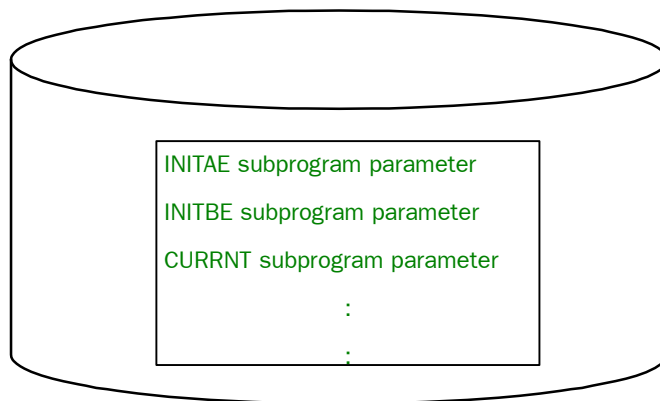


Fig. 1 Condition file concept

The condition files are classified by accelerating voltages.

Condition files for different accelerating voltages are distinguished as different condition files even if their names are the same.

Accelerating voltage: 50 kV



Calibration condition name = TEST

Accelerating voltage: 100 kV



Calibration condition name = TEST

The program distinguishes these two.

3 CALIBRATION WINDOW

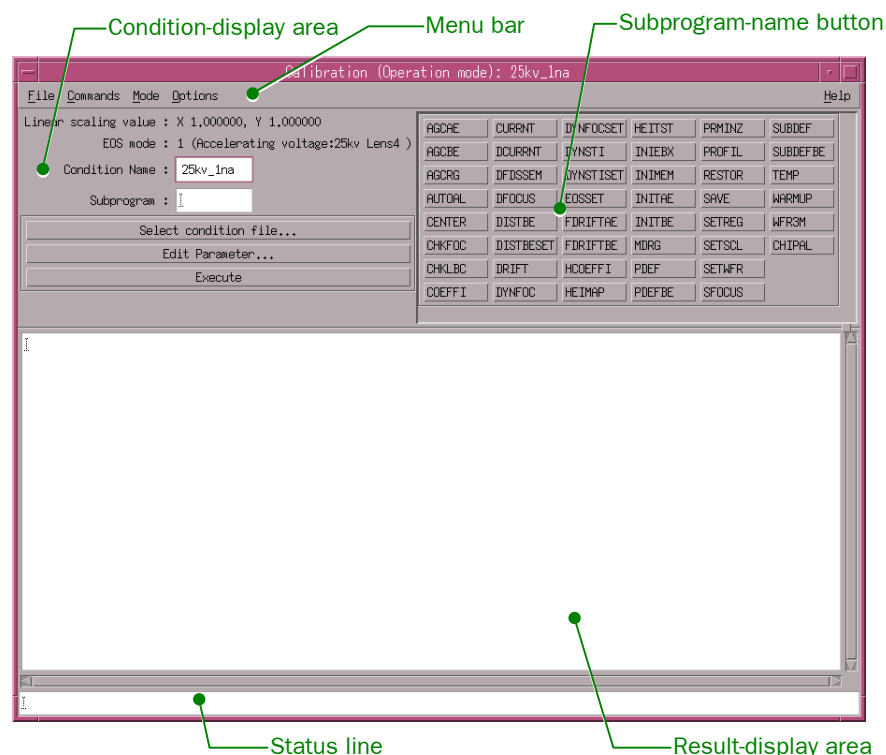


Fig. 2 Calibration window

3.1 Menu Bar

Five menu names, **F**ile, **C**ommand, **M**ode, **O**ptions and **H**elp are on this bar. Selecting a name from this bar using the mouse opens the pull-down menu having that name.

3.1.1 File

Menu item	Function
<u>L</u>oad...	Opens the condition-selection (read) window (Section 3.3.1) to change the condition file to be used. The function is the same as the condition-file select button (Section 3.2).
<u>C</u>opy...	Opens the condition-selection (copy) window (Section 3.3.2) to copy the condition file.
<u>D</u>elte...	Opens the condition-selection (delete) window (Section 3.3.3) to delete the condition file.
<u>E</u>xit...	Terminates the calibration program. This is ineffective while a subprogram is being executed.

3.1.2 Commands

Menu item	Function
Subprogram stop	Suspends the subprogram. This is effective only while the subprogram is being executed. The subprogram may not suspend for a while after you select this item.
Mark detection continues	When the toggle button of this item is pressed (checked), the mark-detection processing is kept continued in the mark-detection subprogram.
<u>C</u>lear...	Erases all information from the result-display area.
<u>B</u>atch...	Opens the Continuous calibration window (Section 4.4).

3.1.3 Mode

Multiple X terminals can open calibration windows; however, system operation can be performed in only one of these windows.

- The right of operation is called “the control privilege.”
- The window that has acquired the control privilege is said to be in the operation mode.
- On the other hand, a window is in the display mode if it has not acquired the control privilege.

If there is no other window that has the control privilege before you open calibration window, the new window will acquire the control privilege and will open in the operation mode. If another window that has the control privilege is already open before you open a calibration window, the new window will open in the display mode.

In window in the operation mode, you can display edit and run the program. On the other hand you cannot run the program is window in display mode.

In a window in the display mode, you can display and edit subprogram parameters but you cannot run the program.

Menu item	Function
<u>O</u>peration mode	Selecting this item acquires the control right. The system switches to the operation mode after the control privilege is acquired. If the acquisition of the control right fails, an error message appears.
<u>D</u>isplay mode	Cancels the control privilege and returns to the display mode.

3.1.4 Options

Menu item	Function
Display detailed results	Lets you select which stage-characteristic data to display in the height measurement.
Logging	When the toggle button of this item is pressed (checked), the contents of the result-display area are output to a file.
Log file ...	Opens the Log File selection window (Section 3.5).
Subprogram buttons customize	Selecting this item opens a window for selecting a subprogram-button customization file.

Menu item	Function
Display all subprogram buttons	Selecting this item displays all the subprogram buttons.
Subprogram buttons customize	Selecting this item opens a window for rearranging the subprogram buttons.

3.1.5 Help

Menu item	Function
Product information	Shows the version number of the program. You can confirm the version.

3.2 Condition Display Area and Operation Buttons

3.2.1 Condition name indication and control buttons

Display item	Function
Linear scaling value	The present linear scaling setting appears here. If you want to change it, use the Linear scaling value window (Section 4.24).
EOS mode	The EOS mode for which you are editing parameters appears here. If you want to change it, use the Condition File to Load window (Section 3.3.1).
Condition Name	Displays the name of the condition file in operation. To change the condition name, use the Condition File to Load window (Section 3.3.1). Since the text-field length is limited, the entire condition name may not be displayed if the name is long. The part that is displayed can be moved using the arrow keys when the cursor is placed in the text field.
Subprogram	Displays the name of the subprogram in operation.
Select condition file... button	Opens the Condition File to Load window (Section 3.3.1). The operation is the same that you select Load ... from the File menu (Section 3.1.1).
Edit Parameter... button	Opens the parameter-setting window of the subprogram to be used. Only the parameter-setting window for editing condition file corresponding to the accelerating voltage to be used. An error will occur if no subprogram has been specified.
Execute button	Executes the specified subprogram. An error will occur if the subprogram has not been specified. The values saved in the condition file whose condition name is displayed are used as the parameters of the object subprogram.

3.2.2 Subprogram-name buttons

These subprogram buttons are used to select the subprogram to be used. Double-clicking the button of a subprogram opens the Parameter-setting window for the subprogram.

3.2.3 Result-display area

Displays the results of the subprogram execution. The user cannot edit these results. The window scrolls to display new results when the cursor is positioned on the bottom line of the window. The oldest results will be erased when too many results have been displayed. Selecting Delete from the Command menu (Section 3.1.2) on the menu bar erases all the results.

3.2.4 Status line

Display messages telling related to startup and termination of the subprogram, receiving intermediate results, and errors that have been detected. All messages on this line will be erased when the subprogram execution starts.

3.3 Condition-Selection Windows

Here are windows for selecting, copying, and deleting condition files.

3.3.1 Condition File to Load window

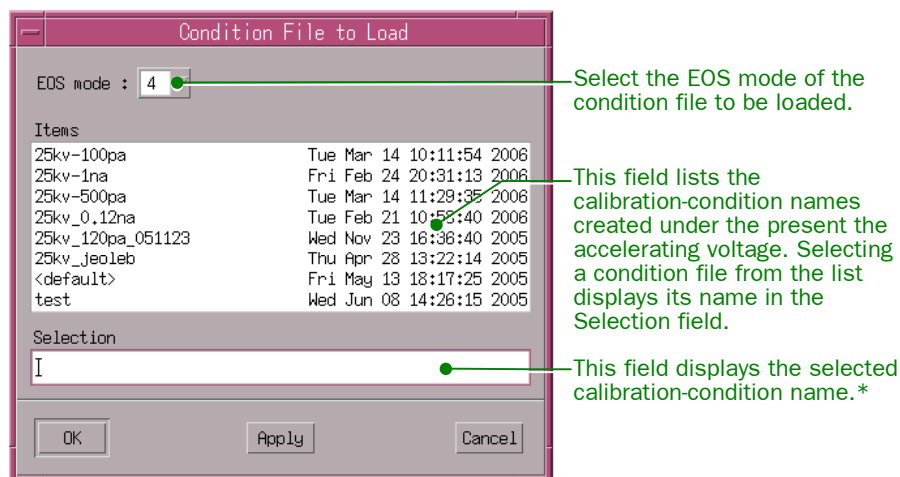


Fig. 3 Condition File to Load window

Button name	Function
OK	Terminates the loading process** and closes the window.
Apply	Executes the loading process**.
Cancel	Closes the window.

* Entered the condition name in this field when you create a condition file.

** Loading file

A warning window will appear if the Selection field is blank.

This condition file is applied to the Condition window as the operation window when the selection field is not blank and the condition name in the Selection field is in the condition-file list on this window.

A window asking whether or not to create a new condition file will appear when the Selection field is not blank and the condition name in the selection field does not exist in the condition-file list.

3.3.2 Condition File to Copy window

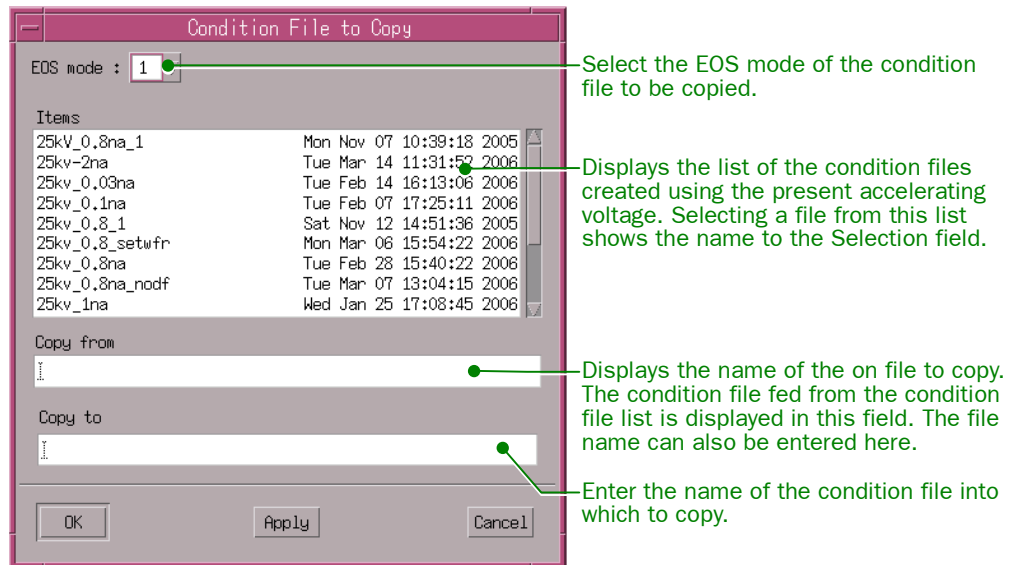


Fig. 4 Condition File to Copy window

Button name	Function
OK	Executes the copying* and closes the window.
Apply	Executes the copying*. The window does not close.
Cancel	Closes the window.

* Copying

The program confirms that the copy from and copy to fields are not empty. If either one or both fields are empty, a window appears to tell you that.

The program confirms that the source and the destination are not the same. If the source and the destination are the same, a window appears to tell you that the same name is specified for the source and the destination.

The process checks if the copy destination is the name of an existing condition file. If a file exists with this condition name, an overwrite confirmation window appears.

3.3.3 Condition File to Delete window

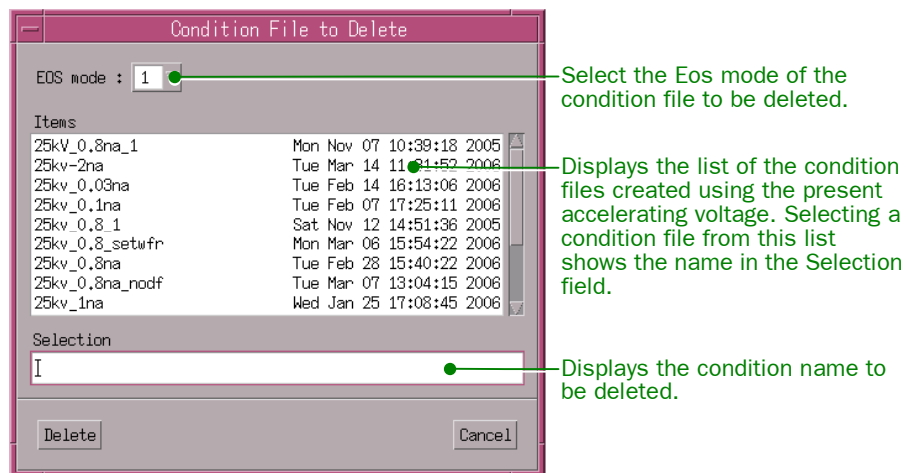


Fig. 5 Condition File to Delete window

Button name	Function
Delete	Deletes* the condition file in the Selection field.
Cancel	Closes the window.

3.4 Product Information Window

-  The version number and creation date and time in the above Product information window are examples for use in this manual, so they might differ from those of your system.

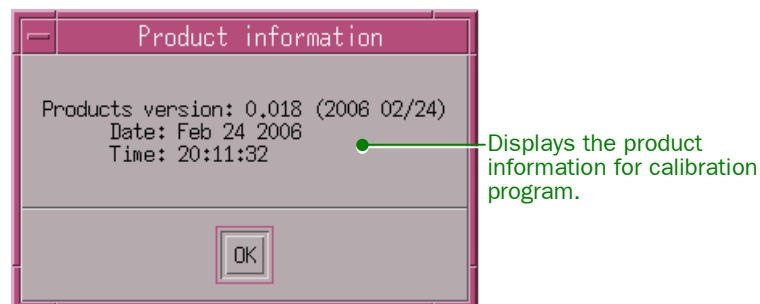


Fig. 6 Product Information window

Button name	Function
OK	Closes the window.

* Deleting

If the Selection field is empty, a window will open, telling you that the field is empty.

If the condition name in the Selection field is not in the Items list, a window will open, telling you that the name is not in the list.

If the condition name in the Selection field is included in the item list, a confirmation window appears, asking your permission to delete the condition file.

3.5 Log File Selection Window

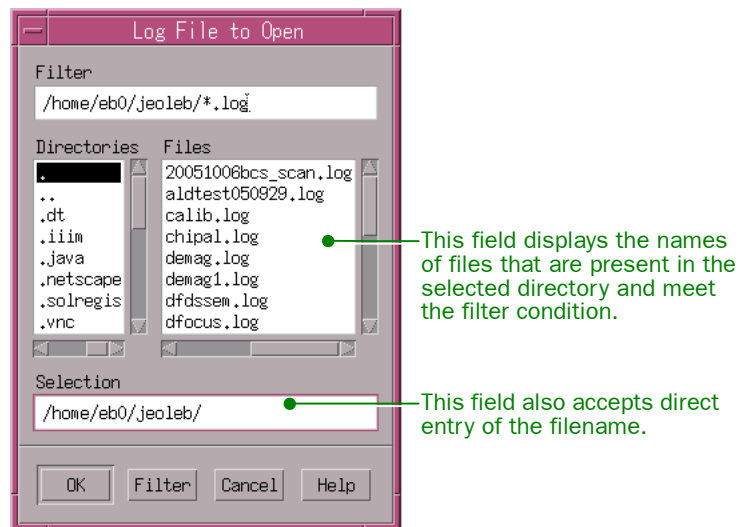


Fig. 7 Log File Selection window

Button name	Function
OK	This button applies your selection in the window and closes the window.
Filter	Selects files that match the condition specified in the filter field and updates the file list.
Cancel	Closes the window.
Help	Not used.

3.6 Subprogram Buttons Customize (Load) Window

This window lets you select any file from the list of subprogram-button customization files previously created in the Subprogram buttons customize window.

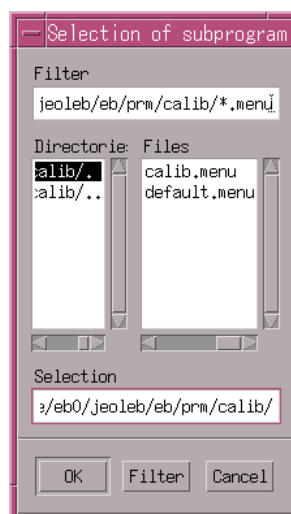


Fig. 8 Subprogram Buttons Customize (Load) Window

3.7 Subprogram Buttons Customize Window

In this window you can customize the layout of the subprograms you want to use. Selecting a subprogram name in the left field and then pressing a button in the right field assigns the subprogram name to the button.

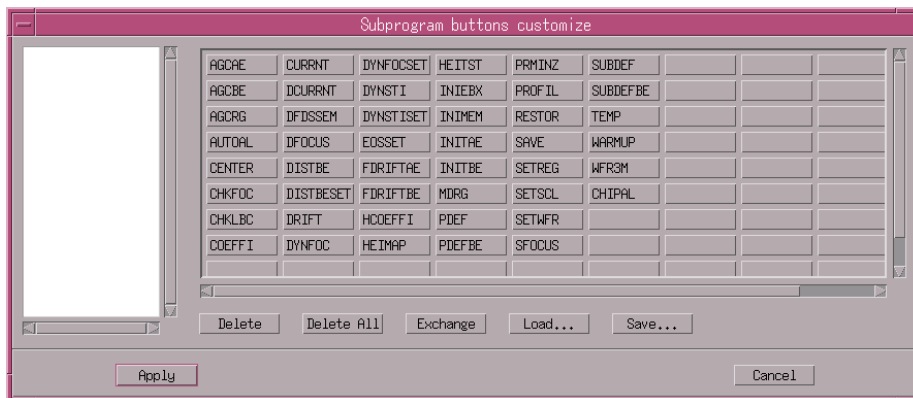


Fig. 9 Subprogram buttons customize Window

Button name	Description
Delete	Selecting a subprogram button and then pressing this button deletes the selected subprogram button and moves the subprogram name to the left field.
Delete All	Pressing this button deletes all the subprogram buttons and moves the subprogram names to the left field.
Exchange	Selecting two subprogram buttons and pressing this button interchanges the two buttons.
Load...	Pressing this button opens the Subprogram buttons customize (load) window, where you select an existing subprogram-button customization file.
Save...	Pressing this button opens the Subprogram buttons customize (save) window, where you save a new subprogram-button customization file.
Apply	This button applies the results of deletion/replacement to the subprogram layout in the Calibration window.
Cancel	This button closes the window without applying the results of deletion/replacement.

4 PARAMETER-SETTING WINDOWS

The window for setting the parameters of each subprogram is the parameter-setting window. Each subprogram having parameters for program execution has a parameter-setting window.

The following specifications apply to each parameter-setting window.

- The parameter names are displayed on the left side of the window and the parameter value follows the parameter name.
- The parameter unit, follows the parameter name.
- Since certain parameters are limited in value, the corresponding field will be shown in red if the value is outside the limits.
- With a red input field, you cannot save parameters or run subprograms.
- In some subprograms, parameter saving and program execution are not allowed.

The following button names are shown at the bottom of the parameter-setting window for the subprograms described in this chapter. The functions of the buttons are as follows. However, there are some subprograms such as BATCH whose buttons have different functions. For those subprograms, the functions of the buttons are described individually.

Button name	Function
Save	Saves the parameter settings displayed on the window.
Execute	Executes the subprogram using the parameter settings displayed on the window.
Reset	Discards the parameter settings displayed on the window; then reloads the parameters from the condition file and sets them.
Cancel	Discards the parameter settings displayed on the window and closes the window.

4.1 AGCAE

AGCAE automatically adjusts the preamplifier constants for AE mark detection and displays the results. Also, you can apply the preamplifier constants calculated in AGCAE to other subprograms that use the AE mark.

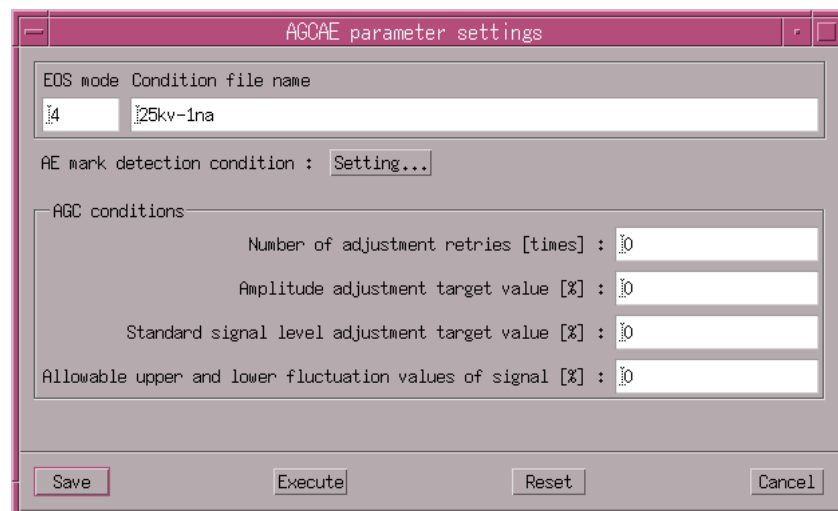
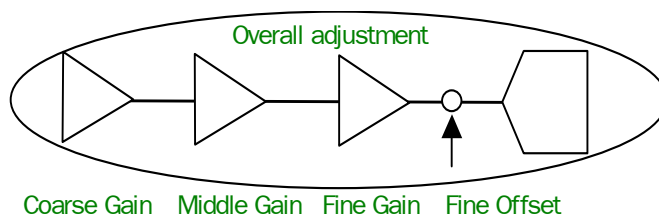


Fig. 10 AGCAE parameter setting Window

■ Descriptions of parameters

- AE mark detection condition (Section 4.42.1)
This button opens a window for setting AE mark-detection conditions.
- Number of adjustment retries [times]
This field lets you specify the number of retries in automatic gain control.
- Amplitude adjustment target value [%]
This field lets you specify the wave height to be adjusted in automatic gain control as a percentage of the amplitude. The recommended setting is about 90 %.
- Standard signal level adjustment target value [%]
This field lets you specify the lower limit of the waveform to be adjusted in automatic gain control as a percentage of the amplitude. The recommended setting is about 10 %.
- Allowable upper and lower fluctuation values of signal [%]
This field lets you set the tolerance for the Amplitude adjustment target value and the Standard signal level adjustment target value. When the waveform falls within this tolerance range, the program ends. With an Amplitude adjustment target value of 90 %, if you specify a tolerance of 10 %, the adjustment ends when the wave height falls within the range 80 to 100 %; with a Standard signal level adjustment target value of 10 %, the adjustment ends when the minimum waveform falls within the range 0 to 20 %.

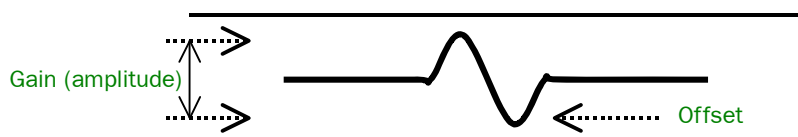
■ AE mark detection system



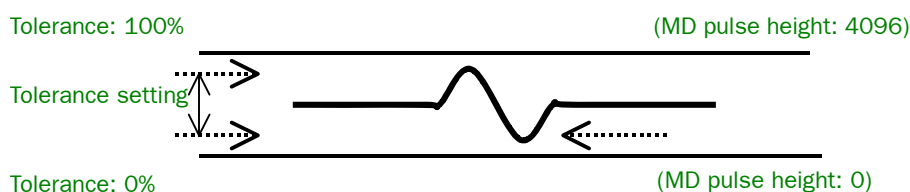
The Gains (Coarse, Middle, Fine) and the Fine Offset are adjusted automatically.

Coarse Gain	Middle Gain	Fine Gain	Fine Offset
1, 100, 10000	0~15	0~255	0~255

● Description of gain/offset



● Description of adjustment tolerance



■ Updating the amplifier constants

If AGCAE completes normally, the following inquiry window appears. You can update the preamplifier constants by clicking on the subprograms to which you want to apply the update in the subprogram list and then clicking on **OK**. Selecting subprograms in the list highlights them. Normally, click on **All Select** below the subprogram list and then click on **OK** to update the preamplifier constants for all the subprograms that use the AE mark.

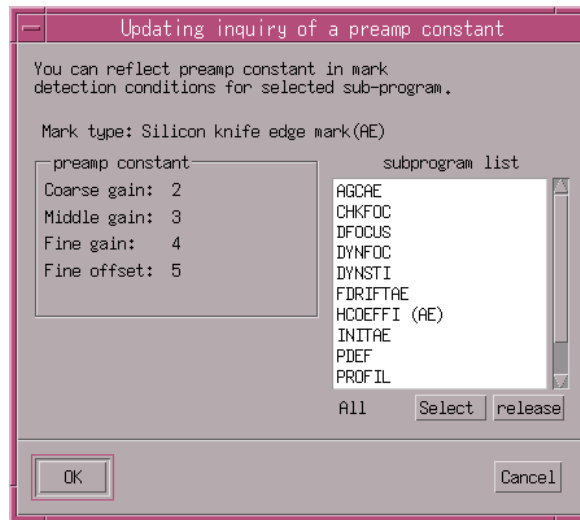


Fig. 11 Updating inquiry a preamp constant

4.2 AGCBE

AGCBE automatically adjusts the amplifier settings (gain, offset) for BE mark signal detection and displays the results. You can apply the amplifier settings calculated in AGCBE to other subprograms that use the BE mark.

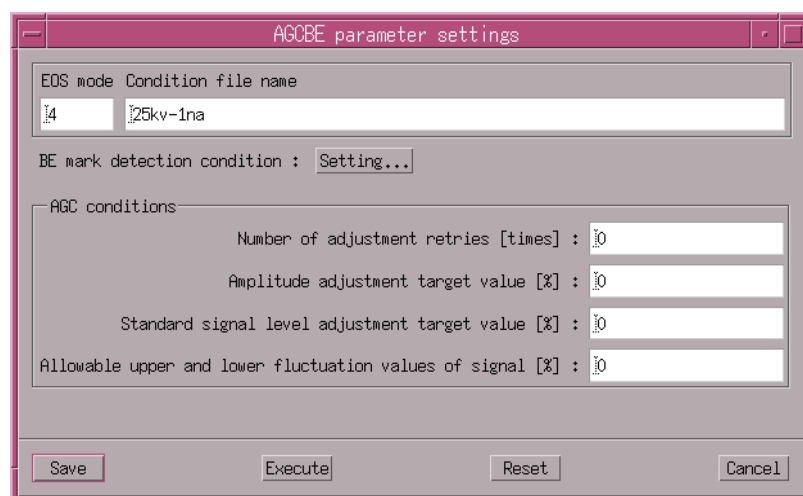
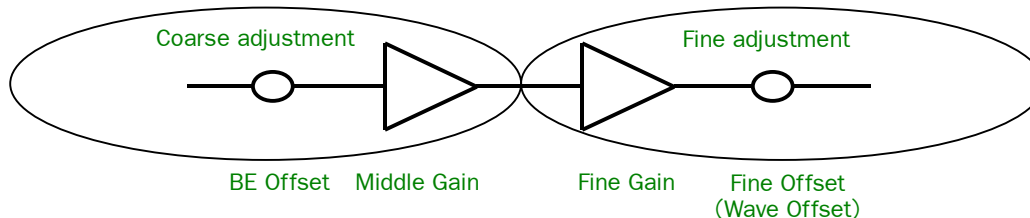


Fig. 12 AGCBE parameter setting Window

■ Descriptions of parameters

- BE mark detection condition (Section 4.42.2)
This button opens a window for setting BE mark-detection conditions.
- Number of adjustment retries [times]
This field lets you specify the number of retries in automatic gain control.
- Amplitude adjustment target value [%]
This field lets you specify the wave height to be adjusted in automatic gain control as a percentage of the amplitude. The recommended setting is about 90 %.
- Standard signal level adjustment target value [%]
This field lets you specify the lower limit of the waveform to be adjusted in automatic gain control as a percentage of the amplitude. The recommended setting is about 10 %.
- Allowable upper and lower fluctuation values of signal [%]
This field lets you set the tolerance for the Amplitude adjustment target value and the Standard signal level adjustment target value. When the waveform falls within this tolerance range, the program ends. With an Amplitude adjustment target value of 90 %, if you specify a tolerance of 10 %, the adjustment ends when the wave height falls within the range 80 to 100 %; with a Standard signal level adjustment target value of 10 %, the adjustment ends when the minimum waveform falls within the range 0 to 20 %.

■ BE mark/RG mark detection system



BE Offset	Middle Gain	Fine Gain	Fine Offset
0-4095	0-15	0-255	0-255

■ Updating the preamplifier constants

If AGCBE completes normally, the following inquiry window appears. You can update the amplifier constants by clicking on the subprograms to which you want to apply the update in the subprogram list and then clicking on **OK**. Selecting subprograms in the list highlights them. Normally, click on **All Select** below the subprogram list and then click on **OK** to update the preamplifier constants for all the subprograms that use the BE mark.

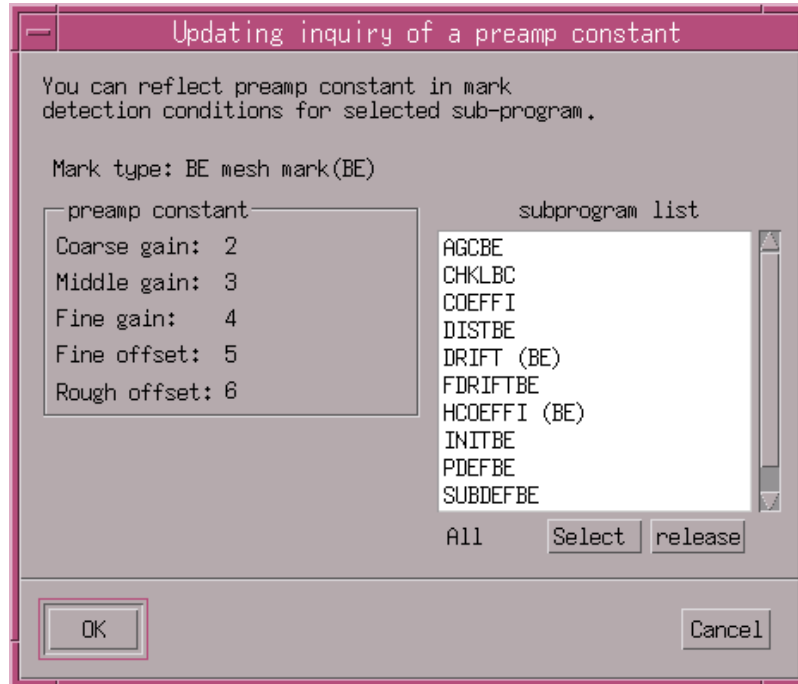


Fig. 13 Updating inquiry a preamp constant

4.3 AGCRG

AGCRG automatically adjusts the amplifier settings (gain, offset) for RG mark signal detection and displays the results. You can apply the amplifier settings calculated in AGCRG to other subprograms that use the BE mark.

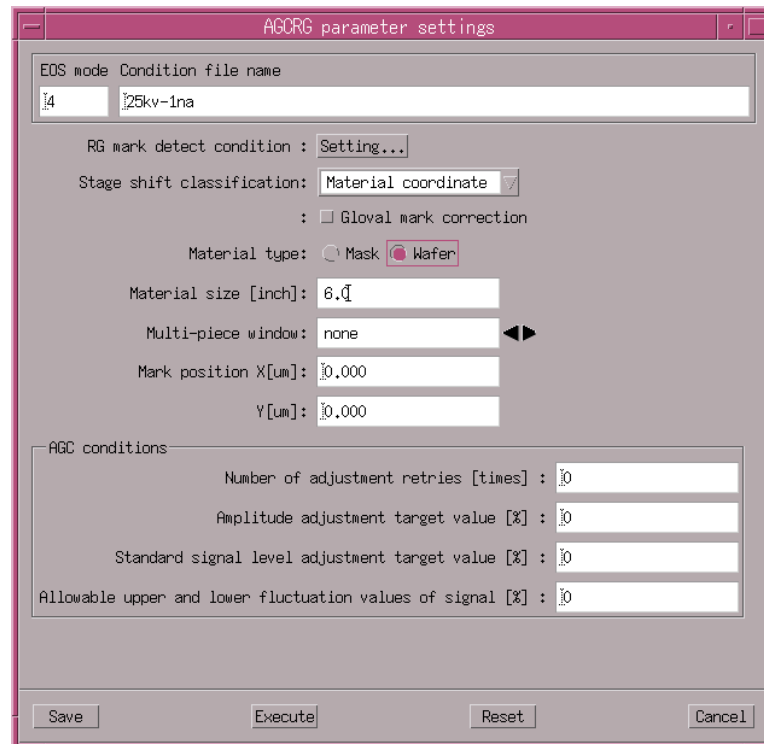


Fig. 14 AGCRG parameter setting Window

■ Descriptions of parameters

- RG mark detection condition (Section 4.42.3)
This button opens a window for setting registration mark-detection conditions.
- Stage shift classification
This field lets you specify the position of the registration mark in the stage-coordinate system.
- Global mark correction
This button lets you specify whether or not to move the stage with respect to the substrate rotation correction performed by SETWFR.
- Material type
Here you can select Mask or Wafer.
- Material size [inch]
This field lets you specify the substrate size in inches.
- Multi-piece window
For a multi-wafer cassette, this field lets you specify which piece is the wafer you want to measure. According to the Material type, Material size [inch], and Multi-piece window setting, the following Mark position is checked if it is in range.
- Mark position X, Y [μm]
These fields let you specify the position of the registration mark in the coordinate system specified as the Stage shift classification.

- Number of adjustment retries [times]
This field lets you specify the number of retries in automatic gain control.
- Amplitude adjustment target value [%]
This field lets you specify the wave height to be adjusted in automatic gain control as a percentage of the amplitude. The recommended setting is about 90 %.
- Standard signal level adjustment target value [%]
This field lets you specify the lower limit of the waveform to be adjusted in automatic gain control as a percentage of the amplitude. The recommended setting is about 10 %.
- Allowable upper and lower fluctuation values of signal [%]
This field lets you set the tolerance for the Amplitude adjustment target value and the Standard signal level adjustment target value. When the waveform falls within this tolerance range, the program ends. With an Amplitude adjustment target value of 90 %, if you specify a tolerance of 10 %, the adjustment ends when the wave height falls within the range 80 to 100 %; with a Standard signal level adjustment target value of 10 %, the adjustment ends when the minimum waveform falls within the range 0 to 20 %.

■ BE mark/RG mark detection system

Same as the BE mark/RG mark detection system of AGCBE. See the description of AGCBE.

■ Updating the preamplifier constants

If AGCRG completes normally, the following inquiry window appears. You can update the amplifier constants by clicking on the subprograms to which you want to apply the update in the subprogram list and then clicking on **OK**. Selecting subprograms in the list highlights them. Normally, click on **All Select** below the subprogram list and then click on **OK** to update the preamplifier constants for all the subprograms that use the registration mark.

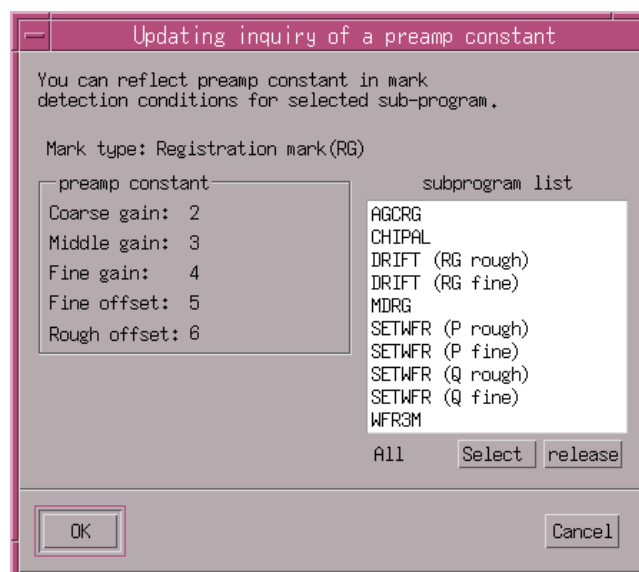


Fig. 15 Updating inquiry a preamp constant

4.4 BATCH

The BATCH subprogram is used to execute the CALIB subprograms set on the subprogram list shown below one by one from the top of the list. When you select **BATCH**, all the subprograms shown in the subprogram list are executed like the execution in the AUTO. The AUTO subroutine executes each subprogram only one time; however, the BATCH subroutine repeats subprogram execution from the top of the list by the specified cycle.

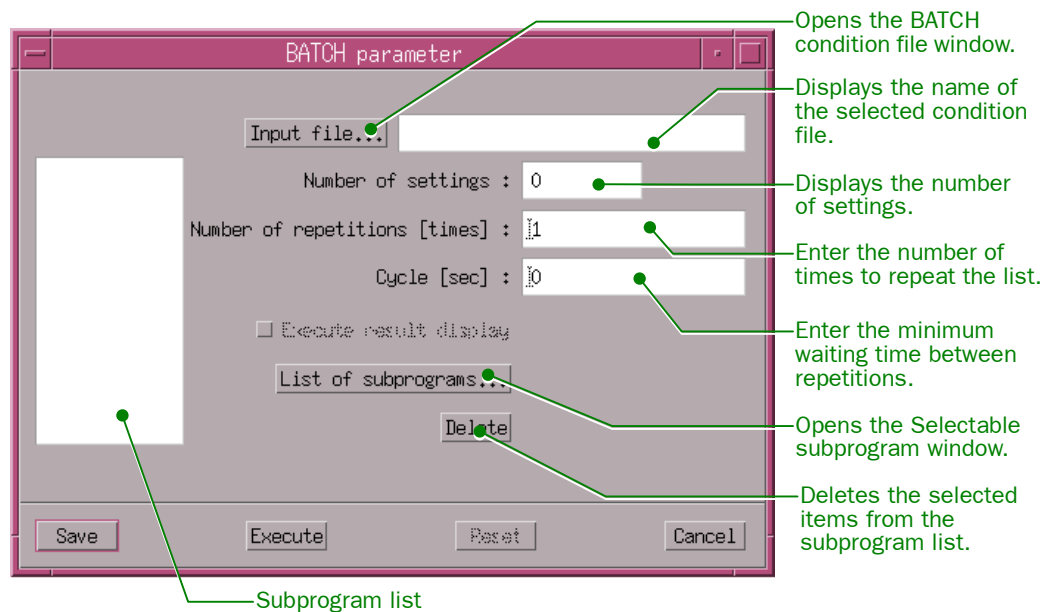


Fig. 16 BATCH parameter setting Window

Button name	Function
Save	Saves the contents of the subprogram list and stored in the present input file. Execution is not performed when the field of the input file is empty.
Execute	Performs the BATCH processing according to the contents of the present subprogram list.
Reset	Resets the parameters that you are editing.
Cancel	Closes the window.

- **Number of settings**

Displays the number of subprograms in the subprogram list. This field cannot be edited.

- **Number of repetitions and cycle**

Example:

Subprograms in the list

- INITAE
- INITBE
- CURRNT

Number of repetitions = 3 (times)

Cycle = 300 (sec)

The BATCH subroutine performs as follows under these conditions.

First cycle	INITAE execution INITBE execution CURRNT execution
Second cycle	INITAE execution INITBE execution CURRNT execution
Third cycle	INITAE execution INITBE execution CURRNT execution

-  If the first cycle ends within 300 seconds, the program waits 300 seconds after starting the first cycle before starting the second cycle. The second cycle does not start until at least 300 seconds have passed from the start of the first cycle.
-  If this cycle lasts more than 300 seconds, the next cycle starts immediately after the termination of this cycle

■ BATCH condition file window

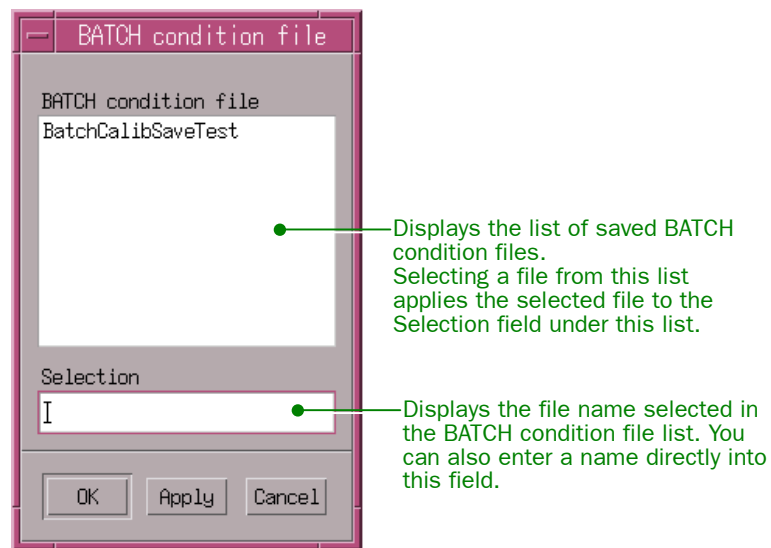


Fig. 17 BATCH condition file window

Button name	Function
OK	The subprogram list is replaced with the contents of the BATCH condition file in the Selection field and the window is closed.
Apply	The subprogram list is replaced with the contents of the BATCH condition file in the Selection field.
Cancel	Closes the window.

To create a condition file, enter a condition-file name (that is not found in the list) directly in the Selection field and press the **OK** or **Apply** button. The subprogram list is not changed in this case.

■ Selectable subprogram window

This window lists the subprograms that you can run in BATCH. Pressing the button of a subprogram name appends it to the subprogram list.



Fig. 18 Selectable subprogram window

Button name	Function
Close	Closes the window.

 The subprogram AUTOAL is optional, so it functions only for customers who have purchased it.

4.5 CENTER

CENTER centers the deflector. In the specified Measurement mode, CENTER moves the mark according to the Measurement field size and detects the mark; then, from the difference between the mark position directly under the beam and the mark position with respect to the field size, the amount of shift correction is calculated for the deflector.

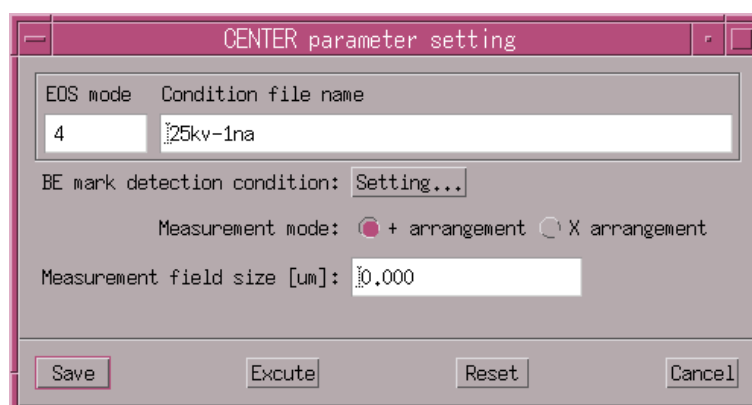


Fig. 19 CENTER parameter setting

- BE mark detection condition (Section 4.42.2)
This button opens a window for setting the registration mark detection conditions.
- Measurement mode
+ arrangement: The measurement sequence is center-> left-> right-> top-> bottom.
X arrangement: The measurement sequence is center-> top left-> top right-> bottom right-> bottom left.
- Measurement field size [μm]
This field lets you specify the field size for centering the deflector.

4.6 CHIPAL (Optional)

Based on the substrate's shift, expansion/contraction, and rotation obtained by SETWFR, CHIPAL detects the chip marks on the substrate and calculates the chip distortion. This command is useful when you purchase the optional 4 chip marks alignment system.

Fig. 20 CHIPAL parameter window

■ Descriptions of parameters

- RG mark detection condition (Section 4.42.3)
This button opens a window for setting registration mark-detection conditions.
- Measurement mode
This field lets you select and specify the chip-mark detection mode and the writing correction mode.
This table describes each mode.

Mode	Description
mode 0	In writing, chip marks are not detected, but the displacement (shift amount) at the chip-mark center and the size (gain) and rotation of the beam deflection region are corrected according to the global mark's displacement (shift amount) obtained from global-mark detection, and the expansion/contraction and rotation correction amount of the substrate.
	The Calibration Program does not perform program operations such as mark detection even if it performs internal processing.
mode 1	In writing, chip mark M1 is detected to correct for the displacement (shift amount) at the chip-mark center.
	In calibration, the displacement is measured and the result is displayed.
mode 4	In writing, chip marks M1, M2, M3, and M4 are detected to correct the displacement (shift amount) at the chip-mark center, the size (gain) and rotation of the beam deflection region, and the chip distortion.
	In calibration, the displacement at the chip center and the size and rotation of the beam deflection region are measured and displayed.
mode S	In writing, the position of chip mark M1 is detected, using SEM observation, to correct for the displacement (shift amount) at the chip-mark center.
	In calibration, the stage moves to chip mark M1, and the operator is prompted to continue the operation with the SEM image. After the operator finds M1 and registers it, the displacement is measured and the result is displayed.
mode V1	In writing at the position of chip mark M1, only the height is measured, without detecting the mark, to correct the size (gain) and rotation of the beam deflection region and the focus value of the objective lens.
	In calibration, the height is measured at the position of chip mark M1 and the result is displayed.
mode V4	In writing at the positions of chip marks M1, M2, M3, and M4, only the height is measured, without detecting the marks, to correct the size (gain) and rotation of the beam deflection region and the focus value of the objective lens.
	In calibration, the height is measured at the positions of chip marks M1, M2, M3, and M4, and the measured heights and the average of the heights at the four points are displayed.

- AGCRG execution

With a check in the Execute box next to AGCRG, the system performs AGC (Automatic Gain Control) for mark detection during the execution of CHIPAL.

 This parameter applies also to the writing.

- Height measurement execution

With a check in the Execute box next to Height measurement execution, the system measures the height during the execution of CHIPAL.

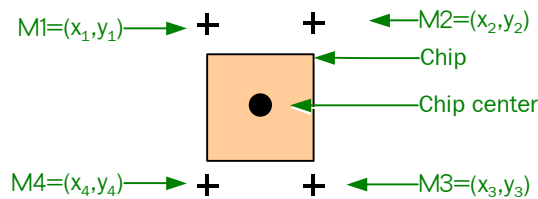
- Chip center coordinate position X, Y [μm]

This field lets you specify the chip center coordinate position in the substrate coordinate system.

- Chip mark M1 design position X, Y [μm]

This field lets you specify the position of chip mark M1 in the above-mentioned coordinate system where the chip center is at the origin (chip center coordinate system; X increases in the right direction, Y increases in the upward direction).

- Chip mark M2 design position X, Y [μm]
This field lets you specify the position of chip mark M2 in the above-mentioned coordinate system where the chip center is at the origin (chip center coordinate system; X increases in the right direction, Y increases in the upward direction).
 - Chip mark M3 design position X, Y [μm]
This field lets you specify the position of chip mark M3 in the above-mentioned coordinate system where the chip center is at the origin (chip center coordinate system; X increases in the right direction, Y increases in the upward direction).
 - Chip mark M4 design position X, Y [μm]
This field lets you specify the position of chip mark M4 in the above-mentioned coordinate system where the chip center is at the origin (chip center coordinate system; X increases in the right direction, Y increases in the upward direction).
-  If the coordinates of M1, M2, M3, and M4 are $M1=(x_1, y_1)$, $M2=(x_2, y_2)$, $M3=(x_3, y_3)$, and $M4=(x_4, y_4)$, the relationships among them are as follows:
- $x_2 - x_1 > 0$
 - $x_3 - x_1 > 0$
 - $y_1 - y_3 > 0$ and $y_2 - y_3 > 0$
 - $x_3 - x_4 > 0$ and $x_2 - x_4 > 0$
 - $y_1 - y_4 > 0$ and $y_2 - y_4 > 0$
- The positional relationship is as follows.



- Rotation allowance [degree]
This field lets you specify the tolerance for the rotation of the chip marks.

■ Descriptions of parameters in the HEIGHT section

Use the HEIGHT parameters if you want to check the value of the writing job command HEIGHT before writing patterns on a substrate whose height cannot be measured or cannot be measured precisely. Normally, remove the check from the Height measurement execution box before executing the subprogram.

- The amount of minute compensation of a focus lens [point]
If you want to add an offset to the focus lens value of the objective lens, set the offset here.
- The amount of minute compensation of beam position shift X, Y [nm]
This field lets you specify the beam shift amounts for changing the focus lens value of the objective lens.
- The amount of minute compensation of a main deflector gain X, Y [point]
This field lets you specify the fine gain compensation amounts of the main deflector for the difference between the height reference plane and the substrate height.
- The amount of minute compensation of a main deflector rotation X, Y [point]
This field lets you specify the amount of minute compensation of a main deflector rotation between the height reference plane and the substrate height.

■ Descriptions of parameters in the SUBHEI section

Use the SUBHEI parameters if you want to check the value of the writing job command SUBHEI before writing patterns on a substrate whose height cannot be measured or cannot be measured precisely. Accordingly, in general, remove the check from Height measurement execution before execution.

- The amount of minute compensation of a sub deflector shift X, Y [point]
This field lets you specify the fine shift compensation amounts of the sub-deflector for the difference between height reference plane and the height of the substrate.
- The amount of minute compensation of a sub deflector gain X, Y [point]
This field lets you specify the fine gain compensation amounts of the sub-deflector for the difference between height reference plane and the height of the substrate.
- The amount of minute compensation of a sub deflector rotation X, Y [point]
This field lets you specify the fine rotation compensation amounts of the sub-deflector for the difference between height reference plane and the height of the substrate.

4.7 CHKFOC (Optional)

According to the amount of change of the beam position, tolerance for the beam size, and tolerance for the amount of distortion, CHKFOC determines whether the dynamic focus correction value is effective in the present situation.

This command is useful when you purchase the optional DF/DS system.

CHKFOC parameter settings

EOS mode: 4 Condition file name: 25kv-1na

AE mark detection condition: Setting...

Correction memory: ☐ Invalid

Zigzag: ☒ X->Y ☐ Y->X

Measurement start point: Top left

Measurement field size [um]: 120,000

Number of first quadrant measurement points [point]: 5

First quadrant measurement pitch: Setting...

Allowable value of FD center beam size drift[um]: 0.000

Allowable value of beam size difference with respect to FD center [um]: 0.000

Allowable value of deformation difference with respect to FD center [um]: 0.000

Save Execute Reset Cancel

Fig. 21 CHKFOC parameter settings window

- **AE mark detection condition (Section 4.42.1)**
This button opens a window for setting the AE mark detection conditions.
- **Correction memory**
This button lets you select whether to enable or disable the dynamic focus correction during measurement.
- **Zigzag**
Here you can specify the measurement sequence. Normally, specify X->Y.

Measurement sequence	Description
X->Y	In the field, zigzag measurement proceeds in the x direction.
Y->X	In the field, zigzag measurement proceeds in the y direction.

- **Measurement start point**
This field lets you select the measurement start point. Normally, select Top left.

Measurement start point	Description
Top left	Measurement starts from the top left of the field.
Top right	Measurement starts from the top right of the field.
Bottom right	Measurement starts from the bottom right of the field.
Bottom left	Measurement starts from the bottom left of the field.

- **Measurement field size [μm]**
This field lets you specify the measurement field size.
- **Number of first quadrant measurement points [point]**
This field lets you specify the number of measurement points in the first quadrant, with the field center at the origin. The points include the ones on the axes.
 For example, if you want to measure 5×5 points in the deflection field, specify 3.
- **First quadrant measurement pitch**
This button lets you specify the measurement pitches. Since the first measurement point is on an axis and the last measurement point is at a corner of the measurement field, the number of points (pitches) you specify is the number of first-quadrant measurement points minus two; the rest will be calculated automatically.
- **Allowable value of FD center beam size drift [μm]**
This field lets you specify the allowance for the difference between the beam positions at the field center and on the measurement field.
- **Allowable value of beam size difference with respect to FD center [μm]**
This field lets you specify the allowance for the difference between the beam sizes at the field center and on the measurement field.
- **Allowable value of deformation difference with respect to FD center [μm]**
This field lets you specify the allowance for the difference between the amounts of distortion at the field center and on the measurement field.

■ First quadrant measurement pitch settings window

Clicking **First quadrant measurement pitch** opens this window. The measurement pitch is the pitch from the measurement point outside the field to the next measurement point. The number of measurement pitches that you can set is two less than the number of measurement points in the first quadrant. Select the measurement pitch that you want to change, and type a value. Clicking **Apply–Close** puts the changes to the first-quadrant measurement pitches into effect.

- ✎ When you click **Save** or **Execute** in the parameter setting window, the program does not use the settings presently displayed in the First quadrant measurement pitch settings window, but uses the settings that you have saved in the program by clicking **Apply**.

Fig. 22 First quadrant measurement pitch settings window

Button	Description
Apply	Saves the settings in the program.
Reset	Discards the settings that you are editing, and displays the settings stored in the program in the text input fields.
Close	Closes the window without saving the settings that you are editing.

4.8 CHKLBC

The CHKLBC subprogram automatically measures the scaling coefficients of the LBC (laser beam controller for positioning).

Fig. 23 CHKLBC parameter setting window

- BE mark detection condition (Section 4.42.2)
This button opens a window for setting BE mark-detection conditions.
- Measurement field size [μm]
This field lets you specify the measurement field size.
 Since the LBC correction amount is $\pm 10 \mu\text{m}$, $10 \mu\text{m}$ is recommended.
- Zigzag
You can specify the measurement sequence. Normally, specify X->Y.

Measurement sequence	Description
X->Y	In the field, zigzag measurement is made in the x direction.
Y->X	In the field, zigzag measurement is made in the y direction.

- Measurement starting point
This field lets you select the measurement starting point. Normally, select **Top left**.

Measurement starting point	Description
Top left	Measurement starts from the top left of the field.
Top right	Measurement starts from the top right of the field.
Bottom right	Measurement starts from the bottom right of the field.
Bottom left	Measurement starts from the bottom left of the field.

- Number of back-and-forth motions [times]
This field lets you specify the number of back-and-forth measurements.
- Measurement points in first quadrant [point]
Here you can specify the number of measurement points in the first quadrant, with the field center at the origin. The measurement points include the ones on the axes.
 For example, if you want to measure 5×5 points in the deflection field, specify 3.
- First quadrant measurement pitch
This button lets you specify the measurement pitch. Since the first measurement point is on an axis and the last measurement point is at a corner of the measurement field, the number of points (pitches) you specify is the number of measurement points in the first quadrant minus two; the rest will be calculated automatically.
 CHKLBC measures 3×3 points because the measurement field size is small. Accordingly, the measurement pitch in the first quadrant is half of the measurement field size; you do not have to change this parameter.
- Number of one-side measurements [times]
This field lets you specify the number of changes of the scaling value in the measurement in which the LBC scaling value changes.
- Scaling conversion step X, Y [%]
This field lets you specify the step size for changing the scaling value in the measurement in which the LBC scaling value changes.
If the measuring conditions, for example, are,
 - Scaling value: 1.0,
 - The number of times in one-way measurement: 2,
 - Step X and Y: 1.00 [%],
 the measurement is performed after changing the scaling value by 0.01 on both sides of the scaling value 1.0.
The LBC correction coefficient measurement is performed by changing the scaling value to 0.98, 0.99, 1.00, 1.01 and 1.02 in this case.
While changing the LBC scaling value, CHKLBC measures the distortion and automatically measures the LBC correction coefficient.
- Maximum permissible distortion of back-and-forth measurement average values [μm]
This field lets you specify the permissible value for averaging the results of back-and-forth distortion measurements at each point.
- Allowable positional difference of identical point back-and-forth motion [μm]
This field lets you specify the tolerance among the positions resulting from back-and-forth positional measurements at each point.
- Tolerance between right and left points [μm]
This field lets you specify the tolerance for the difference between the two values measured at two points that are symmetric with respect to the y-axis, with the measurement field center at the origin.
- Tolerance between upper and lower points [μm]
This field lets you specify the tolerance for the difference between the two values measured at two points that are symmetric with respect to the x-axis, with the measurement field center at the origin.
- Allowable difference from the scale set before measurement [%]
This field lets you specify the tolerance for the difference between the scaling value before CHKLBC execution and the scaling value measured this time.

■ First quadrant measurement pitch setting window

Clicking **First quadrant measurement pitch** opens this window. The window is the same as with CHKFOC. The measurement pitch is the pitch from the measurement point outside the field to the next measurement point. The number of measurement pitches that you can set is two less than the number of measurement points in the first quadrant. Select the measurement pitch that you want to change, and enter a value. Clicking **Apply–Close** puts the changes to the first-quadrant measurement pitches into effect.

 The values used when you click the **Save** or **Execute** button on the CHKLBC parameter window are not those shown on this window but those applied and stored in the program.

Button name	Function
Apply	This button saves your settings in the program.
Reset	This button discards the parameters you are editing, and displays the settings stored in the program in the text input fields.
Close	This button closes the window, discarding the parameters you are editing.

4.9 COEFFI

COEFFI automatically measures the main/sub deflection correction (GAIN, ROT, SHIFT) weighting factor (DAC value point→amount of correction [μm] conversion factor).

 The results of executing this subprogram are used when the subprogram PDEFBE, PDEF, SUBDEFBE, or SUBDEF is executed.

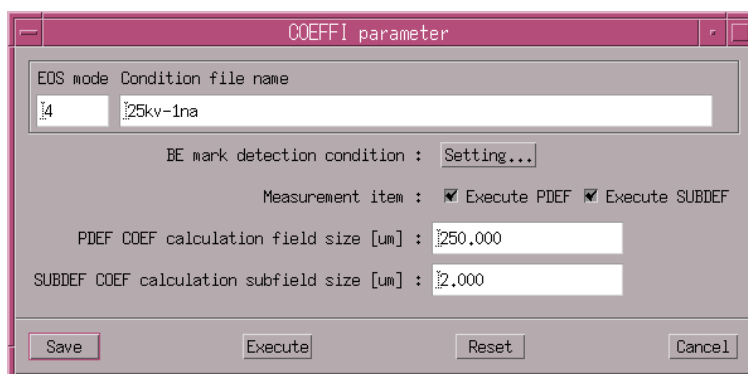


Fig. 24 COEFFI parameter setting window

- BE mark detection condition (Section 4.42.2)
This button opens a window for setting AE mark-detection conditions.
- BE mark detection condition (Section 4.42.2)
This button opens a window for setting BE mark-detection conditions.
- Measurement item
You can specify measurement items by clicking on either or both of the Execute PDEF and Execute SDEF boxes.
 Measurement is performed only for the selected items. An error occurs when no measurement items are selected, preventing parameter execution or subprogram execution.

- PDEF COEF calculation field size [μm]

This field lets you specify the value obtained by subtracting the scan width from the maximum field size of the main deflector.

 The maximum field size of the main deflector is as follows:

Accelerating voltage	4th lens mode	5th lens mode
25 kV	2000 μm	250 μm
50 kV	1000 μm	125 μm
100 kV	500 μm	62.5 μm

- SDEF COEF calculation subfield size [μm]

This field lets you specify the maximum subfield size of the sub-deflector.

 The maximum subfield size of the sub-deflector is as follows:

Accelerating voltage	4th lens mode	5th lens mode
25 kV	16 μm	2 μm
50 kV	8 μm	1 μm
100 kV	4 μm	0.5 μm

4.10 CURRNT

The CURRNT subprogram measures the beam current.

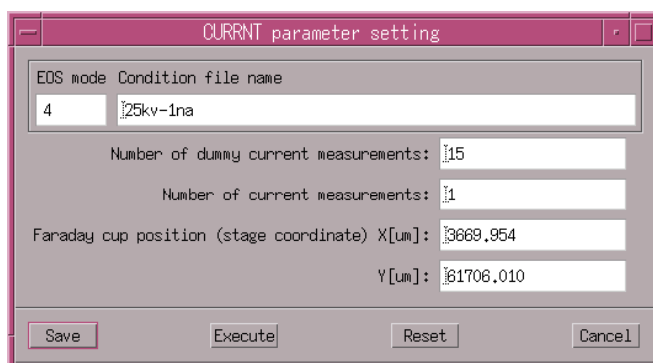


Fig. 25 CURRNT parameter setting window

- Number of dummy current measurements [times]

This field lets you set the number of dummy current measurements. Due to a hardware limit, the number must be greater than or equal to five.

 The recommended number of dummy measurements is about five.

- Number of current measurements [times]

This field lets you set the number of current measurements.

 The average of the currents measured will be used in writing, so the recommended number is about five.

- Faraday cup position (stage coordinate) X, Y [μm]

These fields let you specify the Faraday cup position as stage coordinates.

4.11 DCURRNT

DCURRNT automatically measures the dark current. It has no parameter window.

4.12 DFDSSEM (Optional)

Using the SEM image, you can measure and set the dynamic focus (DF) and dynamic astigmatism (DS) correction values manually, write in the correction memory, and create and save a correction file.

This command is useful when you purchase the optional DF/DS system.

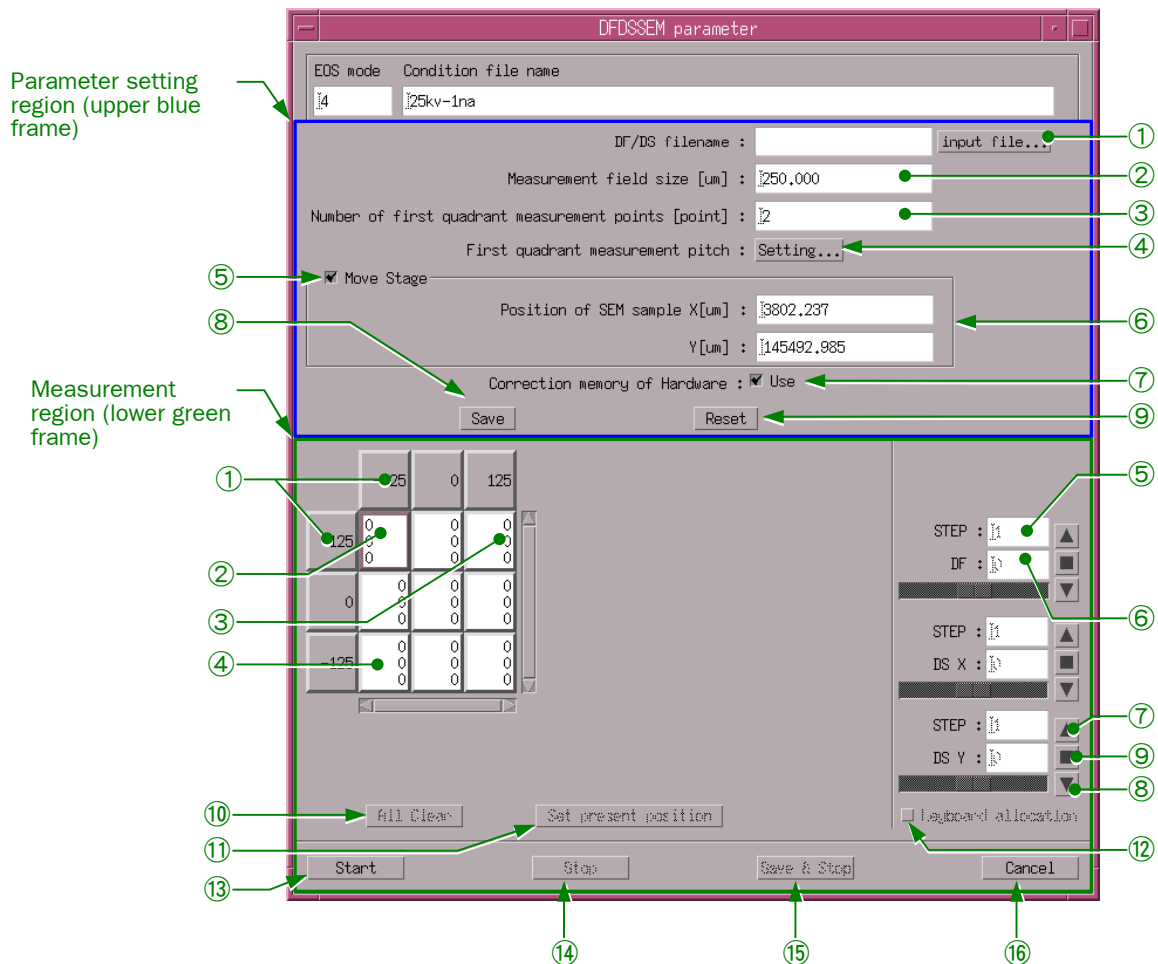


Fig. 26 DFDSSEM parameter setting window

■ Parameter setting region (upper blue frame)

① DF/DS filename

You can browse to select an existing correction filename. Clicking on the **input file...** button opens a selection window.

DF/DS files are present in the directory /usr/ebsys/log/, and the format of their names is year_month_day_hour_minute_second.dfocuse.ic.

② Measurement field size [μm]

This field lets you specify the measurement field size.

③ Number of first quadrant measurement points [point]

Here you can specify the number of measurement points per quadrant.

 If you want to measure 5 × 5 points in the field plane, enter 3.

- ④ **First quadrant measurement pitch**
This button lets you specify or check the measurement pitches. Clicking here opens a window for entering the measurement pitches. The window is the same as with CHKFOC. The directional sequence for entering pitches is from the outside to the center of the field. If you equalize the measurement pitches, entering the field size and the number of first-quadrant measurement points automatically calculates the measurement pitches.
- ⑤ **Move Stage**
Use this feature if you want to use the same SEM sample position continuously for DF, DSX, and DSY measurements in the field plane. Putting a check here using the mouse will automatically move the stage according to the deflection, maintaining the same measurement or observation location.
- ⑥ **Position of SEM sample**
This field displays the SEM sample position for astigmatism adjustment. The SEM sample position for astigmatism adjustment in the Stage GUI is used here.
- ⑦ **Correction memory of hardware**
Putting a check in the Use box will read and display the present correction memory of the hardware in the table in the Measurement region when the measurement starts. This will enable you to start the measurement using the present correction memory conditions.
- ⑧ **Save**
This button saves the Measurement field size (②), the Number of first quadrant measurement points (③), and the First quadrant measurement pitch (④).
- ⑨ **Reset**
This button resets the Measurement field size (②), the Number of first quadrant measurement points (③), and the First quadrant measurement pitch (④).

■ Measurement region (lower green frame)

- ① **Coordinates in the field**
Coordinates in the measurement field appear in μm . The center of the measurement field is the origin.
- ② **Points already measured (blue)**
The blue part is already measured.
- ③ **Point being measured (yellow)**
The yellow part is being measured.
- ④ **Points not yet measured (white)**
The white part is not yet measured.
- ⑤ **STEP**
This field lets you enter the step by which you can change the DF value. Place the mouse pointer in the box, and enter a value using the keyboard. The same applies to DSX and DSY.
 The recommended values are 50 points for DF and 5 points for DSX and DSY.
- ⑥ **Register value**
This field displays the DF register value at the point presently measured. The same applies to DSX and DSY.

- ⑦ ▲ button
Each time you click on this button, the value you set as STEP is added to the present DF register value. The same applies to DSX and DSY.
- ⑧ ▼ button
Each time you click on this button, the value you set as STEP is subtracted from the present DF register value. The same applies to DSX and DSY.
- ⑨ ■ button
Clicking on this button sets the DF register value to zero. The same applies to DSX and DSY.
- ⑩ All Clear
Clicking on this button sets the DF, DSX, and DSY register values at all the measurement points to zero.
- ⑪ Set present position
Use this feature if you want to enter a new present stage position. Moving to the next measurement point displays the SEM image at the new position.
- ⑫ Keyboard allocation
Put a check here if you want to change the DF, DSX, and DSY register values using the numeric keypad without using the ⑦, ⑧, and ⑨ buttons.
 The numbers in the table indicate keys on the numeric keypad.
- | | + Direction value change | - Direction value change | Zero value setup |
|--------------------|--------------------------|--------------------------|------------------|
| DF register value | 7 | 8 | 9 |
| DSX register value | 4 | 5 | 6 |
| DSY register value | 1 | 2 | 3 |
- ⑬ Start
Check the field size, number of measurement points, and measurement pitch in the Parameter setting region, and then press the **Start** button to start the measurement. If the Use box of Correction memory of hardware has a check, a reading time of about one minute is required. When the reading of the correction memory is completed, the DF, DSX, and DSY values at each point appear.
- ⑭ Stop
After starting the measurement, if you want to stop it, press this button. If you stop the measurement, the program does not create a DF/DS correction file.
- ⑮ Save & Stop
After the measurement is completed at all of the points, pressing this button creates and saves a DF/DS correction file in the directory /usr/ebsys/log/ under the name year_month_day_hour_minute_second.dfocuse.ic.
- ⑯ Cancel
This button closes the DFDSSEM parameter setting window.

■ Procedure

1. Open the DFDSSEM parameter setting window.
2. To start the measurement, select one of the following:

- Specify an existing DF/DS correction file
The measurement field size, number of measurement points, measurement pitch, and each register value appear. Changing the measurement field, number of measurement points, or measurement pitch sets all the DF/DS correction values to zero. Specifying a correction filename does not write data in the correction memory.
 - Use the present DF/DS correction memory
You can specify any field size and number of measurement points. Each correction value read from the correction memory appears.
 - Specify the default (zero)
Specifying only the measurement field size, number of measurement points, and pitch sets all the correction values to zero.
3. Press **Save** to save the measurement field size, number of measurement points, and measurement pitch at startup.
If you modify the field size, number of measurement points, or measurement pitch after DFDSSEM starts, pressing **Reset** resets them to the startup conditions.
 4. Put a check in the **Move Stage** box (recommended), and press **Start**.
This moves the stage to the SEM sample position, displaying the SEM image in the Image GUI.
 5. Click on a measurement position in the field.
The SEM image at the specified position remains displayed after the stage moves. The part not yet specified is displayed in white, the part presently specified is in yellow, and the part already specified is in blue.
 6. Specify the steps by which to change the dynamic focus (DF), dynamic astigmatism X (DSX), and dynamic astigmatism Y (DSY) correction values.
 7. Observe the SEM image while changing the DF, DSX, and DSY correction values, and determine the optimum correction values.
 8. Operate the ▲, ▼, and ■ buttons to change the values.
The ▲ button changes the value in the + direction, the ▼ button changes the value in the – direction, and the ■ button sets the value to zero. Also, the Keyboard allocation feature enables you to assign these operations to keys on the numeric keypad.
 9. During measurement, if you want to set all the measured values to zero, press **All Clear**.
 10. If you have changed the observation position using the Stage GUI, press **Set present position**.
You can change the observation position. However, the SEM sample position coordinates do not change.
 11. During measurement, press **Stop**.
This enables you to change the measurement conditions and start the measurement again.
 12. After the measurement is completed, press **Save & Stop**.
This writes the DF/DS correction values in the correction memory and creates a correction file.
 13. 13. Press **Cancel** to exit DFDSSEM.
You can write data in the correction memory and create a correction file in this procedure.

4.13 DFOCUS (Optional)

DFOCUS performs the automatic measurement of the dynamic focus (DF) and dynamic astigmatism (DS) correction files, creates a correction file, and sets up the system. The following directory will contain the correction file under the name year_month_day_hour_minute_second.dfocustic:

/usr/ebsys/log/

✍ For example, if DFOCUS starts at 13:15:45 on December 14, 2005, in the EOS mode 3, the filename is as follows:

20051214131545.3.dfocustic

This command is useful when you purchase the optional DF/DS system.

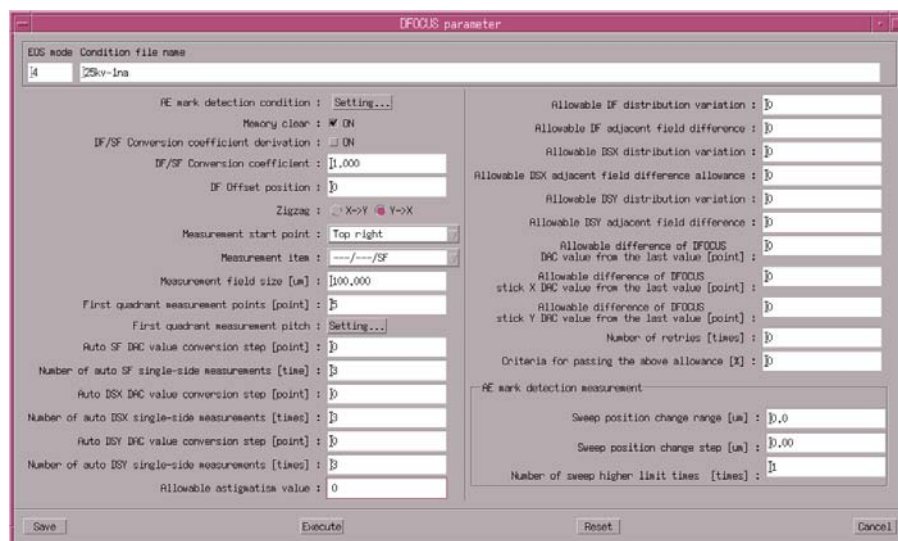


Fig. 27 DFOCUS parameter setting window

- AE mark detection condition (Section 4.42.1)
This button opens a window for setting AE mark-detection conditions.
- Memory clear
If you put a check in the ON box, the measurement will start with all the present hardware DF/DS correction memory cleared (to zero).
- DF/SF Conversion coefficient derivation
Selecting ON measures and calculates the DF/objective lens correction conversion coefficient, and performs measurement while changing the focus value (Static Focus) of the objective lens instead of that of the DF lens.
✍ As compared to the objective lens, the DF lens has a narrower correction range, so it is recommended that you select DF/SF Conversion coefficient derivation.
- DF/SF Conversion coefficient
If you enable the above parameter DF/SF conversion coefficient derivation and perform the measurement, the result will apply here after the measurement is completed.
✍ When you execute DFOCUS for the first time, this field might contain zero. With zero, an error occurs during execution. Thus, enter a convenient number such as 1, save it, and then execute DFOCUS. After the measurement, the result appears.

- DF Offset position

DF is usable in the range of ± 2048 (point). However, by the nature of DF, the values are positive in the direction from the center to the outside of the field. Therefore, if you want to utilize DF in the entire range of ± 2048 (point) for a large field, enter -2000 in the DF offset position field.

 If you want to use this parameter, be sure to enable the above Memory clear. With Memory clear disabled, the DF offset position is not used in measurement.

- Zigzag

You can specify the measurement sequence. Normally, specify X->Y.

Measurement sequence	Description
X->Y	In the field, zigzag measurement is made in the x direction.
Y->X	In the field, zigzag measurement is made in the y direction.

- Measurement start point

This field lets you select the measurement start point. Normally, select **Top left**.

Measurement start point	Description
Top left	Measurement starts from the top left of the field.
Top right	Measurement starts from the top right of the field.
Bottom right	Measurement starts from the bottom right of the field.
Bottom left	Measurement starts from the bottom left of the field.

- Measurement item

This field lets you select the items you want to measure. Normally, select **DF/DSX/DSY**.

Item	Description
DF/---/---	FOCUS
---/DSX/---	STIG-X
DF/DSX/---	FOCUS, STIG-X
---/---/DSY	STIG-Y
DF/---/DSY	FOCUS, STIG-Y
---/DSX/DSY	STIG-X, STIG-Y
DF/DSX/DSY	FOCUS, STIG-X, STIG-Y

- Measurement field size [μm]

This field lets you specify the measurement field size.

- For 100 kV, the maximum field size is 500 μm . Thus, if the scan width is 8 μm , it is recommended that you specify 492 (μm), which is obtained by subtracting the scan width from the maximum field size.
- For 50 kV, the maximum field size is 1000 μm . Thus, if the scan width is 8 μm , it is likewise recommended that you specify 992 (μm).

- First quadrant measurement points [point]

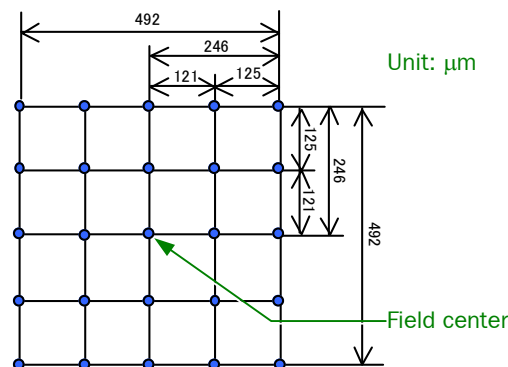
Here you can specify the number of measurement points in the first quadrant. Specifying n results in measuring $(2n-1) \times (2n-1)$ points in the field.

- First quadrant measurement pitch

Clicking on this button opens the window shown below. The window is the same as the input window of CHKFOC. Measurement pitch 1 means the pitch from the point outside the field to the next point. The number of pitches you can set is equal to the number of measurement points in the first quadrant–2. Select the pitch you want to change, and enter a value using the keyboard. Clicking on **Apply–Close** puts the changes in the measurement pitches in the first quadrant into effect.

Button name	Function
Apply	This button saves the values you are editing in the program.
Reset	This button resets the values you are editing to the values stored in the program.
Close	This button resets the values you are editing, and closes the window.

 If the measurement field size is 492 μm ; the scan width is 8 μm ; the number of first quadrant measurement points is 3 (5 $\times$ 5 measurement); and measurement pitch 1 in the first quadrant is 125 μm , then the measurement pitches among 5 $\times$ 5 points are as follows.



- Auto SF DAC value conversion step [point]
This field lets you set the step size for changing the focus value of the objective lens.
- Number of auto SF single-side measurements [time]
This field lets you specify how many points to measure as the focus value of the objective lens changes. Specifying n results in measuring (2n–1) points.
- Auto DSX (DSY) DAC value conversion step [point]
This field lets you set the step size for changing the DSX (DSY) value.
- Number of auto DSX (DSY) single-side measurements [times]
This field lets you specify how many points to measure as the DSX (DSY) value changes. Specifying n results in measuring (2n–1) points.
- Allowable astigmatism value
This field lets you set the allowable value for convergence evaluation in the DSX/DSY measurement.
- Allowable DF (DSX, DSY) distribution variation
Here you specify the tolerance for the variation (p-p) of DF (DSX, DSY) in the field.
- Allowable DF (DSX, DSY) adjacent field difference
This field lets you specify the tolerance for the difference between adjacent measurement points of DF (DSX, DSY).

- Allowable difference of DFOCUS DAC value from the last value [point]
This field lets you specify the tolerance for the difference between the DF values at the start of and after the execution of DFOCUS.
- Allowable difference of DFOCUS stick X (Y) DAC value from the last value [point]
This field lets you specify the tolerance for the difference between the DSX (DSY) values at the start of and after the execution of DFOCUS.
- Number of retries [times]
This field lets you specify the upper limit on the number of retries in DFOCUS execution. DFOCUS will be repeated until DS decreases below the Allowable astigmatism value. Exceeding the upper limit on the number of retries causes an error, terminating the measurement.
- Criteria for passing the above allowance [%]
This field lets you specify the criteria to assess the percentage of passed measurement points after evaluation with tolerances.

■ AE mark detection measurement

During DFOCUS execution, the beam scan contaminates the mark. Because of this, after the beam is scanned the specified number of times, the process moves to the next mark position and measures it. You can set the conditions for this mark movement.

- Sweep position change range [μm]
This field lets you specify the range of the scan position.
 - Sweep position change step [μm]
This field lets you specify the step value for shifting the scan position.
 - Number of sweep higher limit times [times]
This field lets you specify the upper limit on the number of times the same mark position is scanned during DFOCUS execution. When this limit is reached, the process moves to the next scan position and continues the measurement.
-  This parameter depends on the beam current you want to use. For example, if the beam current is 2 nA, the recommended number of times is about 200.

4.14 DISTBE

DISTBE measures the beam deflection distortion in the field plane, writes data in the deflection distortion correction memory, and creates a deflection distortion correction file (the amount of distortion immediately under the beam is assumed to be zero). The following directory will contain the correction file under the name year_month_day_hour_minute_second.distbe.ic:

/usr/ebsys/log/

✂ For example, if DISTBE starts at 13:15:45 on December 14, 2005, in the EOS mode 3, the filename is as follows:

Filename: 20051214131545.3.distbe.ic

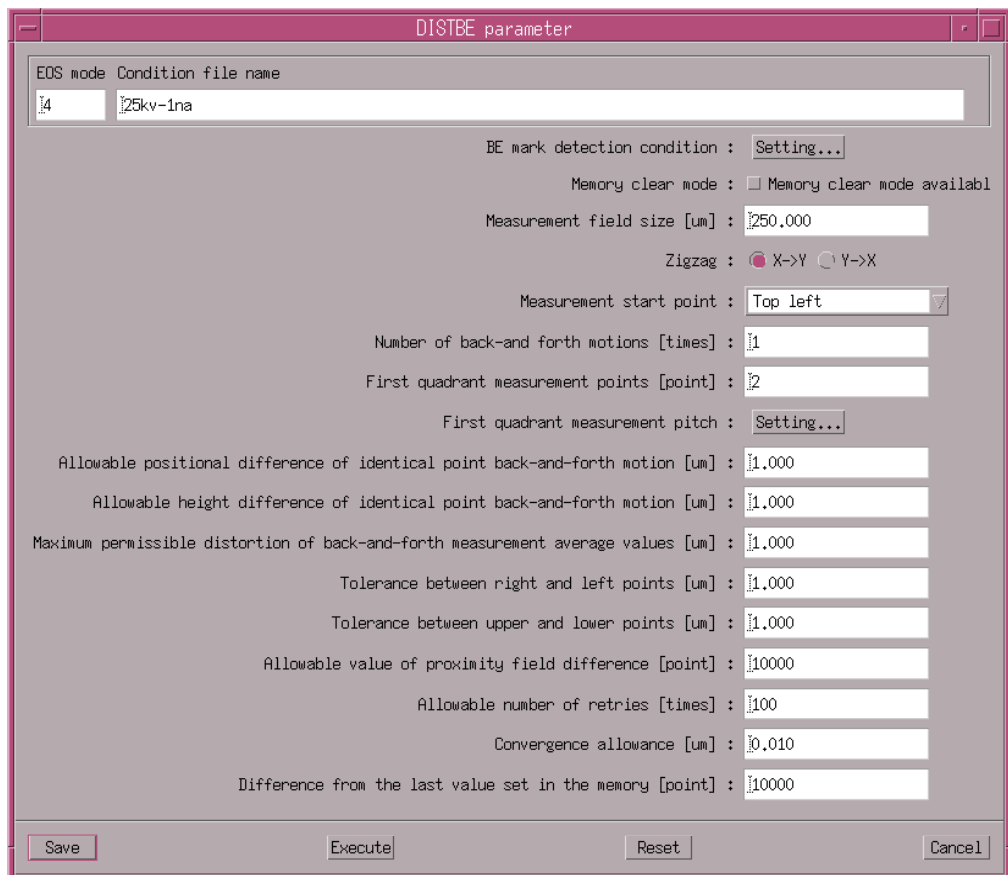


Fig. 28 DISTBE parameter setting window

- BE mark detection condition (Section 4.42.2)
This button opens a window for setting BE mark-detection conditions.
- Memory clear mode
Here you can select whether or not to clear all the distortion correction memory (to zero) before starting the DISTBE measurement. It is recommended that you put a check in the Memory clear mode available box.
- Measurement field size [μm]
This field lets you specify the measurement field size.
 - For 100 kV, the maximum field size is 500 μm . Thus, if the scan width is 8 μm , it is recommended that you specify 492 (μm), which is obtained by subtracting the scan width from the maximum field size.

- For 50 kV, the maximum field size is 1000 μm . Thus, if the scan width is 8 μm , it is likewise recommended that you specify 992 (μm).
- **Zigzag**
Here you can specify the measurement sequence. Normally, specify **X->Y**.

Measurement sequence	Description
X->Y	In the field, zigzag measurement is made in the x direction.
Y->X	In the field, zigzag measurement is made in the y direction.

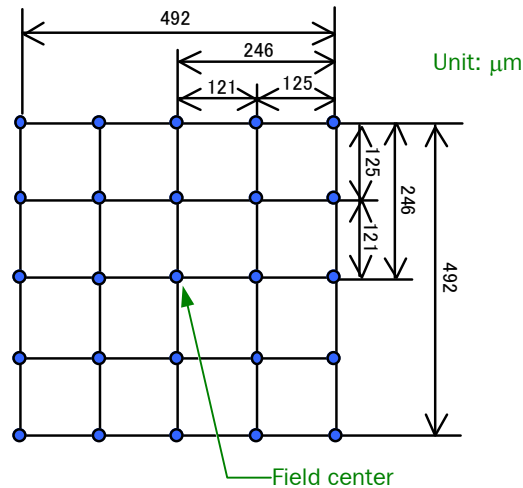
- **Measurement start point**
This field lets you select the measurement start point. Normally, select **Top left**.

Measurement start point	Description
Top left	Measurement starts from the top left of the field.
Top right	Measurement starts from the top right of the field.
Bottom right	Measurement starts from the bottom right of the field.
Bottom left	Measurement starts from the bottom left of the field.

- **Number of back-and-forth motions [times]**
This field lets you specify the number of back-and-forth measurements. If you specify a number greater than or equal to 2, it will take a longer time to complete the measurements. Thus, specify 1 under normal conditions.
- **First quadrant measurement points [point]**
Here you can specify the number of measurement points in the first quadrant. Specifying n results in measuring $(2n-1) \times (2n-1)$ points in the field.
- **First quadrant measurement pitch**
Clicking on this button opens the window shown below. The window is the same as the input window of CHKFOC. Measurement pitch 1 means the distance from the point outside the field to the next point. The number of pitches you can set is the number of measurement points in the first quadrant minus two. Select the pitch you want to change, and enter a value using the keyboard. Clicking on **Apply–Close** puts the changes in the measurement pitches in the first quadrant into effect.

Button name	Function
Apply	This button saves the values you are editing in the program.
Reset	This button resets the values you are editing to the values stored in the program.
Close	This button resets the values you are editing, and closes the window.

-  If the measurement field size is 492 μm ; the scan width is 8 μm ; the number of first quadrant measurement points is 3 (5 $\times$ 5 measurement); and measurement pitch 1 in the first quadrant is 125 μm , then the measurement pitches among 5 $\times$ 5 points are as follows.



- Allowable positional difference of identical point back-and-forth motion [μm]
One back-and-forth measurement measures the same position twice. This field lets you specify the tolerance for the difference between the two positions resulting from the measurement.
- Allowable height difference of identical point back-and-forth motion [μm]
One back-and-forth measurement measures the same position twice. This field lets you specify the tolerance for the difference between the two heights resulting from the measurement.
- Maximum permissible distortion of back-and-forth measurement average values [μm]
One back-and-forth measurement measures the same position twice. This field lets you set the tolerance for the difference between the two average values resulting from the distortion measurement.
- Tolerance between right and left points [μm]
This field lets you set the tolerance for the difference between the distortion correction values at two measurement points that are symmetric with respect to the y-axis, with the measurement field center at the origin.
- Tolerance between upper and lower points [μm]
This field lets you set the tolerance for the difference between the distortion correction values at two measurement points that are symmetric with respect to the x-axis, with the measurement field center at the origin.
- Allowable value of proximity field difference [point]
This field lets you set the tolerance for the difference between the distortion correction DAC values at adjacent measurement points.
- Allowable number of retries [times]
This field lets you set the upper limit on the number of retries in measurement. One back-and-forth measurement is one retry.
- Convergence allowance [μm]
The measurement is repeated until the deflection distortion falls within this allowance.
- Difference from the last value set in the memory [point]
This field lets you specify the tolerance from the distortion correction value when the DISTBE starts.

4.15 DISTBESET

DISTBESET writes the beam deflection distortion correction values stored in a file in the deflection distortion correction memory. The following directory contains the correction files under the name year_month_day_hour_minute_second.distbe.ic.

/usr/ebssys/log/

✂ For example, if DISTBE starts at 16:25:40 on January 5, 2006 in the EOS mode 3, the filename is as follows:

Filename: 20060105162540.3.distbe.ic

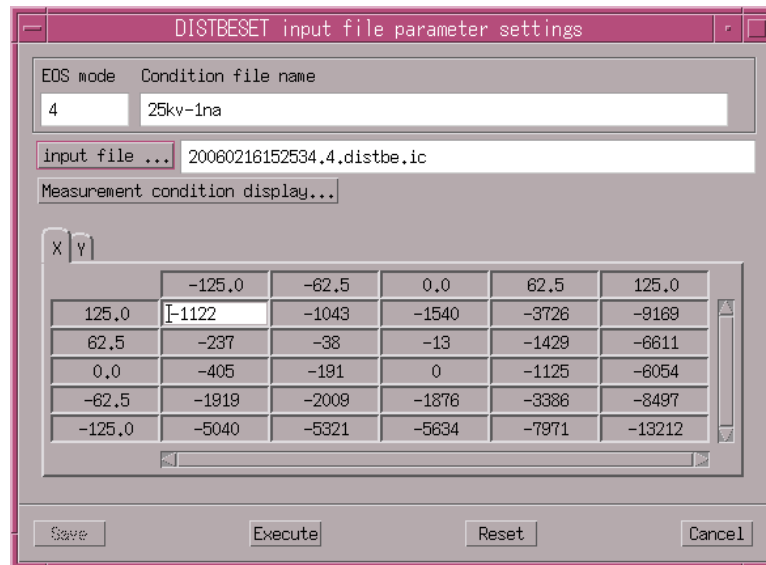


Fig. 29 DISTBESET input file parameter settings window

- **input file...**

Clicking **input file...** opens the DISTBESET final result file selection window. Select the file (*.distbe.ic) that you want to input, and click **OK** and then **CANCEL** to close the window.

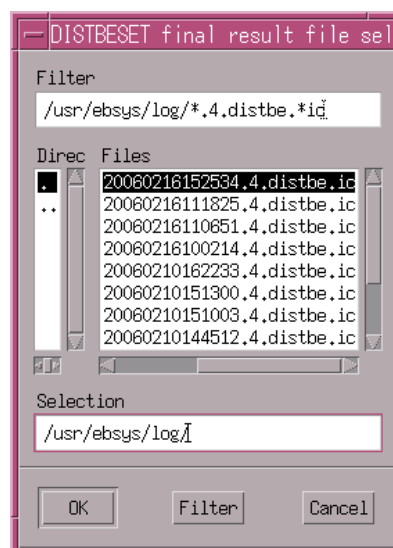
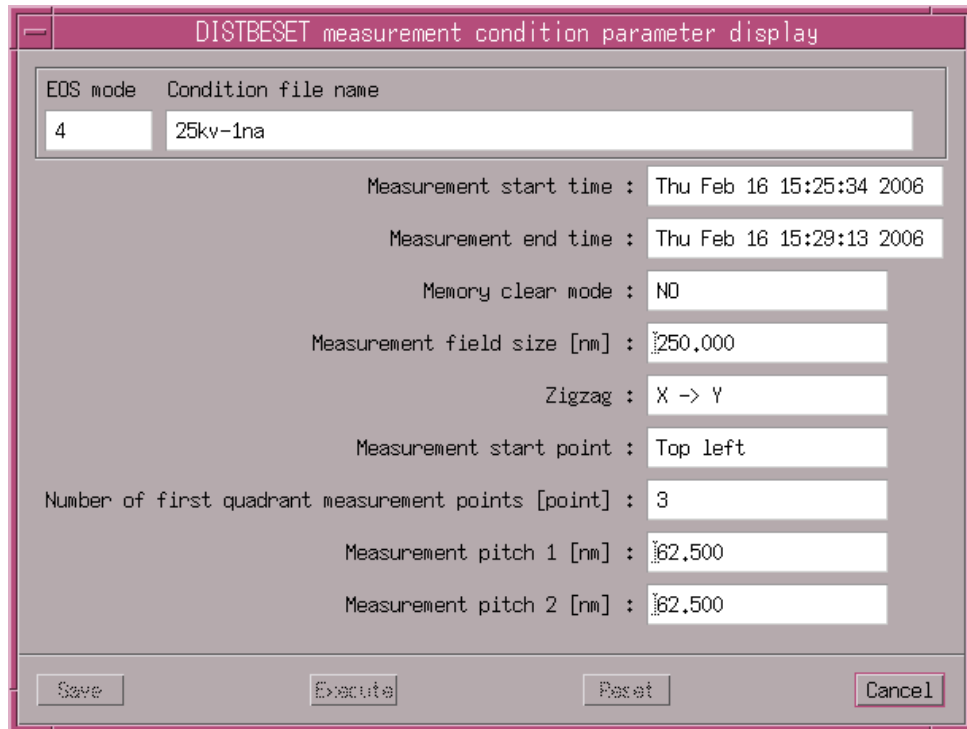


Fig. 30 DISTBESET final result file select window

- **Measurement condition display...**

This button displays the measurement conditions of the correction file that you selected by clicking **input file...**



The image shows a software window titled "DISTBESET measurement condition parameter display". It contains several input fields for measurement parameters. At the top, there are two fields: "EOS mode" with the value "4" and "Condition file name" with the value "25kv-1na". Below these are fields for "Measurement start time" (Thu Feb 16 15:25:34 2006) and "Measurement end time" (Thu Feb 16 15:29:13 2006). Further down are "Memory clear mode" (NO), "Measurement field size [nm]" (250,000), "Zigzag" (X -> Y), "Measurement start point" (Top left), "Number of first quadrant measurement points [point]" (3), "Measurement pitch 1 [nm]" (62,500), and "Measurement pitch 2 [nm]" (62,500). At the bottom of the window are four buttons: "Save", "Execute", "Reset", and "Cancel".

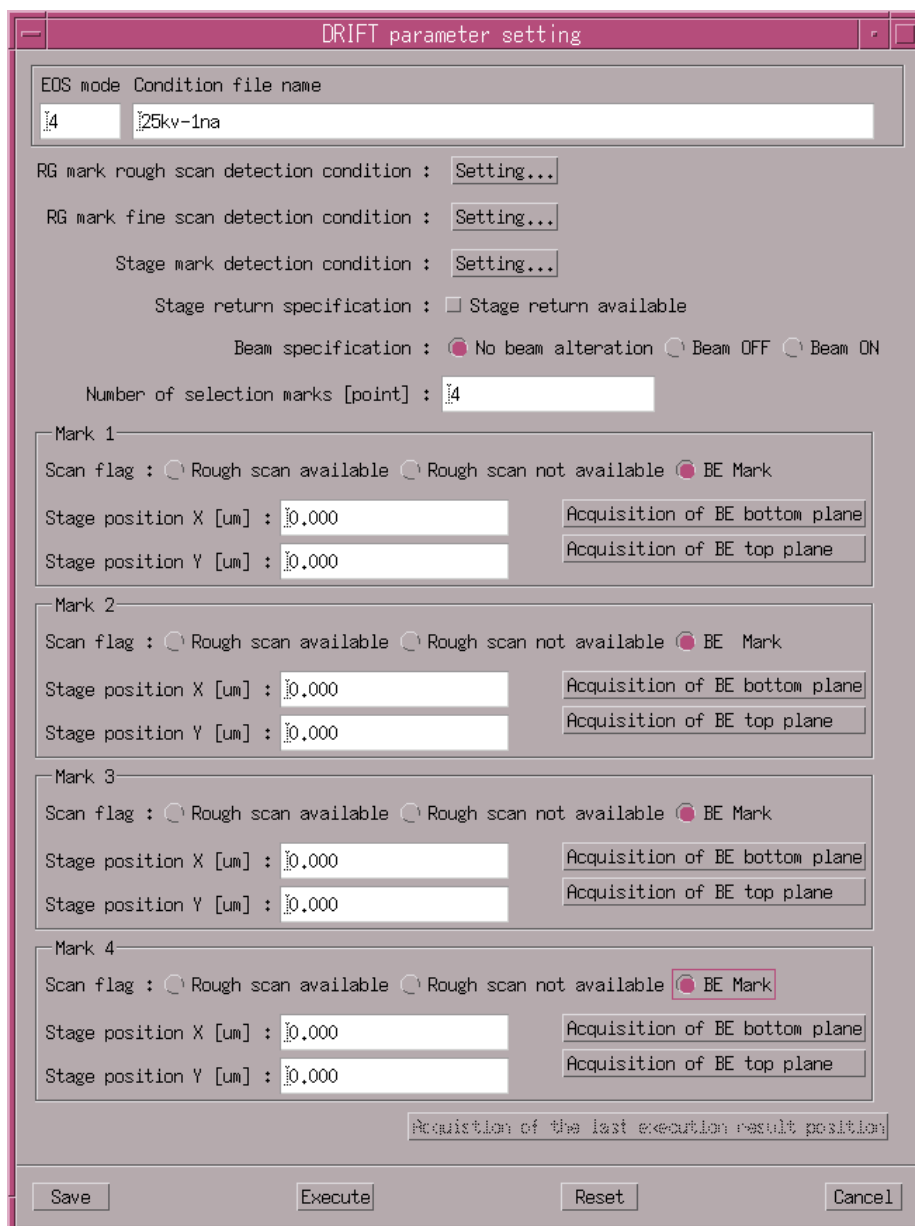
Fig. 31 DISTBESET measurement condition parameter display window

- **Measured value display**

The deflection distortion X/Y correction values are displayed. You can correct the correction values in this window.

4.16 DRIFT

DRIFT measures the amount of drift of the beam position. Also, the parameters you set here will serve as execution parameters if you specify DRIFT in CYCLIC CALIB during writing. The stage mark means the BE mark on the reference mesh, and the RG (registration) mark means the mark on the substrate. The drift measurement that uses the stage mark measures only the beam drift. On the other hand, in the drift measurement that uses the RG mark, the measured amount of drift includes the mark position's drift resulting from the expansion or contraction of the substrate, the displacement of the substrate, and the displacement of the cassette. If you specify multiple drift-mark measurements, the drift correction during writing uses the average of the multiple measured values.



The image shows a software window titled "DRIFT parameter setting". It contains several input fields and buttons for configuring drift measurement parameters.

EOS mode: 4
Condition file name: 25kv-1na

RG mark rough scan detection condition: Setting...
RG mark fine scan detection condition: Setting...
Stage mark detection condition: Setting...
Stage return specification: ☐ Stage return available
Beam specification: ☒ No beam alteration ☐ Beam OFF ☐ Beam ON
Number of selection marks [point]: 4

Mark 1
Scan flag: ☐ Rough scan available ☐ Rough scan not available ☒ BE Mark
Stage position X [um]: 0.000 **Acquisition of BE bottom plane**
Stage position Y [um]: 0.000 **Acquisition of BE top plane**

Mark 2
Scan flag: ☐ Rough scan available ☐ Rough scan not available ☒ BE Mark
Stage position X [um]: 0.000 **Acquisition of BE bottom plane**
Stage position Y [um]: 0.000 **Acquisition of BE top plane**

Mark 3
Scan flag: ☐ Rough scan available ☐ Rough scan not available ☒ BE Mark
Stage position X [um]: 0.000 **Acquisition of BE bottom plane**
Stage position Y [um]: 0.000 **Acquisition of BE top plane**

Mark 4
Scan flag: ☐ Rough scan available ☐ Rough scan not available ☒ BE Mark
Stage position X [um]: 0.000 **Acquisition of BE bottom plane**
Stage position Y [um]: 0.000 **Acquisition of BE top plane**

Acquisition of the last execution result position

Buttons: Save, Execute, Reset, Cancel

Fig. 32 DRIFT parameter setting window

- RG mark rough scan detection condition (Section 4.42.3)
This button opens a window for setting RG mark rough scan detection conditions. The system performs rough scan only for the first RG mark measurement, and performs only fine scan for the subsequent mark-detection measurements.
- RG mark fine scan detection condition (Section 4.42.3)
This button opens a window for setting RG mark fine scan detection conditions.
- Stage mark detection condition (Section 4.42.2)
This button opens a window for setting conditions for detecting the BE mark on the stage. Since a reference mark is used, there are no rough scan detection conditions.
 One DRIFT parameter setting window allows you to open one mark-detection condition setting window. You cannot open multiple mark-detection condition setting windows at the same time.
- Stage return specification
Here you can specify to return the stage to the Faraday cup position after the measurement is completed. If you do not specify it, the stage stops when the measurement is completed and stands still.
 This parameter is not used during writing.
- Beam specification
Here you can specify the beam status (on or off) after the measurement is completed. Selecting Beam ON will turn the blanking off, whereas selecting Beam OFF will turn the blanking on.
 This parameter is not used during writing.
- Number of selection marks [point]
This field lets you specify the number of marks (measurements) for which to perform the drift measurement (you can specify up to four positions).
- Mark 1 (2, 3, 4)
 - Scan flag
Here you can specify the presence or absence of a rough scan at the first (second, third, fourth) mark.
 - Stage position X, Y [μm]
This field lets you specify the position of the first (second, third, fourth) mark in the stage-coordinate system.
 During writing, the measurement conditions, Stage return specification, and Beam specification are not used.
 For the writing, specify either Acquisition of BE bottom plane or Acquisition of BE top plane.
- Acquisition of the last execution result position
This button updates each mark position data to the latest.

4.17 DYNSTISET (Optional)

DYNSTISET writes the dynamic focus and dynamic astigmatism correction values stored in a file in the dynamic focus/astigmatism correction memory. The following directory contains the correction files under the name year_month_day_hour_minute_second.dfocust.ic.

/usr/ebsys/log/

✍ For example, if DFOCUS starts at 11:13:30 on January 5, 2006 in the EOS mode 3, the filename is as follows:

Filename: 20060105111330.3.dfocust.ic

You can correct the existing measurement results.

This command is useful when you purchase the optional DF/DS system.

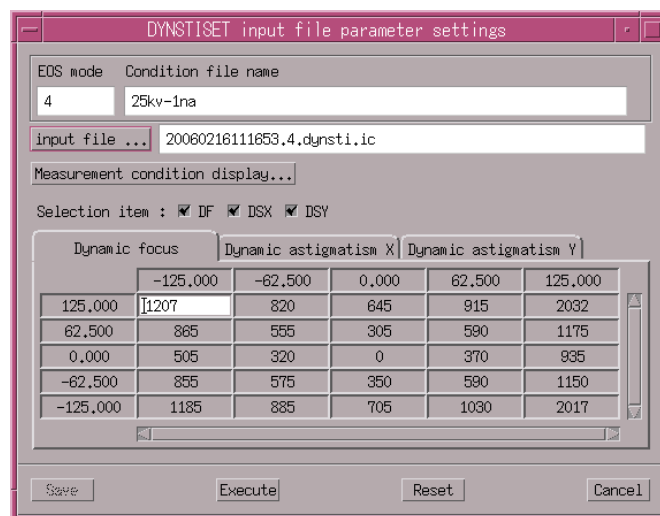


Fig. 33 DYNSTISET input file parameter settings window

• input file...

Clicking **input file...** opens the DYNSTISET final result file selection window. Select the file (*.dfocust.ic) that you want to input, and click OK and then CANCEL to close the window.

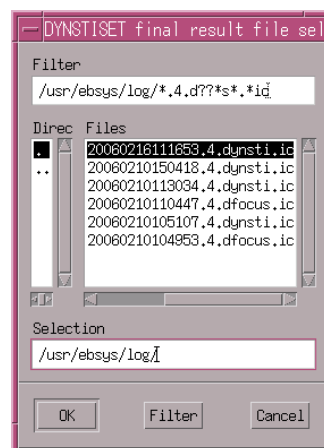


Fig. 34 DYNSTISET final result file select window

- Selection item

Here you can select **DF**, **DSX**, or **DSY** for the item for which you want to rewrite correction values.

- Measured value display

The dynamic focus, dynamic astigmatism X, and dynamic astigmatism Y correction values are displayed. You can correct the correction values in this window.

- Measurement condition display

This button displays the measurement conditions of the correction file that you selected as the input file.

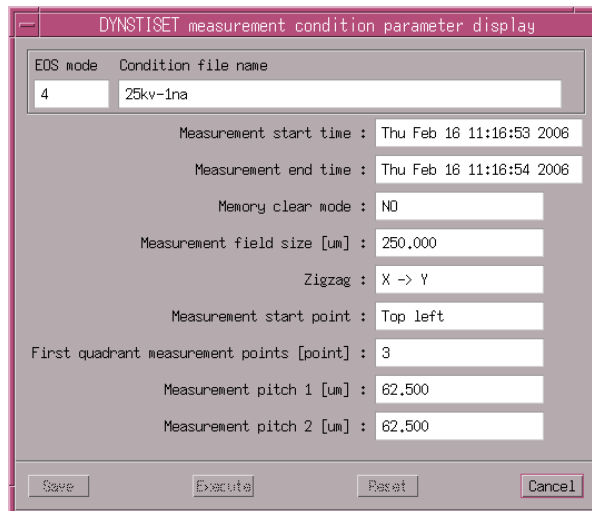


Fig. 35 DYNSTSET measurement condition display window

4.18 EOSSET

EOSSET changes the EOS mode (1-6). EOSSET changes only the system-related parameters. It is necessary to change the accelerating voltage/lens mode manually.

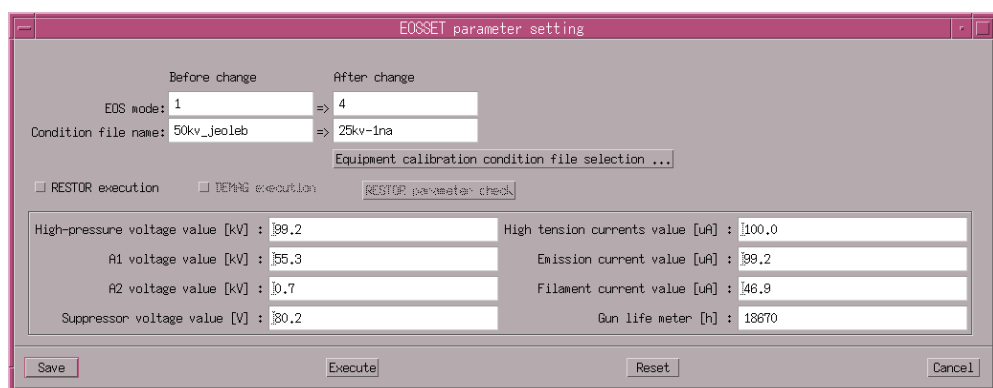


Fig. 36 EOSSET parameter setting window

- Before change/After change

Before change: The present EOS mode and calibration condition file are displayed.

After change: A new EOS mode and a corresponding calibration condition file are specified.

- Equipment calibration condition file selection

This button opens a file-selection window for selecting the calibration condition file to be used after the mode change.

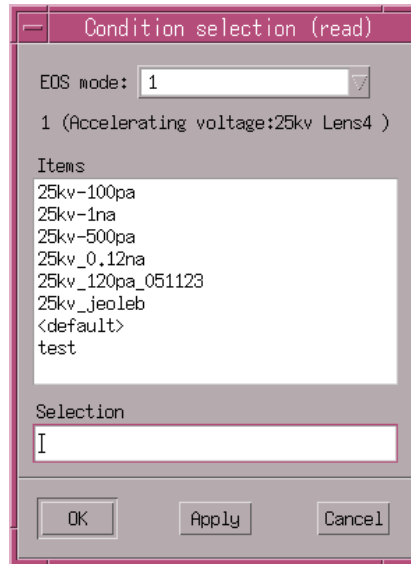


Fig. 37 Condition selection (read)

- RESTOR execution

Select this button if you want to execute RESTOR (☞ 4.44) that corresponds to the calibration conditions to restore the system state when the EOS parameters are set.

- DEMAG execution

Select this button if you want to perform DEMAG (lens demagnetization) when the EOS parameters are set.

- RESTOR parameter check

This button displays the details of RESTOR (☞ 4.44) that corresponds to the calibration conditions.

- High-voltage information display

The high-voltage information after EOSSET execution is displayed. The information consists of the high voltage [kV], high tension current [μ A], A1 voltage [kV], emission current [μ A], A2 voltage [kV], filament current [μ A], suppressor current [μ A], and gun life meter [h].

4.19 FDRIFTAE

Using the AE mark, FDRIFTAE automatically measures the fast-cycle beam drift.

FDRIFTAE parameter settings

Condition name: 25kv-1na Accelerating voltage: 14

AE mark detection condition: Setting...

AE bottom plane Mark position X [um]: 1947.301 Y [um]: 110238.820

Waiting time for a single measurement [msec]: 10

Trigger conditions

- ☐ Blanking ON
- ☐ SUBDEF Beam size change X [um]: Y [um]:
- ☐ PDEF Beam position change X [um]: Y [um]:
- ☐ Stage shift X [um]: Y [um]:
- ☐ Dynamic focus [point]:
- ☐ Dynamic correction X [point]:
- ☐ Dynamic correction Y [point]:

Save Execute Reset Cancel

Fig. 38 FDRIFTAE parameter settings window

- AE mark detection condition (Section 4.42.1)
This button opens a window for setting AE mark-detection conditions.
- AE bottom plane mark position X, Y [μm]
This field lets you specify the position of the mark that you want to measure.
- Waiting time for a single measurement [msec]
This field lets you specify the time to wait before starting the measurement.

■ Trigger conditions

Here you can set the conditions that trigger the drift measurement.

- Blanking ON
Select this button if you want to use blanking on as a trigger.
- SUBDEF beam size change X, Y [μm]
The secondary deflector moves the beam position the specified amount of change, and then the drift is measured (the beam returns to the original position before measurement).

- PDEF beam position change X, Y [μm]
The main deflector moves the beam position the specified amount of change, and then the drift is measured (the beam returns to the original position before measurement).
- Stage shift X, Y [μm]
These fields let you specify the amounts of stage movement to be performed at the midpoint of the number of drift measurements (back and forth between the specified position and the measurement position).
- Dynamic focus [point]
This field lets you specify the amount of change of the dynamic focus to be performed at the midpoint of the number of drift measurements (the focus value returns to the original value before measurement).
- Dynamic correction X (Y) [point]
These fields let you specify the amounts of change of the dynamic astigmatism X (Y) to be performed at the midpoint of the specified number of drift measurements. The astigmatism correction value returns to the original value before measurement.

4.20 FDRIFTBE

Using the BE mark, FDRIFTBE automatically measures the fast-cycle beam drift.

The screenshot shows the 'FDRIFTBE parameter settings' dialog box. The 'Condition name' field contains '25kv-1na' and the 'Accelerating voltage' field contains '4'. The 'BE mark detection condition' is set to 'Setting...'. The 'BE bottom plane Mark position X [um]' field contains '768.506' and the 'Y [um]' field contains '15545.924'. The 'Waiting time for a single measurement [msec]' field contains '0'. The 'Trigger conditions' section is expanded, showing several checkboxes and their corresponding X and Y coordinate input fields: 'Blanking' (unchecked), 'SUBDEF Beam size change' (unchecked) with X and Y fields, 'PDEF Beam position change' (unchecked) with X and Y fields, 'Stage shift' (unchecked) with X and Y fields, 'Dynamic focus' (unchecked) with a '[point]' field, 'Dynamic astigmatism X' (unchecked) with a '[point]' field, and 'Dynamic astigmatism Y' (unchecked) with a '[point]' field. At the bottom of the dialog are four buttons: 'Save', 'Execute', 'Reset', and 'Cancel'.

Fig. 39 FDRIFTBE parameter settings window

- BE mark detection condition (Section 4.42.2)
This button opens a window for setting BE mark-detection conditions.
- BE bottom plane mark position [μm]
This field lets you specify the position of the mark that you want to measure.
- Waiting time for a single measurement [msec]
This field lets you specify the time to wait before starting the measurement.

■ Trigger conditions

Here you can set the conditions that trigger the drift measurement. The conditions are the same as those of FDRIFTAE.

4.21 HCOEFFI

From the deflection gains of the bottom-plane and top-plane AE marks and the rotation correction value, HCOEFFI automatically measures the height-correction coefficient that corresponds to a height change of 1 μm .

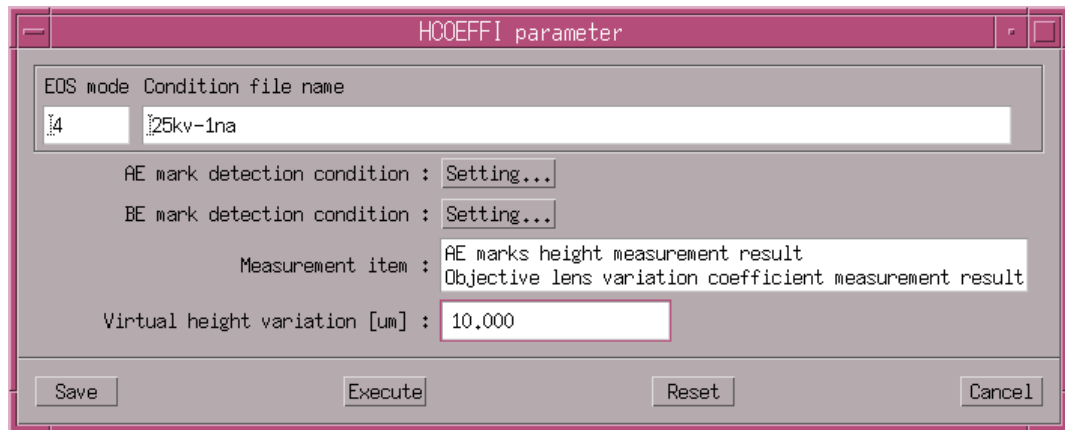


Fig. 40 HCOEFFI parameter setting window

- AE mark detection condition (Section 4.42.1)
This button opens a window for setting AE mark-detection conditions.
- BE mark detection condition (Section 4.42.2)
This button opens a window for setting BE mark-detection conditions.
- Measurement item
Here you can select the items you want to measure. You can select multiple items. Normally, execute HCOEFFI with both “Top and bottom planes AE marks height measurement result” and “Objective lens variation coefficient measurement result” selected.
- Virtual height variation [μm]
The Electron Beam Lithography System does not measure the DF offset variation coefficient, so enter zero.

■ Explanation of the operation

● AE mark top and bottom plane height measurement

1. Set the dynamic focus voltage to zero.
2. Demagnetize the objective lens.
3. Adjust the focus of the objective lens using the bottom-plane AE mark.
4. Execute PDEF, using the bottom-plane AE mark, to determine the deflection gain.
5. Execute PDEF, using the top-plane AE mark, to determine the deflection gain.
6. Execute PDEFBE, using the bottom-plane BE mark, to determine the deflection gain.

At this time, the system measures the dark current for height measurement, measures the height of the reference plane for height measurement, and measures the height of the bottom-plane BE mark. The height of the bottom-plane BE mark is defined by the average resulting from height measurement during PDEFBE execution.
7. Determine the average deflection gain of the top-plane AE mark in the x and y directions, based on the bottom-plane AE mark.
8. Determine the average deflection gain of the bottom-plane BE mark in the x and y directions, based on the bottom-plane AE mark.
9. Using the relationship between the height change determined from PDEFBE and the change of the PDEF gain, determine the height of the top-plane AE mark based on the bottom-plane AE mark.
10. Using the relationship between the height change determined from PDEFBE and the change of the PDEF gain, determine the height between bottom-plane BE marks based on the bottom-plane AE mark.
11. Determine the height of the bottom-plane BE mark and the height of the bottom-plane AE mark, from the reference plane of the bottom-plane AE mark for height measurement.
12. Reset the objective-lens value and the PDEF value to their pre-HCOEFFI execution conditions.

● Objective lens variation coefficient measurement

1. Set the dynamic-focus voltage to zero.
2. Demagnetize the objective lens.
3. Adjust the focus of the objective lens using the bottom-plane AE mark.
4. Execute INITAE, using the bottom-plane AE mark, to detect the mark position.
5. Execute PDEF and SUBDEF using the bottom-plane AE mark.
6. Adjust the focus of the objective lens at the mark position specified by the top-plane AE mark.
7. Using the results of step 3 and step 6, determine the correction coefficient (point/ μm) of the objective-lens value for the case where the height changes from the bottom-plane AE mark to the top-plane AE mark.
8. Execute PDEF and SUBDEF using the top-plane AE mark.

Compare the results with those of step 5 to determine the deflection gains of the main and sub-deflectors and the rotation correction coefficient that correspond to the change of the height.

9. Under this condition, detect the mark position that is the same position as with step 4, using the bottom-plane AE mark.

Compare the results with those of step 4 to determine the correction coefficient of the main deflection beam shift amount that corresponds to the change of the height.

10. Reset the objective-lens value and the PDEF/SUBDEF value to their pre-HCOEFFI execution conditions.

4.22 HEIMAP

HEIMAP automatically measures the height distribution in the material plane. The reference-plane measurement offset position in HEIMAP means the offset of the position of the height-measurement reference plane against the bottom-plane BE mark position. During mask writing, the system measures the height for each writing field, and performs corrections based on the results of the measurement. The height map during the HEIMAP measurement is used, with default values for the case where the height cannot be measured for each field. These parameters take effect only for mask writing.

Fig. 41 HEIMAP parameter setting window

- Material type
Here you can select **Mask** or **Wafer**.
- Material size [inch]
This field lets you specify the material size.
- Material window
This field lets you select the material window. Other than for a multi-cassette, specify 0. You can select one of the following material windows:

0

2A, 2B, 2C, 2D, 2E, 2F, 2G, 2H, 2I, 2J

3A, 3B, 3C, 3D, 3E, 3F, 3G, 3H, 3I, 3J

4A, 4B, 4C, 4D, 4E, 4F, 4G, 4H, 4I, 4J

5A, 5B, 5C, 5D, 5E, 5F, 5G, 5H, 5I, 5J

6A, 6B, 6C, 6D, 6E, 6F, 6G, 6H, 6I, 6J

2A, 2B, ...6J have different material centers for the multi-wafer cassette and the multi-mask cassette in the system parameters (SYSPAR).

 In the above parameters, a 3 inch wafer and the 0 material window are used.

- **Zigzag**

You can select the measurement sequence. Normally, select **X->Y**.

Measurement sequence	Description
X->Y	In the field, zigzag measurement is made in the x direction.
Y->X	In the field, zigzag measurement is made in the y direction.

- **Number of height measurement points X, Y [point]**
This field lets you specify the numbers of height measurement points.
- **Height data measurement pitch X, Y [μm]**
This field lets you specify the height measurement pitches.
- **Offset X, Y [μm]**
This field lets you specify the amount of Offset of a measurement position.
- **Number of height measurements for each measurement point [times]**
This field lets you specify the number of measurements at each measurement point.
For the number of height measurements for each field, the system uses the value set as a system parameter.
- **Maximum allowable height [μm]**
This field lets you specify the allowable height at each measurement point.
- **Allowable value of difference from reference height [μm]**
This field lets you specify the tolerance for the difference between the height at each measurement point and the reference plane.
- **Measurement data average value allowance [μm]**
This field lets you specify the tolerance for the difference between the height at each measurement point and the measured average.
- **Criteria for passing the above allowance [%]**
This field lets you specify the criteria for assessing the fraction of the measurement points that have passed the above allowance requirements.
- **Allowable average difference between all succeeded height data and reference height data [μm]**
This field lets you specify the tolerance for the difference between the average height of the passed measurement points and the height of the reference plane.
- **Criteria for passing the above allowance [%]**
This field lets you specify the criteria for assessing the fraction of the measurement points that have passed the above allowance requirements.
- **Allowable variations from linear approximate calculation data [μm]**
This field lets you specify the tolerance for the difference between the first-order approximation (plane approximation) to the measured height and the measured height.

- Allowable differences between linear approximate calculation and average height data [μm]
This field lets you specify the tolerance for the difference between the first-order approximation (plane approximation) to the measured height and the average height.
- Photoplate tilt allowance [X direction, degree]
This field lets you specify the allowable tilt of the mask plate in the x direction.
- Allowable differences between the values before and after reference height measurements [μm]
This field lets you specify the tolerance for the difference between the reference-plane heights before and after height measurement.
- Reference plane measurement offset position X, Y [μm]
This field lets you specify the positional offset amounts of the height-measurement reference-plane position against the bottom BE mark position.
- LBC value threshold [μm]
This field lets you specify the threshold for checking the LBC correction amount in stage movement during the HEIMAP measurement.
- YAW value threshold [μm]
This field lets you specify the threshold for checking the yaw correction amount in stage movement during the HEIMAP measurement.
- PITCH value threshold [μm]
This field lets you specify the threshold for checking the pitch correction amount in stage movement during the HEIMAP measurement.
- Acceleration threshold [G]
This field lets you specify the threshold for checking the acceleration in stage movement during the HEIMAP measurement.
- Move time threshold [msec]
This field lets you specify the threshold for checking the movement time in stage movement during the HEIMAP measurement.
- Number of retries threshold [times]
This field lets you specify the threshold for checking the number of retries in stage movement during the HEIMAP measurement.
- Measurement value display
This field lets you select **Height measurement, Center of gravity pos measurement, or Area measurement.**

4.23 HEITST

The HEITST subprogram automatically measures the height of the present stage position.

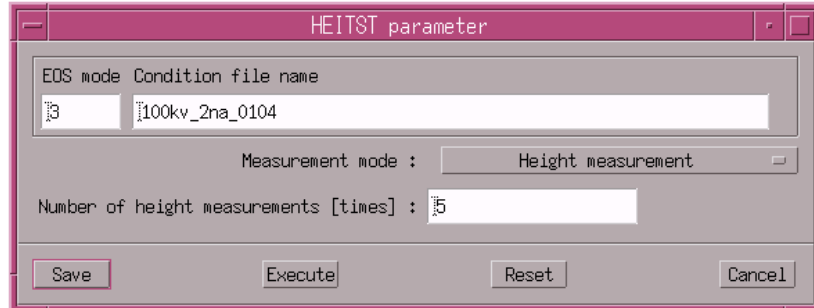


Fig. 42 HEITST parameter setting window

- Measurement mode

This field lets you select **Height measurement, Center of gravity pos measurement, or Area measurement.**

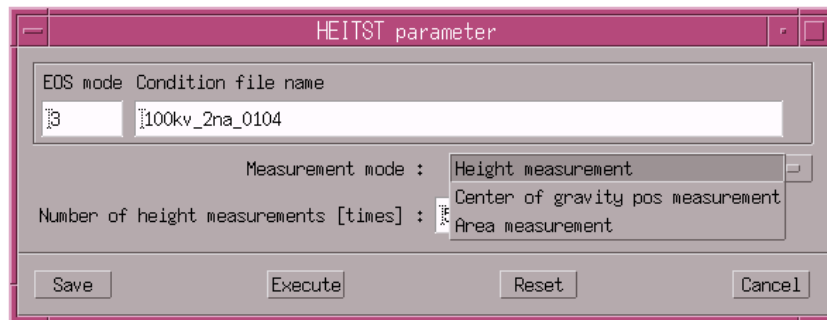


Fig. 43 HEITST parameter setting window (Measurement mode)

- Number of height measurements [times]

This field lets you specify the number of height measurements.

4.24 INIEBX

The subprogram INIEBX enables you to directly set the system conditions and correction coefficients that have been set to the results of calibration. To set an item, turn on its toggle button. You cannot set the items whose toggle buttons are off.

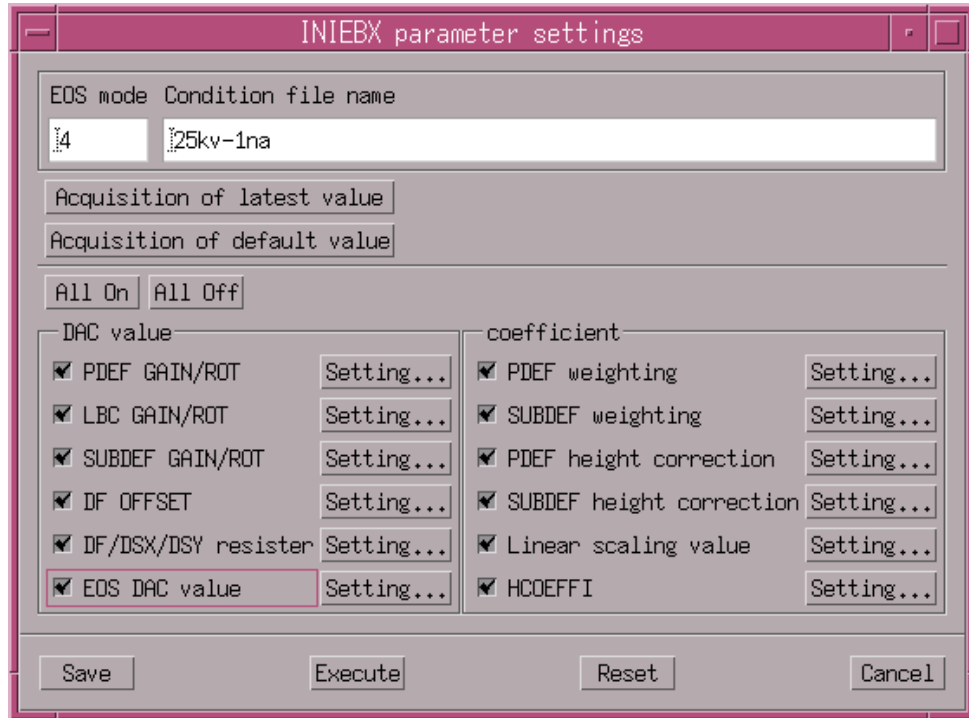


Fig. 44 INIEBX parameter settings window

Here is how each button in the window functions.

- **Acquisition of latest value**

This button reads the present system parameter settings.

- **Acquisition of default value**

This button sets all the parameters to the default values (with all the DAC value/coefficient toggle buttons off).

- **All On/All Off**

On: Selects all the item buttons in the DAC value/coefficient area.

Off: Clears all the item buttons in the DAC value/coefficient area.

■ DAC value

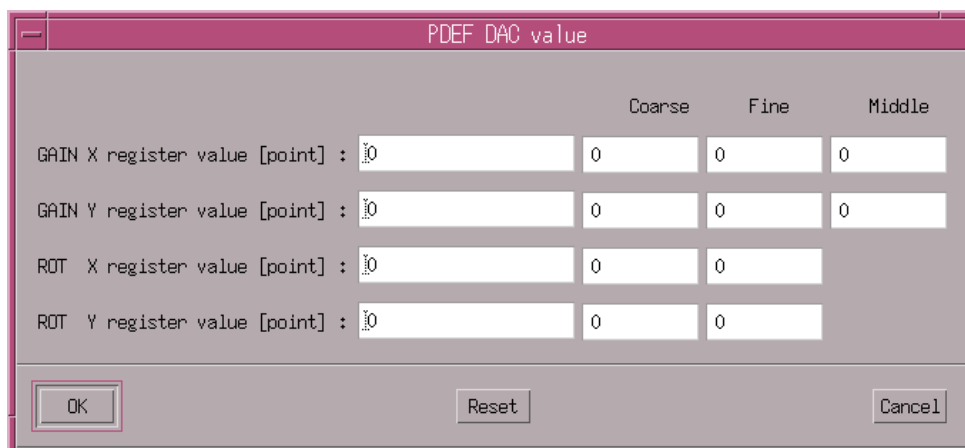
Button	Description
PDEF GAIN/ROT	Opens the PDEF GAIN/ROT DAC value window.
SUBDEF SHIFT/GAIN/ROT	Opens the SUBDEF SHIFT/GAIN/ROT DAC value window.
LBC GAIN/ROT	Opens the LBC GAIN/ROT DAC value window.
DF OFFSET	Opens the DF OFFSET DAC value window.
DF/DSX/DSY resister	Opens the DF/DSX/DSY DAC value window.
EOS DAC value	Opens the EOS window for setting the DAC values.

■ Setting

Clicking the **Setting...** button of each item opens the corresponding setting window, as follows.

● PDEF GAIN/ROT window

This window displays the GAIN/ROT DAC correction values measured by PDEFBE and multiple DAC (Coarse, Middle, Fine) values that comprise the GAIN/ROT DAC values. You can change the leftmost DAC values, but the Coarse, Middle, and Fine values are display only. These values change after INIEBX runs and the DAC values change.



The PDEF DAC value window is a dialog box with a title bar "PDEF DAC value". It contains four rows of input fields for GAIN X, GAIN Y, ROT X, and ROT Y register values. Each row has a text input field followed by three numeric input fields labeled "Coarse", "Fine", and "Middle". The "Coarse" and "Fine" fields are highlighted with a yellow background. At the bottom, there are three buttons: "OK", "Reset", and "Cancel".

		Coarse	Fine	Middle
GAIN X register value [point] :	0	0	0	0
GAIN Y register value [point] :	0	0	0	0
ROT X register value [point] :	0	0	0	
ROT Y register value [point] :	0	0	0	

Fig. 45 PDEF DAC value window

● LBC GAIN/ROT window

This window displays the GAIN/ROT DAC correction values measured by CHKLBC and multiple DAC (Middle, Fine) values that comprise the GAIN/ROT DAC values. You can change the leftmost DAC values, but the Middle and Fine values are display only. These values change after INIEBX runs and the DAC values change.



The LBC DAC value window is a dialog box with a title bar "LBC DAC value". It contains four rows of input fields for GAIN X, GAIN Y, ROT X, and ROT Y register values. Each row has a text input field followed by two numeric input fields labeled "Middle" and "Fine". The "Middle" and "Fine" fields are highlighted with a yellow background. At the bottom, there are three buttons: "OK", "Reset", and "Cancel".

		Middle	Fine
GAIN X register value [point] :	0	0	0
GAIN Y register value [point] :	0	0	0
ROT X register value [point] :	0	0	0
ROT Y register value [point] :	0	0	0

Fig. 46 LBC DAC value window

- **SUBDEF SHIFT/GAIN/ROT window**

This window displays the SHIFT/GAIN/ROT DAC correction values measured by SUBDEFBE and multiple DAC (Coarse, Middle, Fine) values that comprise the SHIFT/GAIN/ROT DAC values. You can change the leftmost DAC values, but the Coarse, Middle, and Fine values are display only. These values change after INIEBX runs and the DAC values change.

Fig. 47 SUBDEF DAC value window

- **DF OFFSET DAC value window**

By default, this window displays the DFOCUS parameter value. You can change this value.

Fig. 48 DF OFFSET DAC value window

- **DF/DSX/DSY DAC value window**

By default, this window displays the correction results of DFOCUS execution. You can change these values.

Fig. 49 DF/DSX/DSY DAC value window

● EOS DAC value window

Here you can set and check the EOS values that you can set using the Column Adjustment GUI.

The screenshot shows a window titled "EOS value" with a list of input fields, each with a value of "0". The fields are:

- Value of Lens 2 :
- Value of Lens 3 :
- Value of Lens 4 :
- Value of Lens 5 :
- value of first alignment X :
- Y :
- Value of second alignment TX :
- TY :
- SX :
- SY :
- Value of third alignment TX :
- TY :
- Value of stig X :
- Y :
- Offset value of Lens 4:
- Offset value of Lens 5:

At the bottom of the window are three buttons: "OK", "Reset", and "Cancel".

Fig. 50 EOS value window

■ Coefficient

Button	Description
Linear scaling value	Opens the Linear scaling value window.
PDEF height correction	Opens the PDEF height correction coefficient window.
SUBDEF height correction	Opens the SUBDEF height correction coefficient window.
PDEF weighting	Opens the PDEF weight coefficient window of COEFFI.
SUBDEF weighting	Opens the SUBDEF weight coefficient window of COEFFI.
HCOEFFI	Opens the HCOEFFI window.

■ Setting

Clicking the **Setting...** button of each item opens the corresponding setting window, as follows.

- **PDEF height correction coefficient window**

This window displays the HCOEFFI-measured amounts of change of the main-deflection DAC values, per μm height, in points. The window enables you to change the values.



PDEF weight coefficient

gain x : 2.100e+00

y : 2.200e+00

rot x : 8.500e-01

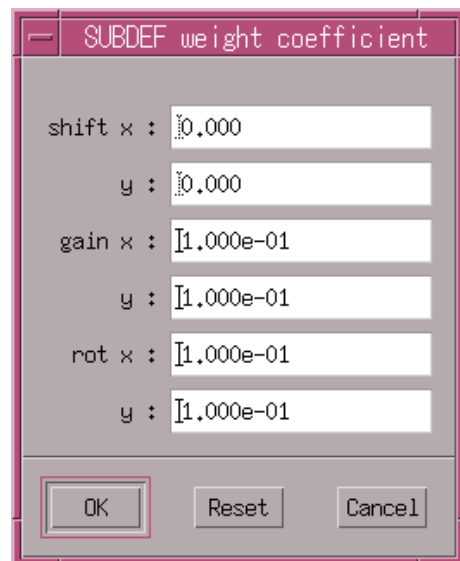
y : 1.000e-01

OK Reset Cancel

Fig. 51 PDEF weight coefficient window

- **SUBDEF height correction coefficient window**

This window displays the HCOEFFI-measured amounts of change of the sub-deflection DAC values, per μm height, in points. The window enables you to change the values.



SUBDEF weight coefficient

shift x : 0.000

y : 0.000

gain x : 1.000e-01

y : 1.000e-01

rot x : 1.000e-01

y : 1.000e-01

OK Reset Cancel

Fig. 52 SUBDEF weight coefficient window

- **PDEF weight coefficient window**

This window enables you to set and check the amounts of change of GAIN and ROT, per DAC value point of the main deflection.



Fig. 53 PDEF weight coefficient window

- **SUBDEF weight coefficient window**

This window enables you to set and check the amounts of change of SHIFT, GAIN, and ROT, per DAC value point of the sub-deflection.

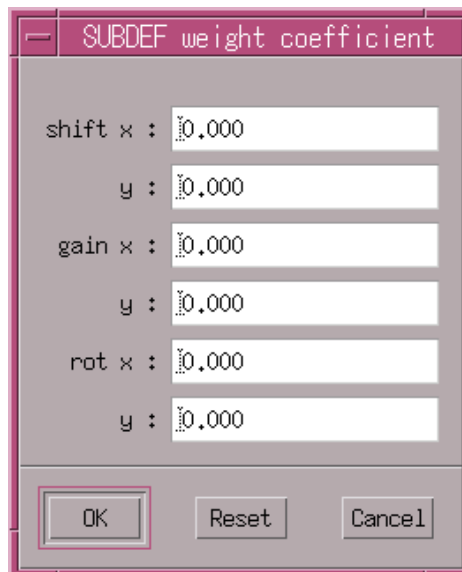


Fig. 54 SUBDEF weight coefficient window

- **Linear scaling value window (Optional)**

This window enables you to check and change the scaling values set by SETSCL. Purchasing the fine pitch/minimum increment feature enables this window.

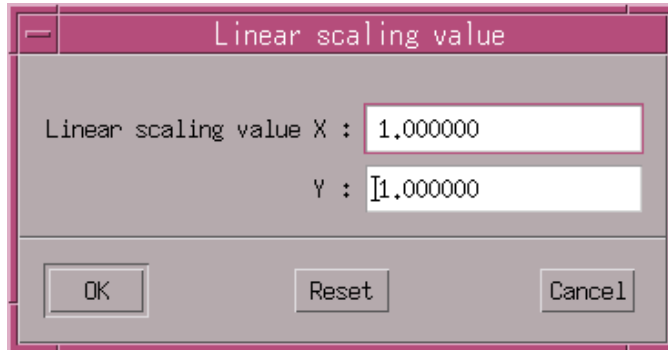


Fig. 55 Linear scaling value window

- **HCOEFFI result window**

This window enables you to set and check the height-conversion coefficients for the objective lens and the main/secondary deflection objective lenses and the height-conversion coefficients for the DF focus lens.

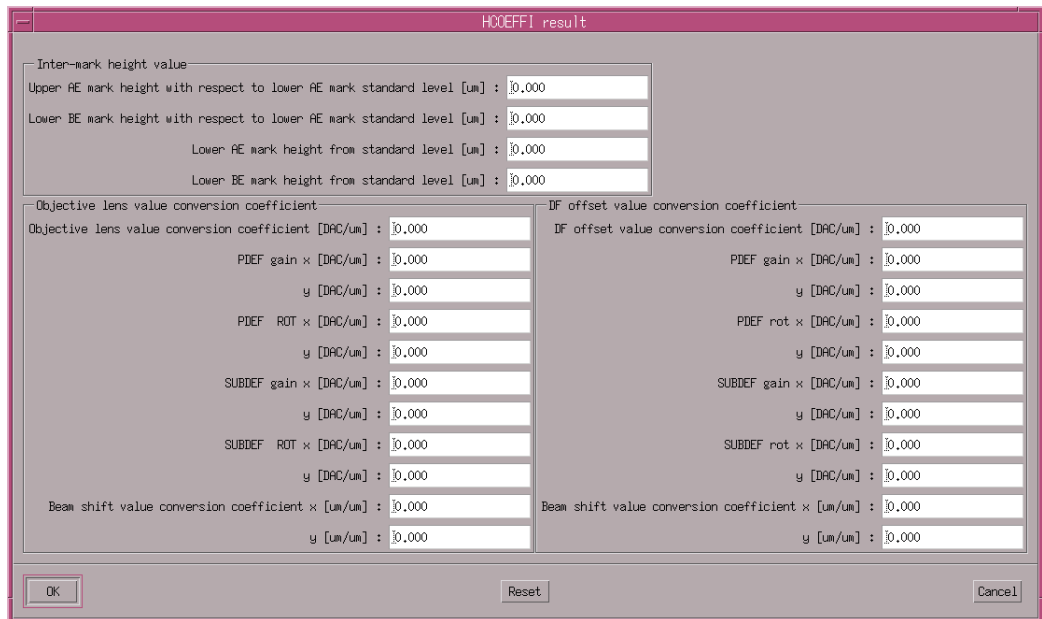


Fig. 56 HCOEFFI result window

The buttons at the bottom of the above windows function as follows.

Button	Description
OK	Saves the values shown in the window in the program.
Reset	Resets the values you are editing, and displays the values stored in the program.
Cancel	Resets the values you are editing, and closes the window.

4.25 INIMEM

INIMEM reads, writes, and clears the main deflector's distortion memory, the dynamic focus memory, and the dynamic astigmatism memory.

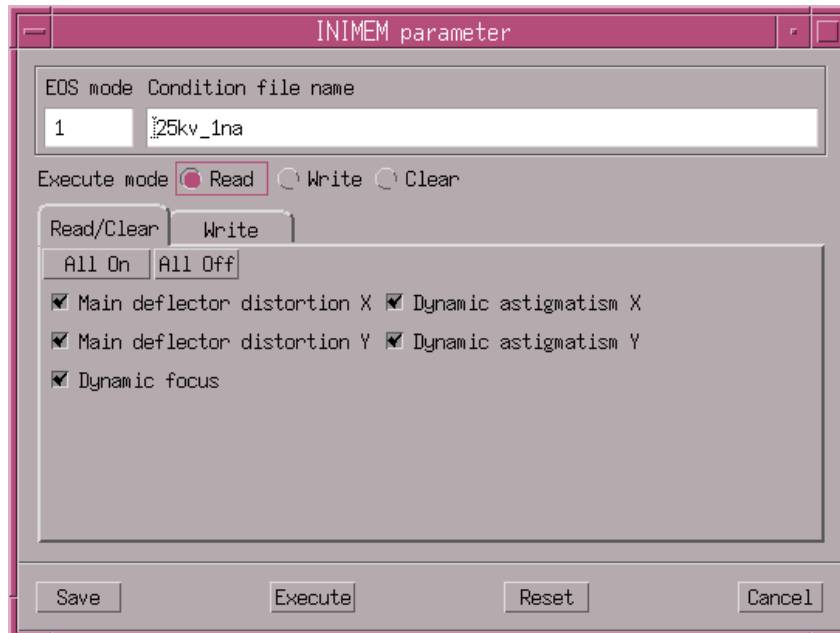


Fig. 57 INIMEM parameter setting window (Read/ Clear)

- Execute mode

Select **Read**, **Write**, or **Clear**. To read the contents of memory, select **Read**; to rewrite part or all of the memory contents, select **Write**; to clear all the memory contents, select **Clear**.

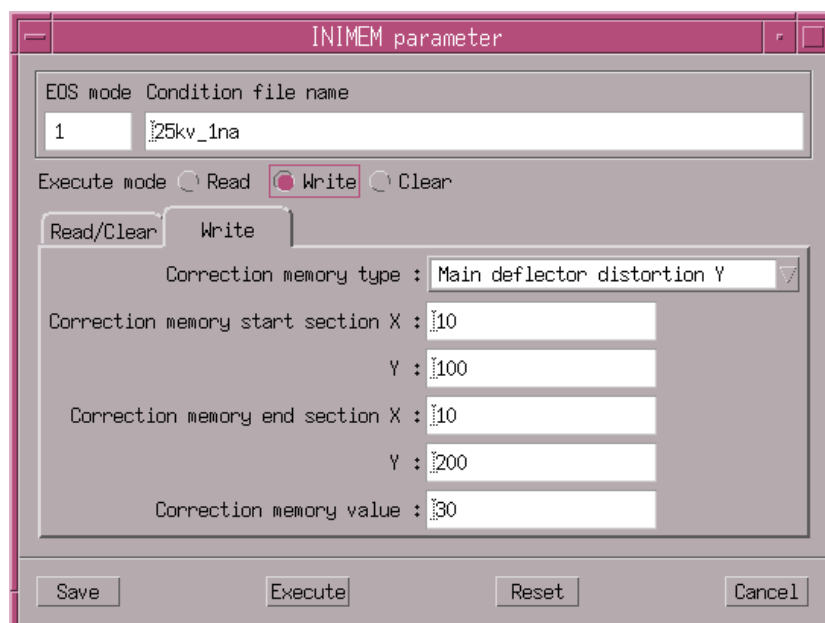


Fig. 58 INIMEM parameter setting window (Write)

- Read/Clear
To read or clear memory, select the Read/Clear tab. Then, select Main deflector distortion, Dynamic focus, and/or Dynamic astigmatism.
- Write
 - Correction memory type
This field lets you select the type of correction memory.
 - Correction memory start section X, Y
These fields let you specify the start point of the correction memory. The input range is 0 to 255.
 - Correction memory end section X, Y
These fields let you specify the end point of the correction memory. The input range is 0 to 255.
 - Correction memory value
This field lets you enter the value of the correction memory that you want to write. The input ranges are as follows.

Item	Input range
Main deflector distortion correction value	± 32767
Dynamic focus correction value	± 2048
Dynamic astigmatism correction value	± 2048

4.26 INITAE

The INITAE subprogram automatically measures the position of the AE mark.

The values on the INITAE parameter window cannot be saved.

For the stage position of the mark, which is the INITAE parameter, the stage position stored in the EWS is displayed when you open the window. The mark position stored in the EWS will be updated to the mark position obtained when you execute the INITAE subprogram; however, the position will not be applied to the INITAE parameter window. It is necessary to click the **Update** button to update the mark position on the window.

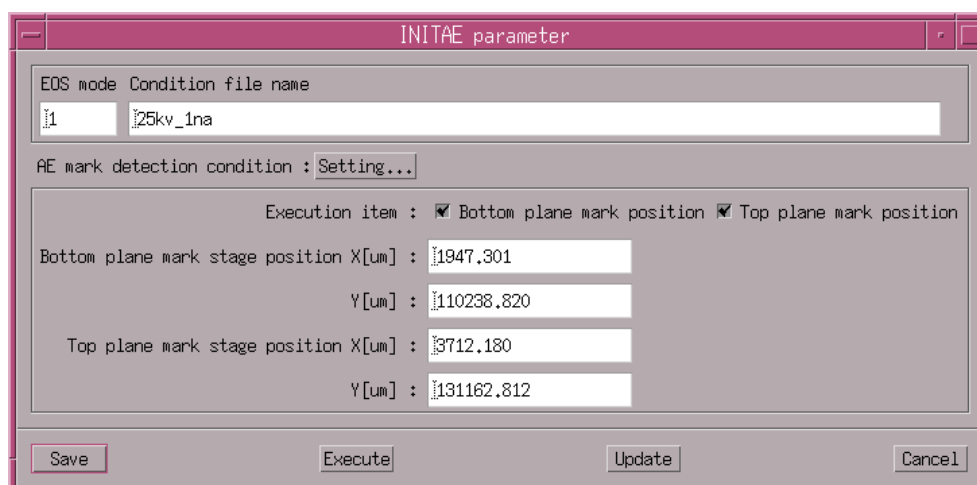


Fig. 59 INITAE parameter setting window

- AE mark detection condition (Section 4.42.1)
This button opens a window for setting AE mark-detection conditions.
- Execution item
Here you can select either or both of Bottom plane mark position and Top plane mark position.
- Bottom plane mark stage position X, Y [μm]
This field displays the bottom-plane AE mark position. You can change the position using the keyboard.
- Top plane mark stage position X, Y [μm]
This field displays the top-plane AE mark position. You can change the position using the keyboard.

4.27 INITBE

The INITBE subprogram automatically measures the BE mark position.

The values on the INITBE parameter window cannot be saved.

For the stage position of the mark, which is the INITBE parameter, the stage position stored in the EWS is displayed when you open the window. The mark position stored in the EWS will be updated to the mark position obtained when you execute the INITBE subprogram; however, the position will not be applied to the INITBE parameter window. It is necessary to click the **Update** button to update the mark position on the window.

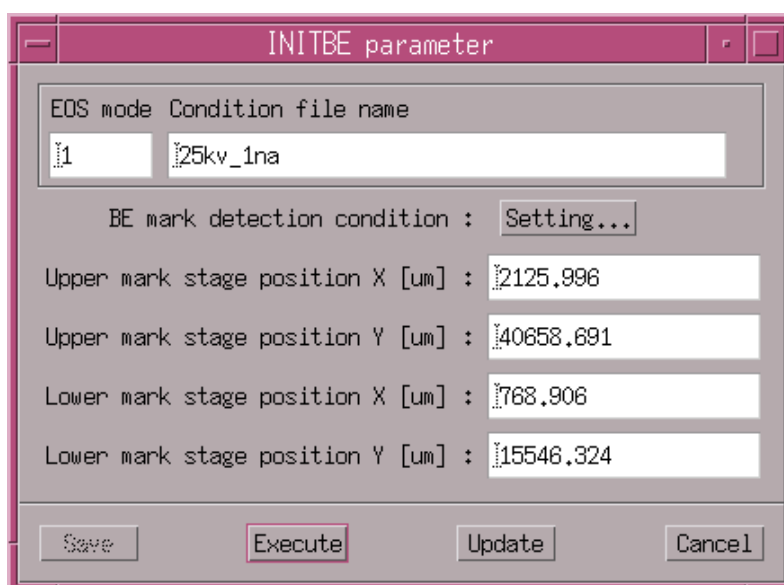


Fig. 60 INITBE parameter setting window

- BE mark detection condition (Section 4.42.2)
This button opens a window for setting BE mark-detection conditions.
- Upper mark stage position X, Y [μm]
This field displays the top-plane BE mark position. You can change the position using the keyboard.
- Lower mark stage position X, Y [μm]
This field displays the bottom-plane BE mark position. You can change the position using the keyboard.

4.28 MDRG

MDRG automatically measures the RG (registration) mark position.

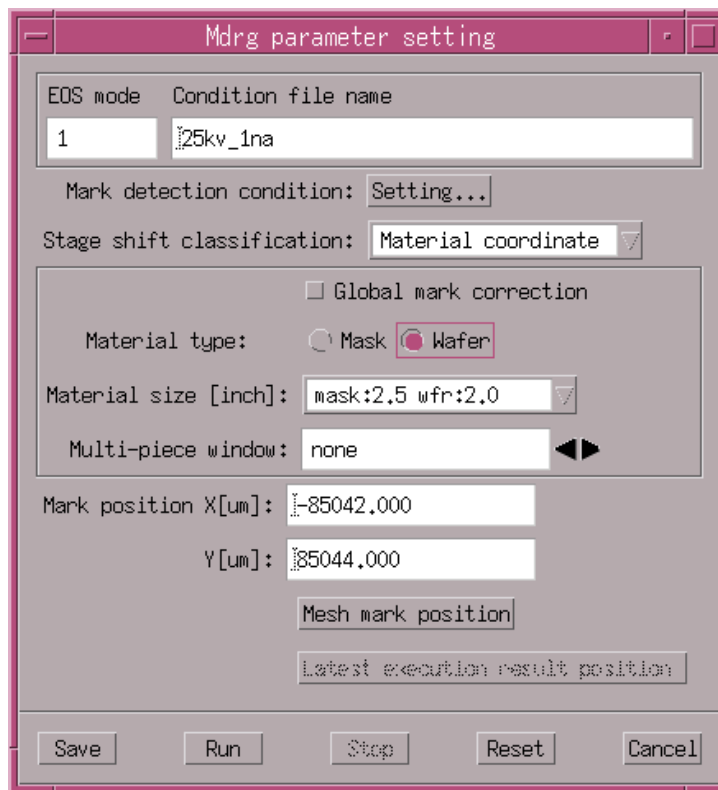


Fig. 61 MDRG parameter setting window

- Mark detection condition (Section 4.42.3)
This button opens a window for setting RG mark-detection conditions.
- Stage shift classification
Select No stage movement, Stage coordinates, or Material coordinates. With No stage movement selected, marks are detected at the present stage position. Selecting Stage/Material coordinates changes the Mark position X, Y readouts to the values in the corresponding coordinate system.
- Global mark correction
This button lets you specify whether to move the stage to the mark position according to the substrate distortion/rotation correction measured by SETWFR.
- Material type
These buttons let you specify whether the RG mark is on the mask or on the wafer.
- Material size [inch]
This field lets you specify the substrate size in inches.
- Multi-piece window
If the multi-piece window is on a multi-piece cassette, specify the parameter A, B, or C as the layout. For information about the layout, see the layout diagram of the multi-piece cassette.
- Mark position X, Y [μm]
These fields display the RG mark position in the stage coordinates. You can change the position by typing values.

- Mesh mark position
If you want to perform mark detection with the BE mark instead of the RG mark, click this button; the mark position is automatically set to the BE bottom plane mark position.
- Latest execution result position
This button automatically specifies the results of the previous MDRG measurement.

4.29 PDEF

Using the AE mark, PDEF automatically corrects the deflection gain and rotation of the main deflector. The parameters of PDEF are also used in the PDEF measurement in HCOEFFI.

Fig. 62 PDEF parameter setting window

- AE mark detection condition (Section 4.42.1)
This button opens a window for setting AE mark-detection conditions.
- Measurement field size [μm]
Here you can specify the size of the field to be measured in the PDEF measurement.

Accelerating voltage	4th lens mode	5th lens mode
25 kV	2000 μm	250 μm
50 kV	1000 μm	125 μm
100 kV	500 μm	62.5 μm

- Number of back-and-forth motions [times]
This field lets you specify the number of back-and-forth measurements.
- Allowable number of retries [times]
This field lets you specify the number of retries until convergence.
- Allowable positional difference of identical point back-and-forth motion [μm]
This field lets you specify the tolerance for the difference between the back and forth positions at each point in the back-and-forth measurement.
- Maximum distortion allowance [μm]
This field lets you specify the tolerance for the difference between the expected and measured mark positions at each measurement point.
 Set this parameter to a large value after starting up the system or changing system conditions drastically, and set it to a small value when the system is running stably.

■ GAIN

- Value added to last convergence value X, Y [μm]
This field lets you specify the correction amounts to add to the PDEF gain correction value after the PDEF convergence.
- Allowable convergence value [μm]
This field lets you specify the tolerance for convergence of the PDEF gain.
- Allowable difference from the last DAC value [point]
This field lets you specify the tolerance for difference between measured and the previous PDEF gain correction value (DAC value).

■ ROT

- Value added to last convergence value X, Y [μm]
This field lets you specify the correction amounts to add to the PDEF rotation correction value after the PDEF convergence.
- Allowable convergence value [μm]
This field lets you specify the tolerance for convergence of the PDEF rotation.
- Allowable difference from the last DAC value [point]
This field lets you specify the tolerance for difference between measured and the previous PDEF rotation correction value (DAC value).

4.30 PDEFBE

Using the BE mark, PDEFBE automatically corrects the deflection gain and corrects the rotation of the main deflector. The PDEFBE parameters apply to the PDEFBE operation in HCOEFFI.

PDEFBE parameter

EOS mode Condition file name

1 25kv_1na

BE mark detection condition : Setting...

Measurement field size [um] : 2000,000

Number of back-and-forth measurements [times] : 1

Value added to last gain convergence value X [um] : 0,000

Y [um] : 0,000

Allowable gain convergence value [um] : 0,010

Allowable difference from the last gain DAC value [point] : 10000

Value added to last rotation convergence value X [um] : 0,000

Y [um] : 0,000

Allowable rotation convergence value [um] : 0,010

Allowable difference from the last rotation DAC value [point] : 10000

Number of retries in case of nonconvergence [times] : 100

Allowable positional difference of identical point back-and-forth motion [um] : 2,000

Allowable height difference of identical point back-and-forth motion [um] : 2,000

Maximum distortion allowance [um] : 2,000

Save Execute Reset Cancel

Fig. 63 PDEFBE parameter setting window

- BE mark detection condition (Section 4.42.2)
This button opens a window for setting BE mark-detection conditions.
- Measurement field size [μm]
If you want to perform the PDEFBE measurement, specify the size of the field to be measured. The EOS mode limitation of field size are same a PDEF
- Number of back-and-forth measurements [times]
This field lets you specify the number of back-and-forth measurements.
- Value added to last gain convergence value X, Y [μm]
This field lets you specify the correction amounts to add to the PDEF gain correction value after the PDEFBE convergence.
- Allowable gain convergence value [μm]
This field lets you specify the tolerance for convergence of the PDEFBE gain.
- Allowable difference from the last gain DAC value [point]
This field lets you specify the tolerance from the previous PDEFBE gain correction value (DAC value).
- Value added to last rotation convergence value X, Y [μm]
This field lets you specify the correction amounts to add to the PDEF rotation correction value after the PDEFBE convergence.

- Allowable rotation convergence value [μm]
This field lets you specify the tolerance for convergence of the PDEFBE rotation.
- Allowable difference from the last rotation DAC value [point]
This field lets you specify the tolerance from the previous PDEFBE rotation correction value (DAC value).
- Number of retries in case of nonconvergence [times]
This field lets you specify the number of retries until convergence.
- Allowable positional difference of identical point back-and-forth motion [μm]
This field lets you specify the tolerance for the difference between the back and forth positions at each point in the back-and-forth measurement.
- Allowable height difference of identical point back-and-forth motion [μm]
This field lets you specify the tolerance for the difference between the back and forth heights at each point in the back-and-forth measurement.
- Maximum distortion allowance [μm]
This field lets you specify the tolerance for the difference between the expected and measured mark positions at each measurement point.
 Set this parameter to a large value after starting up the system or changing system conditions drastically, and set it to a small value when the system is running stably.

4.31 PRMINZ

PRMINZ lets you specify how the stage moves during calibration and set the status after subprogram execution.

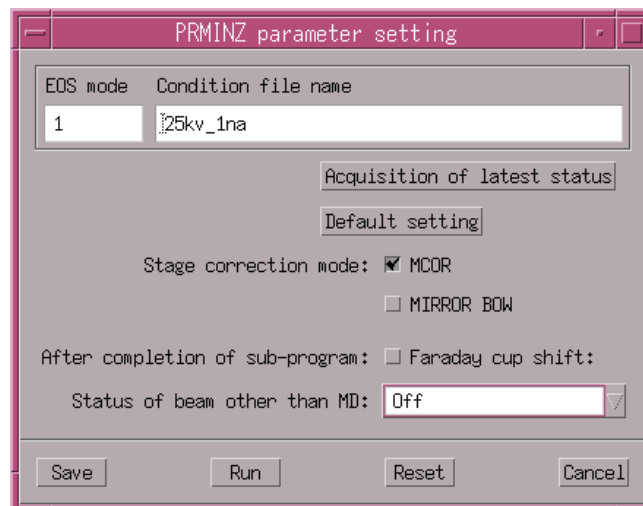


Fig. 64 PRMINZ parameter setting window

- Acquisition of latest status
This button reads the present system parameter settings.
- Default setting
This button sets the **Stage correction mode**, **Faraday cup shift**, and **Status of beam other than MD** parameters to the default values. The default values are as follows: **Stage correction mode**: MCOR enabled, MIRROR BOW enabled; **Faraday cup shift**: enabled; **Status of beam other than MD**: Off.

- Stage correction mode
If you want to move the stage with respect to the mirror-orthogonal correction, select the MCOR box. If you want to move the stage with respect to the mirror-bow correction, select the MIRROR BOW box.
- Faraday cup shift
Select this box if you want to move the stage to the Faraday cup position after executing each calibration subprogram.
- Status of beam other than MD
This field lets you select On, Off, or No change as the beam status setting after subprogram execution. With No change selected, the beam status during subprogram execution remains unchanged.

4.32 PROFIL

The PROFIL subprogram is used as a command to detect the AE marks continuously. To execute PROFIL, click the subprogram-name button on the Calibration window. You cannot execution PROFIL from the PROFIL parameter window.

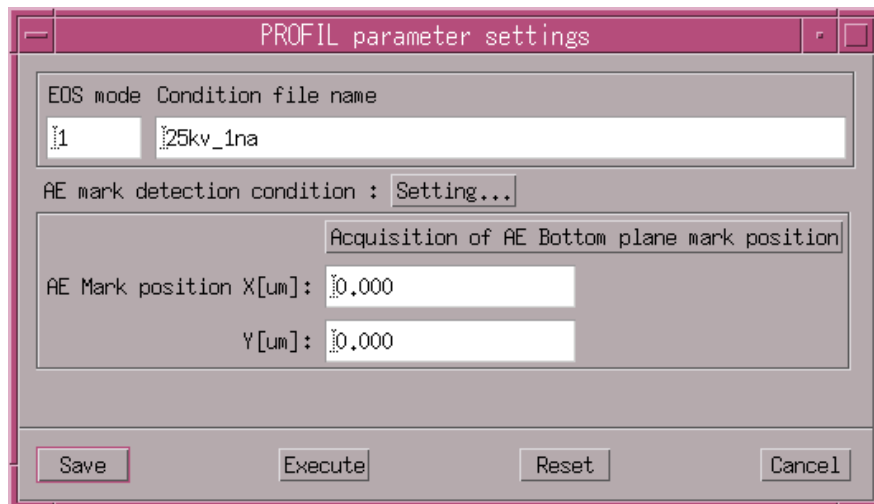


Fig. 65 PROFIL parameter settings window

- AE mark detection condition (Section 4.38.1)
This button opens a window for setting AE mark-detection conditions.
- Acquisition of AE bottom plane mark position
Clicking this button updates the AE mark position X, Y to the latest bottom mark position.

4.33 SETREG

The SETREG subprogram lets you manually set the hardware register values.

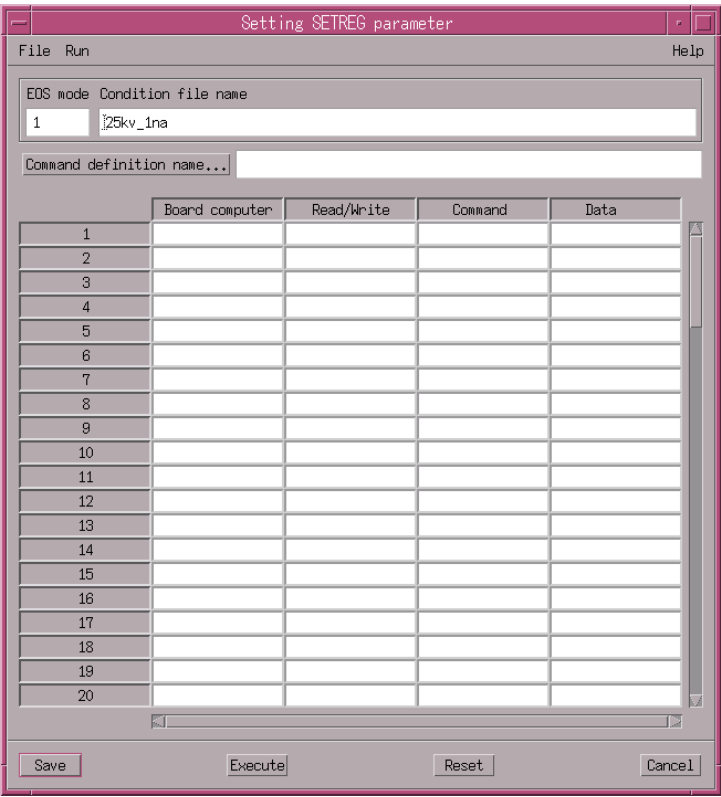


Fig. 66 Setting SETREG parameter

- **File menu**
Use this menu for file management such as reading and saving a register setting command file.
- **Command definition name...**
Use this button to save the parameter settings in a file or read and use them again. Clicking this button opens the following File selection window.

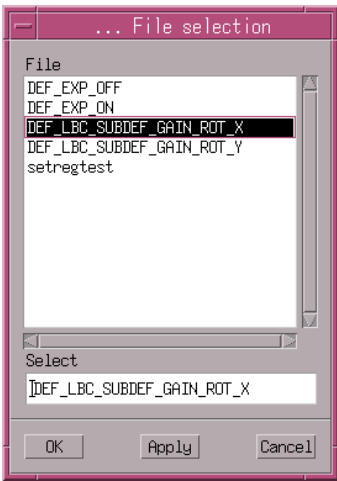


Fig. 67 File selection window

Select a command file from the list in the File area. This is the same as selecting Read on the File menu.

- Board computer
Specify BCE or BCS as the board that enables the register commands.
 - Read/Write
To read from the register, specify R. To write, specify W.
 - Command
Enter the desired command here.
 - Data
If you want to read a command, leave the Data cell blank; to write, enter the desired data here.
-  The results of executing reading will appear in the log of the Calibration GUI.

4.34 SETSCL (Optional)

By performing a series of calibration tasks using the linear-scaling values set here, you can set the correction values in accordance with the scaling values. This command is useful when you purchase the optional fine pitch control or the variable minimum increment writing unit.

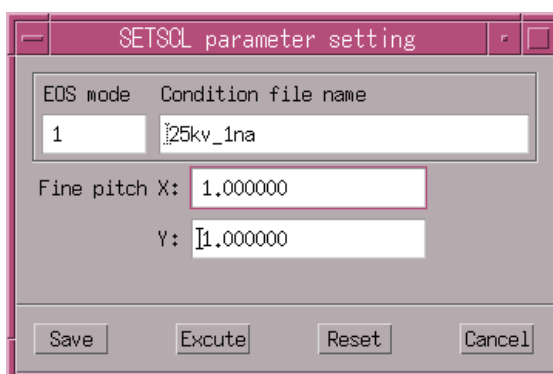


Fig. 68 SETSCL parameter setting window

- Fine pitch
This field lets you enter the linear scaling value.
 The step is 0.000001.

Type	Scaling range
Fine pitch control	0.9375 – 1.0625

Type	Mode	Scaling range
Minimum increment	1	0.500±0.0625
	2	0.625±0.0625
	3	0.750±0.0625
	4	0.875±0.0625
	5	1.000±0.0625

4.35 SETWFR

SETWFR automatically detects global-mark points P and Q.

Fig. 69 SETWFR parameter setting window

- **P-mark rough scan condition** (Section 4.42.3)
This button opens a window for setting P-mark rough scan detection conditions.
- **P-mark fine scan condition** (Section 4.42.3)
This button opens a window for setting P-mark fine scan detection conditions.
- **Q-mark rough scan condition** (Section 4.42.3)
This button opens a window for setting Q, R, S-mark rough scan detection conditions.
- **Q-mark fine scan condition** (Section 4.42.3)
This button opens a window for setting Q, R, S -mark fine scan detection conditions.
- **Measurement mode**
This field lets you select the chip-mark detection mode and the writing-correction mode.
The table below describes each mode.

Mode	Description
A (AUTO)	Selecting this mode automatically detects marks.
S (SEMI)	Selecting this mode automatically detects marks; if the mark detection fails, the program asks the operator what to do next. While viewing the SEM image, the operator moves the stage to find mark positions. After that, the operator can continue detecting marks automatically.

Mode	Description
M (MANUAL)	In this mode, the operator manually performs the same operation (moving the stage while viewing the SEM image) as the automatic mark detection.
C (CANCEL)	Selecting this mode does not detect marks.

- AGC execution flag
Putting a check in the Execute AGC box performs AGC (Automatic Gain Control) for the mark detection during the SETWFR execution.
 This parameter is also used in writing.
- Height measurement flag
Putting a check in the Execute height measurement box measures the height during the SETWFR execution.
- Material type
This field lets you select **Mask** or **Wafer**.
- Material size [inch]
This field lets you specify the material size.
- Material window
This field lets you select the material window. Other than for a multi-cassette, select **0**. You can select one of the following material windows:
0
2A, 2B, 2C, 2D, 2E, 2F, 2G, 2H, 2I, 2J
3A, 3B, 3C, 3D, 3E, 3F, 3G, 3H, 3I, 3J
4A, 4B, 4C, 4D, 4E, 4F, 4G, 4H, 4I, 4J
5A, 5B, 5C, 5D, 5E, 5F, 5G, 5H, 5I, 5J
6A, 6B, 6C, 6D, 6E, 6F, 6G, 6H, 6I, 6J

2A, 2B, ...6J have different material centers for the multi-wafer cassette and the multi-mask cassette in the system parameters (SYSPAR).
 In the above parameters, a 3 inch wafer and the 0 material window are used.
- Allowable rotation value [rad]
This field lets you specify the tolerance for the rotation of the material.
- Material center offset position X, Y [μm]
If you added an offset to the material center, specify the amounts of offset here.
- P-point design position (material system) X, Y [μm]
This field lets you enter the design position of point P in the material-coordinate system.
- Q-point design position (material system) X, Y [μm]
This field lets you enter the design position of point Q in the material-coordinate system.
- R-point design position (material system) X, Y [μm]
This field lets you enter the design position of point R in the material-coordinate system.
- S-point design position (material system) X, Y [μm]
This field lets you enter the design position of point S in the material-coordinate system.

■ Explanation of the parameters in the HEIGHT section

The parameters in the HEIGHT section are used mainly for checking the value of the writing-job command HEIGHT before writing patterns on a substrate whose height cannot be measured or cannot be measured accurately. Accordingly, under normal circumstances, remove the check from the Execute height measurement box before executing the subprogram.

- The amount of minute compensation of a focus lens [point]
This field lets you specify the offset value of the focus value of the objective lens.
- The amount of minute compensation of beam position shift X, Y [nm]
This field lets you specify the beam shift amounts for changing the focus value of the objective lens.
- The amount of minute compensation of a main deflector gain X, Y [point]
This field lets you specify the fine gain compensation amounts of the main deflector for the difference between the height reference plane and the substrate height.
- The amount of minute compensation of a main deflector rotation X, Y [point]
This field lets you specify the fine rotation compensation amounts of the main deflector for the difference between the height reference plane and the substrate height.

■ Explanation of the parameters in the SUBHEI section

The parameters in the SUBHEI section are used mainly for checking the value of the writing-job command SUBHEI before writing patterns on a substrate whose height cannot be measured or cannot be measured accurately. Accordingly, under normal circumstances, remove the check from the Execute height measurement box before executing the subprogram.

- The amount of minute compensation of a sub deflector shift X, Y [point]
This field lets you specify the fine shift compensation amounts of the sub-deflector for the difference between the height reference plane and the substrate height.
- The amount of minute compensation of a sub deflector gain X, Y [point]
This field lets you specify the fine gain compensation amounts of the sub-deflector for the difference between the height reference plane and the substrate height.
- The amount of minute compensation of a sub deflector rotation X, Y [point]
This field lets you specify the fine rotation compensation amounts of the sub-deflector for the difference between the height reference plane and the substrate height.

4.36 SFOCUS

SFOCUS automatically adjusts the focus and astigmatism of the objective lens. The parameters of SFOCUS serve as SFOCUS's measurement parameters to be used in measuring the objective lens change coefficient in HCOEFFFI.

Fig. 70 SFOCUS parameter setting window

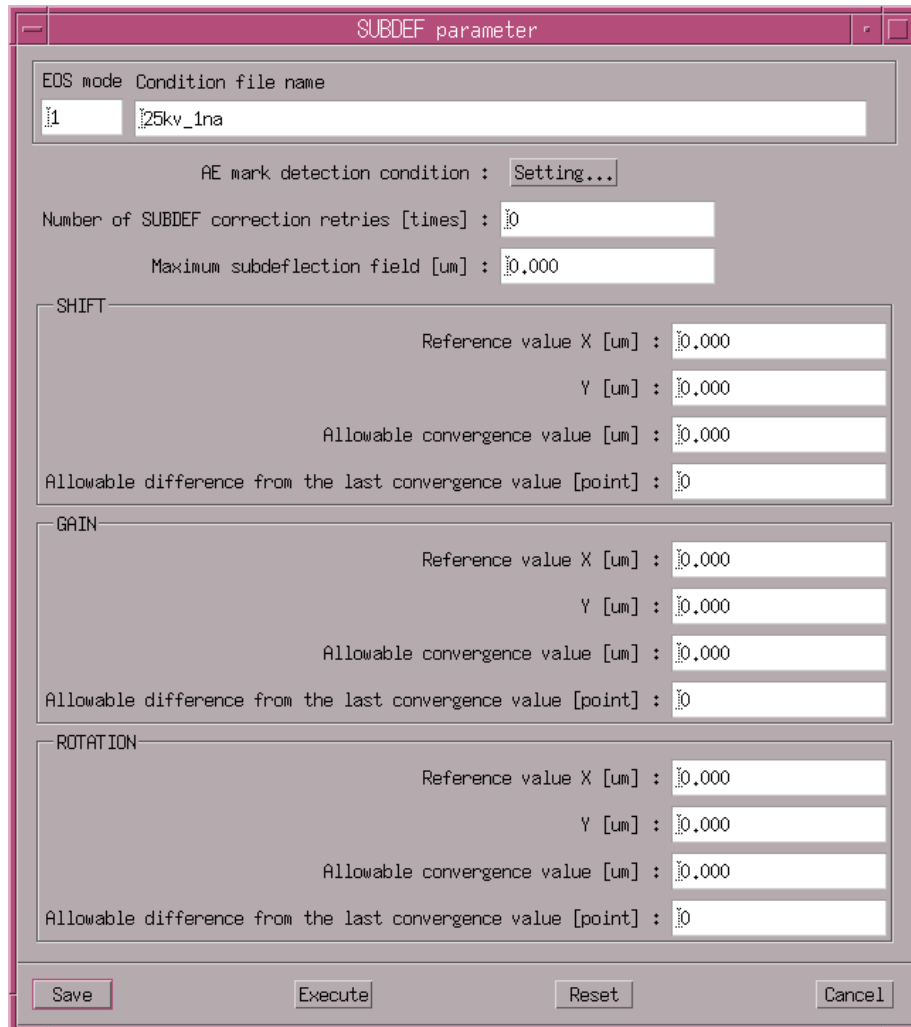
- AE mark detection condition (Section 4.42.1)
This button opens a window for setting AE mark-detection conditions.
- DEMAG execution
Putting a check in the Execute box demagnetizes the objective lens (the fourth lens and fifth lens) before the start of measurement.
 The execution of DEMAG is recommended.
- Measurement item
The items to measure and their descriptions are as follows.

Item	Description
SF/---/---	SFOCUS
---/SSX/---	SFOCUS STIG-X
SF/SSX/---	SFOCUS, SFOCUS STIG-X
---/---/SSY	SFOCUS STIG-Y
SF/---/SSY	SFOCUS, SFOCUS STIG-Y
---/SSX/SSY	SFOCUS STIG-X, SFOCUS STIG-Y
SF/SSX/SSY	SFOCUS, SFOCUS STIG-X, SFOCUS STIG-Y

- Auto SF DAC value conversion step [point]
This field lets you specify the step by which to change the focus value of the objective lens.
- Number of auto SF single-side measurements [times]
This field lets you specify how many points to measure with the focus value of the objective lens changing. Specifying n results in measuring $(2n-1)$ points.
- Auto SSX DAC value conversion step [point]
This field lets you specify the step by which to change the value of the stigmator STIG-X.
- Number of auto SSX single-side measurements [times]
This field lets you specify how many points to measure with the value of the stigmator STIG-X changing. Specifying n results in measuring $(2n-1)$ points.
- Auto SSY DAC value conversion step [point]
This field lets you specify the step by which to change the value of the stigmator STIG-Y.
- Number of auto SSY single-side measurements [times]
This field lets you specify how many points to measure with the value of the stigmator STIG-Y changing. Specifying n results in measuring $(2n-1)$ points.
- Number of retries [times]
This field lets you specify the number of retries in the SFOCUS execution. SFOCUS will be repeated until the STIG-X and STIG-Y values decrease below the Allowable astigmatism value.
- Allowable astigmatism value [point]
This field lets you set the tolerance for convergence of the STIG-X and STIG-Y measurements.
- Allowable difference from the last SFOCUS DAC value [point]
This field lets you specify the tolerance for the difference between the objective lens's focus values at the start of and after execution of SFOCUS.
- Allowable difference from the last SFOCUS STIG-X DAC value [point]
This field lets you specify the tolerance for the difference between the STIG-X values at the start of and after execution of SFOCUS.
- Allowable difference from the last SFOCUS STIG-Y DAC value [point]
This field lets you specify the tolerance for the difference between the STIG-Y values at the start of and after execution of SFOCUS.

4.37 SUBDEF

Using the bottom-plane AE mark, SUBDEF automatically corrects the deflection gain of the sub-deflector and corrects the rotation. The SUBDEF parameters apply to the SUBDEF operation in HCOEFFI.



The image shows a software window titled "SUBDEF parameter". It contains several input fields and buttons. At the top, there are two fields: "EOS mode" with the value "1" and "Condition file name" with the value "25kv_1na". Below these is a button labeled "Setting...". Further down, there are two more input fields: "Number of SUBDEF correction retries [times]" with the value "0" and "Maximum subdeflection field [um]" with the value "0.000". The window is divided into three main sections: "SHIFT", "GAIN", and "ROTATION". Each section contains four input fields: "Reference value X [um]" (0.000), "Y [um]" (0.000), "Allowable convergence value [um]" (0.000), and "Allowable difference from the last convergence value [point]" (0). At the bottom of the window, there are four buttons: "Save", "Execute", "Reset", and "Cancel".

Fig. 71 SUBDEF parameter setting window

- AE mark detection condition (Section 4.42.1)
This button opens a window for setting AE mark-detection conditions.
- Number of SUBDEF correction retries [times]
This field lets you specify the number of retries in the SUBDEF measurement. SUBDEF will be executed until the convergence value decreases below its tolerance or the number of retries exceeds its limit.
- Maximum subdeflection field [μm]
Here you can specify the field size of the sub-deflector for the SUBDEF measurement.
 You can specify up to 4 μm for an accelerating voltage of 100 kV, and up to 8 μm for 50 kV.

■ SHIFT

- Reference value X, Y [μm]
This field lets you specify the reference values for the shift of the sub-deflector.
- Allowable convergence value [μm]
This field lets you specify the tolerance for convergence of the shift of the sub-deflector.
- Allowable difference from the last convergence value [point]
This field lets you specify the tolerance for the difference between the SUBDEF SHIFT DAC values at the start of and after execution of SUBDEF.

■ GAIN

- Reference value X, Y [μm]
This field lets you specify the reference values for the gain of the sub-deflector.
- Allowable convergence value [μm]
This field lets you specify the tolerance for convergence of the gain of the sub-deflector.
- Allowable difference from the last convergence value [point]
This field lets you specify the tolerance for the difference between the SUBDEF GAIN DAC values at the start of and after execution of SUBDEF.

■ ROTATION

- Reference value X, Y [μm]
This field lets you specify the reference values for the rotation of the sub-deflector.
- Allowable convergence value [μm]
This field lets you specify the tolerance for convergence of the rotation of the sub-deflector.
- Allowable difference from the last convergence value [point]
This field lets you specify the tolerance for the difference between the SUBDEF ROTATION DAC values at the start of and after execution of SUBDEF.

4.38 SUBDEFBE

Using the bottom-plane BE mark and the top-plane BE mark, SUBDEFBE automatically corrects the deflection gain of the sub-deflector and corrects the rotation. The SUBDEFBE parameters apply to the SUBDEF operation in HCOEFFI.

SUBDEFBE parameter

EOS mode Condition file name
1 25kv_1na

BE mark detection condition : Setting...

Number of SUBDEFBE correction retries [times] : 100

Maximum subdeflection field [um] : 16.000

SHIFT

Reference value X [um] : 0.000
Y [um] : 0.000
Allowable convergence value [um] : 0.100
Allowable difference from the last convergence value [point] : 10000

GAIN

Reference value X [um] : 0.000
Y [um] : 0.000
Allowable convergence value [um] : 0.010
Allowable difference from the last convergence value [point] : 10000

ROTATION

Reference value X [um] : 0.000
Y [um] : 0.000
Allowable convergence value [um] : 0.010
Allowable difference from the last convergence value [point] : 10000

Save Execute Reset Cancel

Fig. 72 SUBDEFBE parameter setting window

- BE mark detection condition (Section 4.42.2)
This button opens a window for setting BE mark-detection conditions.
- Number of SUBDEFBE correction retries [times]
This field lets you specify the number of retries in the SUBDEFBE measurement. SUBDEFBE will be executed until the convergence value decreases below its tolerance or the number of retries exceeds its limit.
- Maximum subdeflection field [μm]
Here you can specify the field size of the sub-deflector for the SUBDEFBE measurement.
 You can specify up to 4 μm for an accelerating voltage of 100 kV, and up to 8 μm for 50 kV.

■ SHIFT

- Reference value X, Y [μm]
This field lets you specify the reference values for the shift of the sub-deflector.
- Allowable convergence value [μm]
This field lets you specify the tolerance for convergence of the shift of the sub-deflector.
- Allowable difference from the last convergence value [point]
This field lets you specify the tolerance for the difference between the SUBDEFBE SHIFT DAC values at the start of and after execution of SUBDEFBE.

■ GAIN

- Reference value X, Y [μm]
This field lets you specify the reference values for the gain of the sub-deflector.
- Allowable convergence value [μm]
This field lets you specify the tolerance for convergence of the gain of the sub-deflector.
- Allowable difference from the last convergence value [point]
This field lets you specify the tolerance for the difference between the SUBDEFBE GAIN DAC values at the start of and after execution of SUBDEFBE.

■ ROTATION

- Reference value X, Y [μm]
This field lets you specify the reference values for the rotation of the sub-deflector.
- Allowable convergence value [μm]
This field lets you specify the tolerance for convergence of the rotation of the sub-deflector.
- Allowable difference from the last convergence value [point]
This field lets you specify the tolerance for the difference between the SUBDEFBE ROTATION DAC values at the start of and after execution of SUBDEFBE.

4.39 TEMP

TEMP measures the temperature of each part. This subprogram does not have a parameter file.

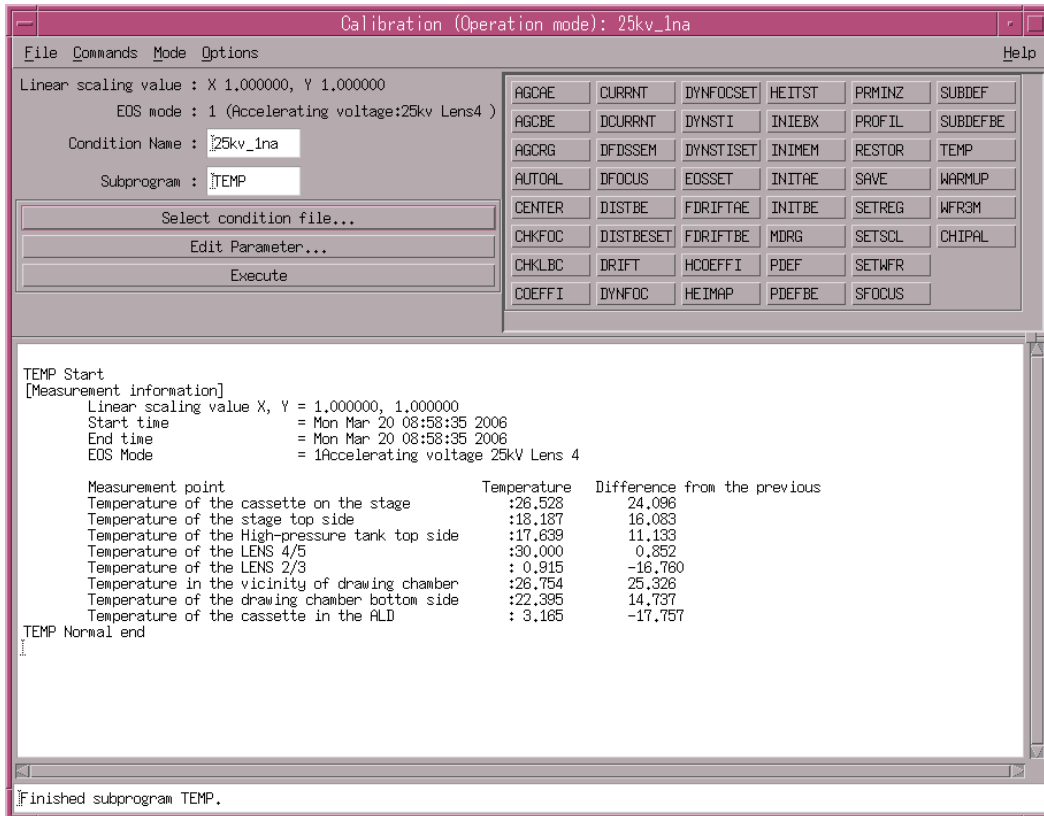


Fig. 73 Calibration (Operation mode) window

- **Temperature measurement points**
 - Temperature of cassette on the stage
 - Temperature of the stage top side
 - Temperature of High-pressure tank top side
 - Temperature of the LENS 4/5
 - Temperature of the LENS 2/3
 - Temperature of vicinity of drawing chamber
 - Temperature of the drawing chamber bottom side
 - Temperature of the cassette in the ALD

4.40 WARMUP

WARMUP performs warm up before writing. Normally, WARMUP is used during writing.

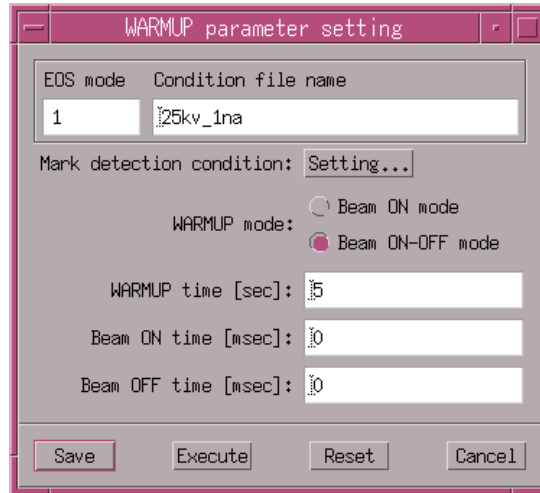


Fig. 74 WARMUP parameter setting window

- BE mark detection condition (Section 4.38.2)
This button opens a window for setting BE mark-detection conditions.
- Warm-up mode
This field lets you select the operation mode. Select **Beam ON** or **Beam ON-OFF** mode.
- Warm-up time [sec]
This field lets you specify the amount of time to warm up.
- Beam ON time [msec]
This field lets you specify the beam-on time in the beam ON-OFF mode.
- Beam OFF time [msec]
This field lets you specify the beam-off time in the beam ON-OFF mode.

4.41 WFR3M

From the global correction values obtained by the subprogram SETWFR, WFR3M detects chip marks on the substrate according to the substrate's shift, expansion/contraction, and rotation, and calculates the chip distortion. CHIPAL uses 4 mark points, whereas WFR3M calculates correction values with 3 mark points.

WFR3M parameter settings

EOS mode Acceleration Voltage:
1 25kv_1na

RG mark detection condition : Setting...

Measurement mode: 3-point mark PDEF Gain, Rotation convergence

Execution item: ☒ AGCRG ☒ Height measurement ☒ SUBDEF measurement

Mark field center position (material coordinate) X[um]: 0.000
Y[um]: 0.000

M1-mark position (relative distance chip center) X[um]: 0.000
Y[um]: 0.000

M2-mark position (relative distance chip center) X[um]: 0.000
Y[um]: 0.000

M3-mark position (relative distance chip center) X[um]: 0.000
Y[um]: 0.000

PDEF SUBDEF HEIGHT SUBHEI

Number of PDEF convergence retries [times]: 100

Gain standard value X[um]: 0.000
Gain standard value Y[um]: 0.000
Gain allowable convergence value [um]: 0.000
Rotation standard value X[um]: 0.000
Rotation standard value Y[um]: 0.000
Rotation allowable convergence value [um]: 0.000

Save Execute Reset Cancel

Fig. 75 WFR3M parameter settings window

■ Parameter description

- RG detection condition ( Section 4.42.2)
This button opens a window for setting the conditions for detecting registration marks.
- Measurement mode
This field lets you select the chip-mark detection mode and the writing-correction mode.

This table describes each mode.

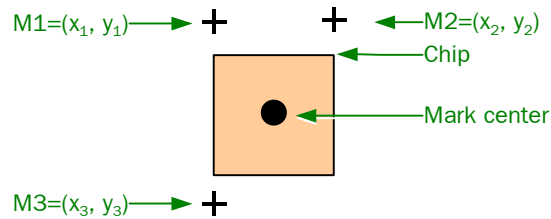
Mode	Description
1-point mark shift measurement	In writing, the position of chip mark M1 is detected to correct the displacement (shift) of the chip-mark center.
	In calibration, the displacement is measured and the results are displayed.
3-point mark PDEF Gain Rotation convergence	In writing, the positions of chip marks M1, M2, and M3 are detected to correct the main deflection's gain/rotation and the objective lens's focus value.
	In calibration, the positions of chip marks M1, M2, and M3 are detected, and the main deflection's gain/rotation and the objective lens's focus value are displayed.
3-point mark PDEF Gain convergence	In writing, the positions of chip marks M1, M2, and M3 are detected to correct the main deflection's gain and the objective lens's focus value.
	In calibration, the positions of chip marks M1, M2, and M3 are detected, and the main deflection's gain and the objective lens's focus value are displayed.
1-point mark PDEF Gain Rotation convergence	In writing, the position of chip mark M1 is detected to correct the main deflection's gain/rotation and the objective lens's focus value.
	In calibration, the position of chip mark M1 is detected, and the main deflection's gain/rotation and the objective lens's focus value are displayed.
1-point mark PDEF Gain convergence	In writing, the positions of chip marks M1 is detected to correct the main deflection's gain and the objective lens's focus value.
	In calibration, the position of chip mark M1 is detected, and the main deflection's gain and the objective lens's focus value are displayed.
1-point mark SEM image mark search	In writing, the position of chip mark M1 is detected by SEM observation to correct the displacement (shift) of the chip mark center.
	In calibration, the stage moves to chip mark M1; the operator is prompted to use the SEM image to continue the operation; the operator searches for M1 and registers it; then, the displacement is measured and the results are displayed.
Without marks	In writing, chip marks are not detected. Based on the global mark's displacement (shift) and the substrate's expansion/contraction and rotation correction, the chip mark center's displacement (shift) and the beam deflection region's size (gain) and rotation are corrected.
	The calibration program performs internal processes, but does not perform operations such as mark detection.

- Execution item

- AGCRG

With AGCRG selected, the amplifier parameters for the RG marks are automatically adjusted (Auto Gain Control) during WFR3M execution.

- Height measurement
With Height measurement selected, the height is measured during WFR3M execution. The results affect the writing.
- SUBDEF measurement
With SUBDEF measurement selected, the sub-deflector's gain, rotation, and shift are corrected with respect to chip mark M1 after PDEF measurement. The results affect the writing.
- Mark field center position X, Y [μm]
The positions of the 3 mark points are defined as distances from this center position. Specify the center coordinate position in the substrate-coordinate system.
- M1 (M2, M3) mark position (relative distance chip center) X, Y [μm]
Specify the position of chip mark M1 (M2, M3) in the coordinate system with the above mark-field center position at the origin (chip center coordinate system: x increases in the right direction, y increases in the upward direction).
 Given that the coordinates are $M1=(x_1, y_1)$, $M2=(x_2, y_2)$, $M3=(x_3, y_3)$, the relationships among M1, M2, and M3 are as follows:
 - $x_2-x_1>0$
 - $x_3-x_1>0$
 - $y_1-y_3>0$ and $y_2-y_3>0$
 The positional relationships are as follows.



- ROT tolerance [degree]
Specify the tolerance for the amount of rotation of the chip mark.

■ Description of the PDEF parameters

- Number of PDEF convergence retries [times]
Specify the number of retries until convergence.
- Gain standard value X, Y [μm]
Set the default gain correction values for PDEFBE measurement.
- Gain allowable convergence value [μm]
Specify the tolerance for convergence of PDEFBE gain.
- Rotation standard value X, Y [μm]
Set the default rotation correction values for PDEFBE measurement.
- Rotation allowable convergence value [μm]
Specify the tolerance for convergence of PDEFBE rotation correction.

■ Description of the SUBDEF parameters

Fig. 76 WFR3M parameter settings (SUBDEF) window

- Maximum sub-deflection field [μm]
Specify the field size for SUBDEF measurement.
- Number of SUBDEF convergence retries [times]
Specify the number of retries until convergence.
- Shift standard value X, Y [μm]
Set the default shift correction values for SUBDEFBE measurement.
- Shift allowable convergence value [μm]
Specify the tolerance for convergence of SUBDEFBE shift.
- Gain standard value X, Y [μm]
Set the default gain correction values for SUBDEFBE measurement.
- Gain allowable convergence value [μm]
Specify the tolerance for convergence of SUBDEFBE gain.
- Rotation standard value X, Y [μm]
Set the default rotation correction values for SUBDEFBE measurement.
- Rotation allowable convergence value [μm]
Specify the tolerance for convergence of SUBDEFBE rotation correction.

■ Description of the HEIGHT parameters

WFR3M parameter settings

EOS mode Acceleration Voltage:
1 25kv_1na

RG mark detection condition : Setting...

Measurement mode: 3-point mark PDEF Gain, Rotation convergence

Execution item: ☒ AGCRG ☒ Height measurement ☒ SUBDEF measurement

Mark field center position (material coordinate) X[um]: 0.000
Y[um]: 0.000

M1-mark position (relative distance chip center) X[um]: 0.000
Y[um]: 0.000

M2-mark position (relative distance chip center) X[um]: 0.000
Y[um]: 0.000

M3-mark position (relative distance chip center) X[um]: 0.000
Y[um]: 0.000

PDEF SUBDEF **HEIGHT** SUBHEI

The amount of minute compensation of a focus lens [point]: 0

The amount of minute compensation of beam position shift X[nm]: 0.000
Y[nm]: 0.000

The amount of minute compensation of a main deflector gain X[point]: 0
Y[point]: 0

The amount of minute compensation of a main deflector rotation X[point]: 0
Y[point]: 0

Save Execute Reset Cancel

Fig. 77 WFR3M parameter settings (HEIGHT) window

- The amount of minute compensation of a focus lens [point]
Specify the value to add to the focus lens value for height measurement.
- The amount of minute compensation of beam position shift X, Y [μm]
Specify the beam shift values to add for height measurement.
- The amount of minute compensation of a main deflector gain [point]
Specify the values to add to the main deflector's gain for height measurement.
- The amount of minute compensation of a main deflector rotation [point]
Specify the values to add to the main deflector's rotation for height measurement.

■ Description of the SUBHEI parameters

WFR3M parameter settings

EOS mode Acceleration Voltage:
1 25kv_1na

RG mark detection condition : Setting...

Measurement mode: 3-point mark PDEF Gain, Rotation convergence

Execution item: ☒ AGDRG ☒ Height measurement ☒ SUBDEF measurement

Mark field center position (material coordinate) X[um]: 0.000
Y[um]: 0.000

M1-mark position (relative distance chip center) X[um]: 0.000
Y[um]: 0.000

M2-mark position (relative distance chip center) X[um]: 0.000
Y[um]: 0.000

M3-mark position (relative distance chip center) X[um]: 0.000
Y[um]: 0.000

PDEF SUBDEF HEIGHT **SUBHEI**

The amount of minute compensation of a sub deflector shift X[point]: 0
Y[point]: 0

The amount of minute compensation of a sub deflector gain X[point]: 0
Y[point]: 0

The amount of minute compensation of a sub deflector rotation X[point]: 0
Y[point]: 0

Save Execute Reset Cancel

Fig. 78 WFR3M parameter settings (SUBHEI) window

- The amount of minute compensation of a sub deflector shift X, Y [point]
Specify the sub-deflector's shift values to add for height measurement.
- The amount of minute compensation of a sub deflector gain [point]
Specify the values to add to the sub-deflector's gain for height measurement.
- The amount of minute compensation of a sub deflector rotation [point]
Specify the values to add to the sub-deflector's rotation for height measurement.

4.42 Mark Detection Condition Setting Window

4.42.1 Setting the AE mark-detection conditions

The AE mark parameter compilation window lets you set the AE mark-detection conditions. It consists of these five windows:

- Scan
- offset
- Gain setting
- Allowable values
- Raster scan

■ Applies to another subprogram

Use this feature if you want to apply the mark-detection conditions to the subprograms that use the AE mark.

1. In the item that is Set item box, click on the item to which you want to apply the subprograms.
2. In the Subprogram that is Mark type box, click on the subprogram to which you want to apply the mark-detection condition.
Selecting some subprograms, click on the subprograms while you press **ctrl** key.
Selecting all subprograms click on the All select button.
3. After selecting the subprogram, press the **Subprogram that can be set** button to apply the detecting condition: then to close the window, click on the **Close** button.

Completing the above procedure applies the mark-detection conditions to the subprograms that use the AE mark.

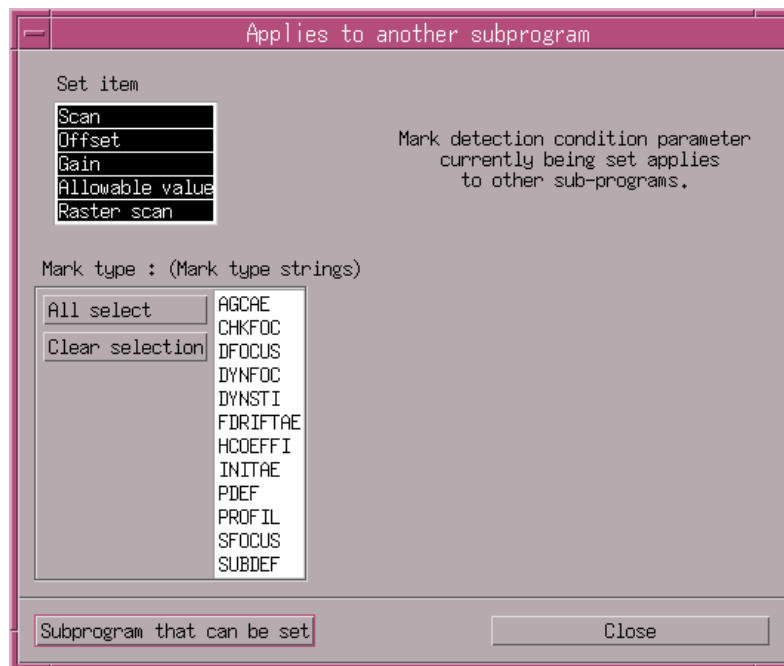


Fig. 79 Applies to another subprogram window

■ Window for setting scan conditions

Here you can set the parameters related to mark scan.

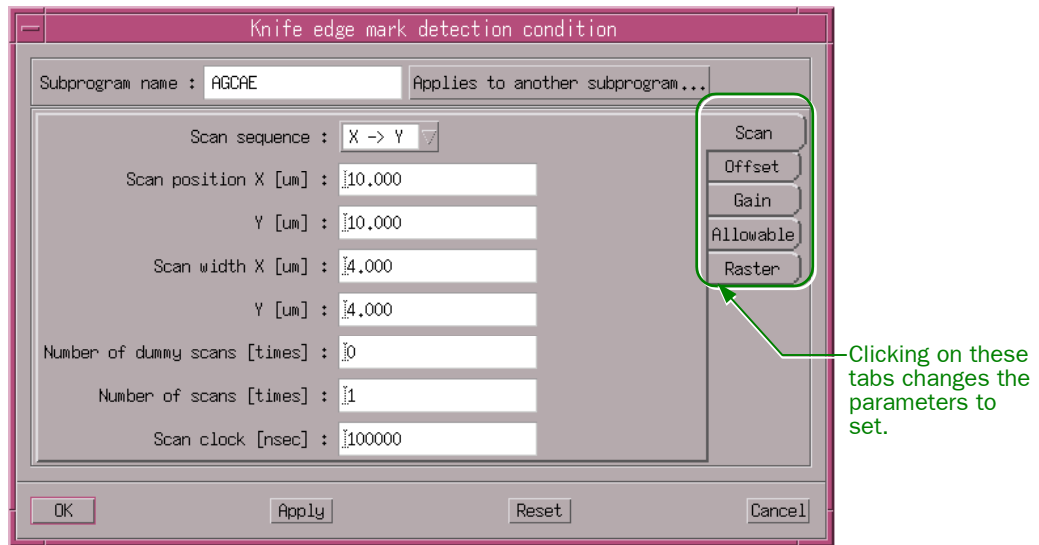


Fig. 80 Knife edge mark detection condition (Scan) window

- Sweep sequence

The sweep sequences, which can be selected, and their descriptions are as follows.

Sweep sequence	Description
X -> Y	First, X scan, then Y scan is performed.
Y -> X	First, Y scan, then X scan is performed.
X only	Only X scan is performed.
Y only	Only Y scan is performed.

- Sweep position X, Y [μm]

This field lets you set the scan position from the mark center.

 For the BOX scan, you cannot specify zero.

- Sweep width X, Y [μm]

This field lets you specify the scan width.

 Setting the sweep width to 4 μm or less results in measuring 4000 or fewer points, reducing the mark-detection time.

 Setting the sweep width to 4 μm or longer results in measuring 4000 points, making the scan pitch coarse.

- Number of dummy sweeps [times]

This field lets you specify the number of dummy scans at one scan position.

- Number of scans [times]

This field lets you specify the number of actual scans at one scan position.

- Scan clock [nsec]

This field lets you specify the travel time per step for the beam to scan.

■ Offset

By specifying the offset of the scan position, you can change the scan position on the mark.

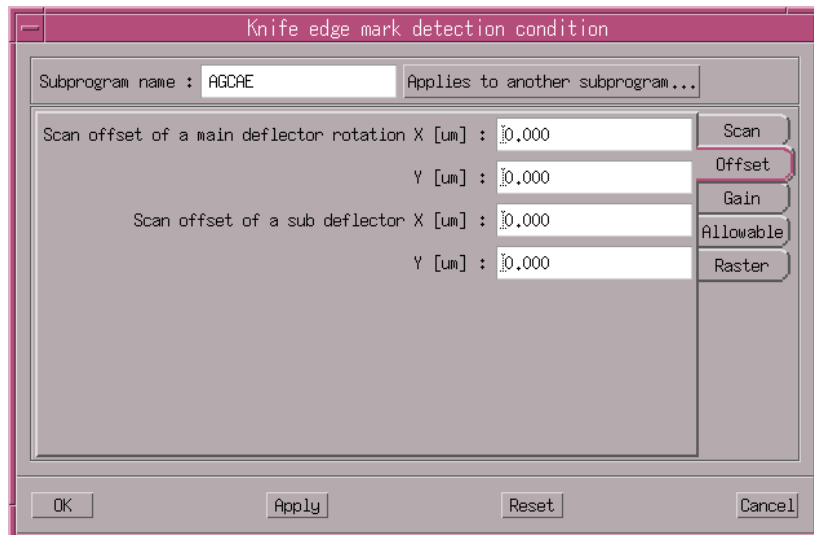


Fig. 81 Knife edge mark detection condition (Offset) window

- Scan offset X, Y [μm]

Setting the scan offset to zero will perform a scan with the deflection center of the main/sub deflector at the center. On the other hand, if the scan offset is set to a certain position, the scan will be performed with the position at the center.

 Normally, set the scan offset to zero. However, before performing DFOCUS or other tasks, you can check to see if INITAE can run normally by setting the scan offset to half of the maximum field size.

■ Window for setting gain

Here you can set the gain and offset of the amplifier in the signal-processing system.

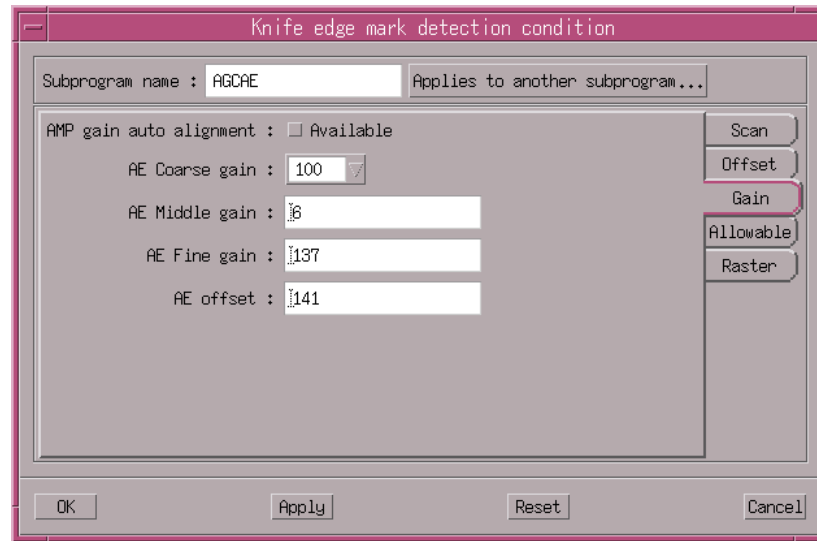
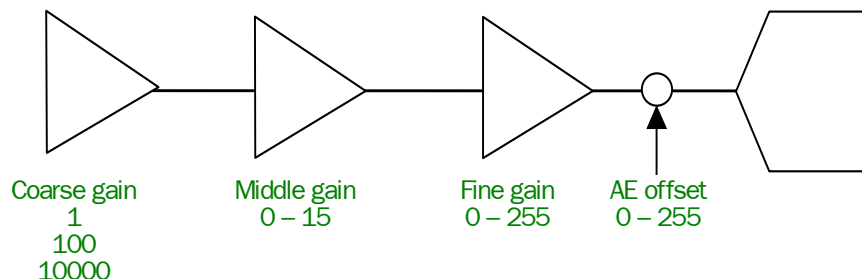


Fig. 82 Knife edge mark detection condition (Gain) window

- AMP gain auto alignment
The Coarse/Middle/Fine gain and the Fine offset are automatically calculated and set with consideration given to current fluctuation.
 - AE Coarse gain
This field lets you select the value of the coarse gain adjustment of the AE amplifier.
 - AE Middle gain
This field lets you specify the value of the middle gain adjustment of the AE amplifier.
 - AE Fine gain
This field lets you specify the value of the fine gain adjustment of the AE amplifier.
 - AE Fine offset
This field lets you specify the fine-offset value of the AE amplifier.
- ✂ The figure below illustrates the gain settings. You can set the parameters in the following ranges.



- ✂ In normal use, the results of running AGCAE are used as the AE-mark gain parameter settings, so it is rare to change the gain parameters manually.

■ Window for setting allowable values

Here you can set allowable values for the mark-detection calculation.

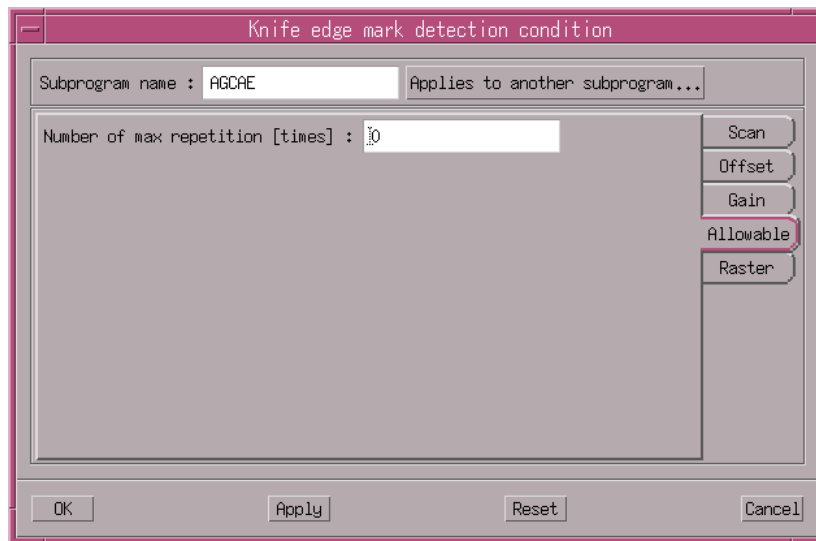


Fig. 83 Knife edge mark detection condition (Allowable) window

- Maximum repetition number [times]
This field lets you specify the maximum number of repetitions of the fitting calculation.

■ Window for setting raster scan conditions

Here you can set the parameters related to the raster scan.

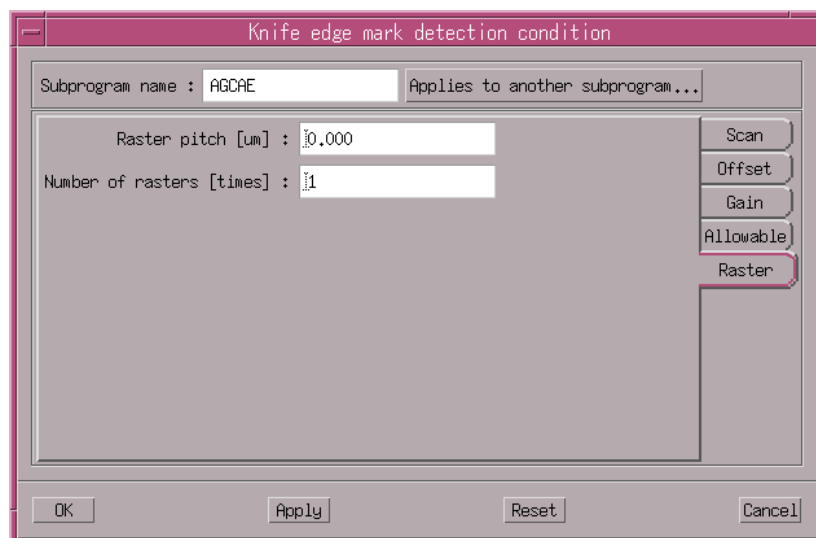


Fig. 84 Knife edge mark detection condition (Raster) window

- Raster pitch [μm]
This field lets you specify the raster-scan pitch.
- Number of rasters [times]
This field lets you specify the number of raster scans.

4.42.2 Revising BE mark parameters

The BE mark detection condition window lets you set the BE mark-detection conditions. It consists of these seven windows:

- Scan
- Offset
- Scan type
- Gain setting
- Correlation
- Allowable values
- Raster scan

■ Applies to another subprogram

Use this feature if you want to apply the mark-detection conditions to the subprograms that use the BE mark.

1. In the item that is Set item box, click on the item to which you want to apply the subprograms.
2. In the Subprogram that is Mark type box, click on the subprogram to which you want to apply the mark-detection condition.
Selecting some subprograms, click on the subprograms while you press **ctrl** key. Selecting all subprograms click on the All select button.
3. After selecting the subprogram, press the **Subprogram that can be set** button to apply the detecting condition: then to close the window, click on the **Close** button.

Completing the above procedure applies the mark-detection conditions to the subprograms that use the BE mark.

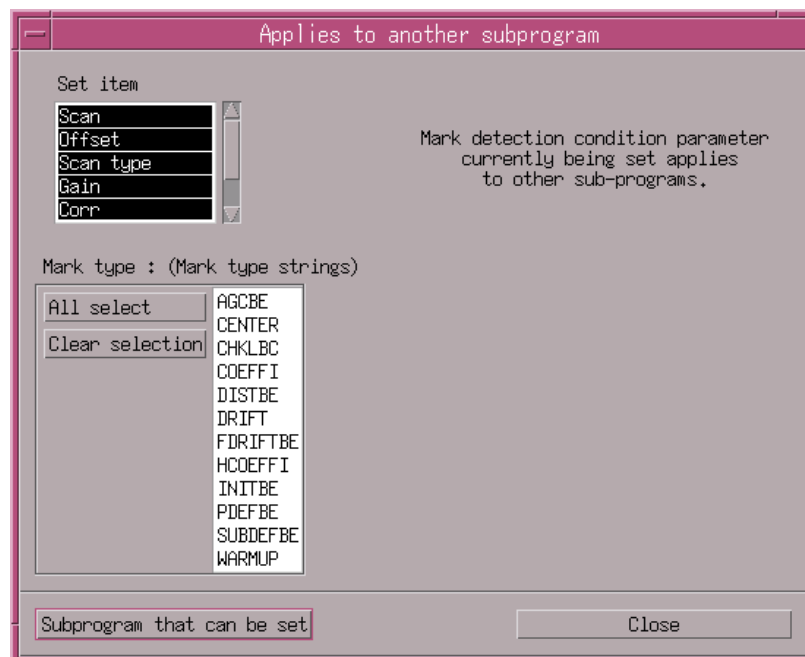


Fig. 85 Applies to another subprogram

■ Window for setting scan conditions

Here you can set the parameters related to the mark scan.

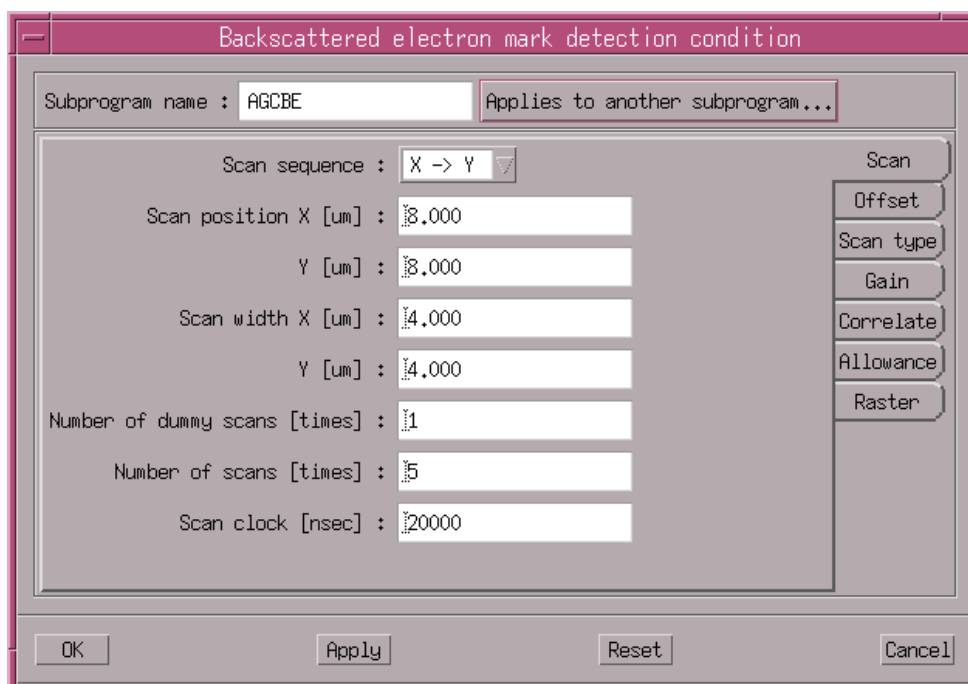


Fig. 86 Backscattered electron mark detection condition (Scan) window

- Sweep sequence

The selectable sweep sequences are as follows.

Sweep sequence	Description
X -> Y	The X scan is performed and then the Y scan is performed.
Y -> X	The Y scan is performed and then the X scan is performed.
X only	Only the X scan is performed.
Y only	Only the Y scan is performed.

- Sweep position X, Y [μm]

This field lets you set the scan position from the mark center.

 For the BOX scan, you cannot specify zero.

- Sweep width X, Y [μm]

This field lets you specify the scan width.

 Setting the sweep width to 4 μm or less results in measuring 4000 or fewer points, reducing the mark-detection time.

 Setting the sweep width to 4 μm or longer results in measuring 4000 points, making the scan pitch coarse.

- Number of dummy sweeps [times]

This field lets you specify the number of dummy scans at one scan position.

 The dummy sweeps are performed only for the first scan position.

- Number of scans [times]

This field lets you specify the number of actual scans at one scan position.

- Scan clock [nsec]
This field lets you specify the travel time per step for the beam to scan.
- Scan offset X, Y [μm]
Setting the scan offset to zero will perform a scan with the deflection center of the main deflector at the center. On the other hand, setting the scan offset to a certain position will perform the scan with the position at the center.
 Normally, set the scan offset to zero.

■ Offset

By specifying the offset of the scan position, you can change the scan position on the mark.

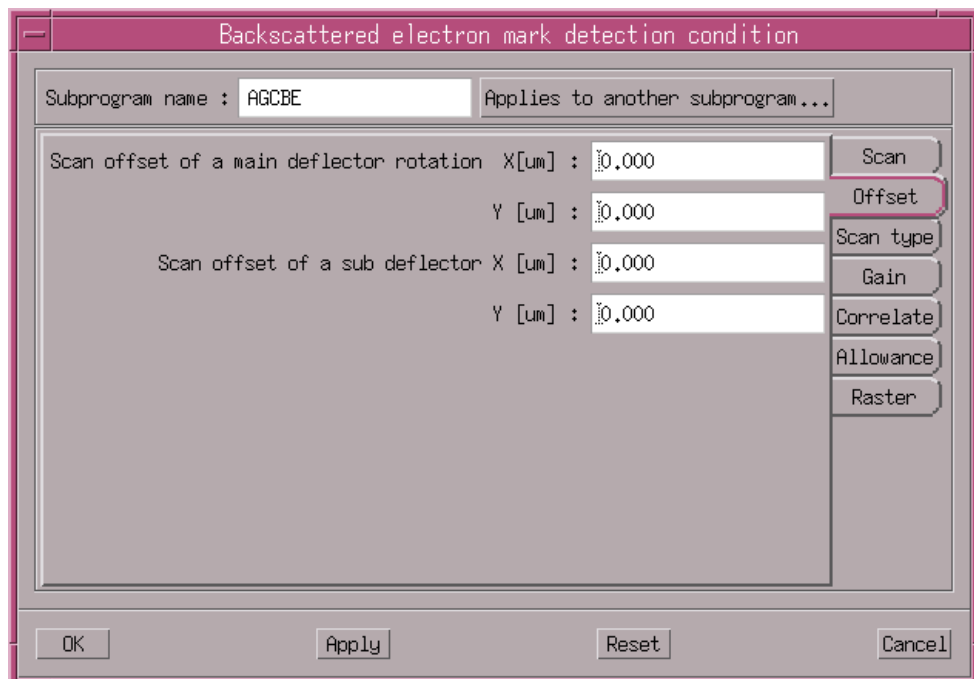


Fig. 87 Backscattered electron mark detection condition (Offset) window

- Scan offset of a main (sub) deflector X, Y [μm]
With the scan-offset position specified as zero, the scan is performed around the deflection center of the main (sub) deflector. If you enter any values in these boxes, scan is performed with the specified position at the center.

■ Window for setting scan type

Here you can set the parameters related to the type of mark scan.

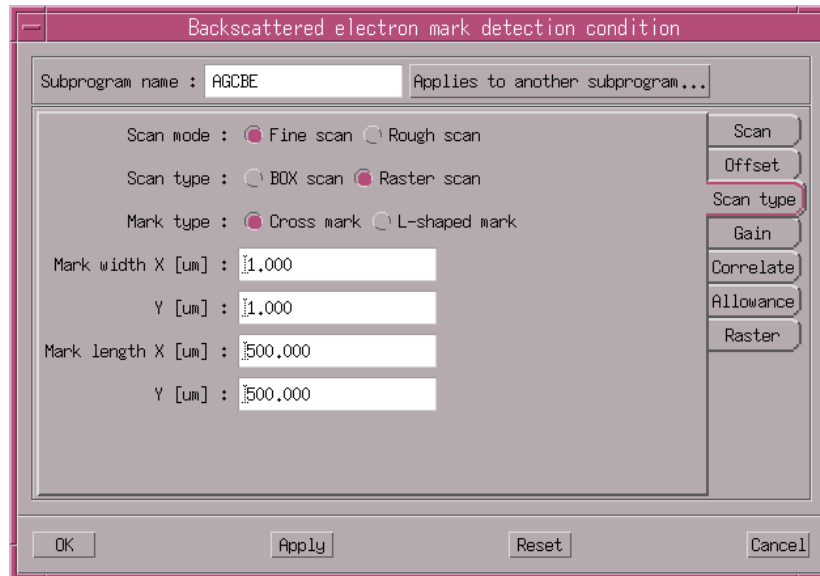


Fig. 88 Backscattered electron mark detection condition (Scan type) window

- Scan mode
You can select Fine scan or Rough scan.
- Scan type
You can select BOX scan or Raster scan.
- Mark type
You can select Cross mark or L-shaped mark.
- Mark width X, Y [μm]
This field lets you specify the mark width.
- Mark length X, Y [μm]
This field lets you specify the mark length.

■ Window for setting gain

Here you can set the gain and offset of the amplifier in the signal-processing system.

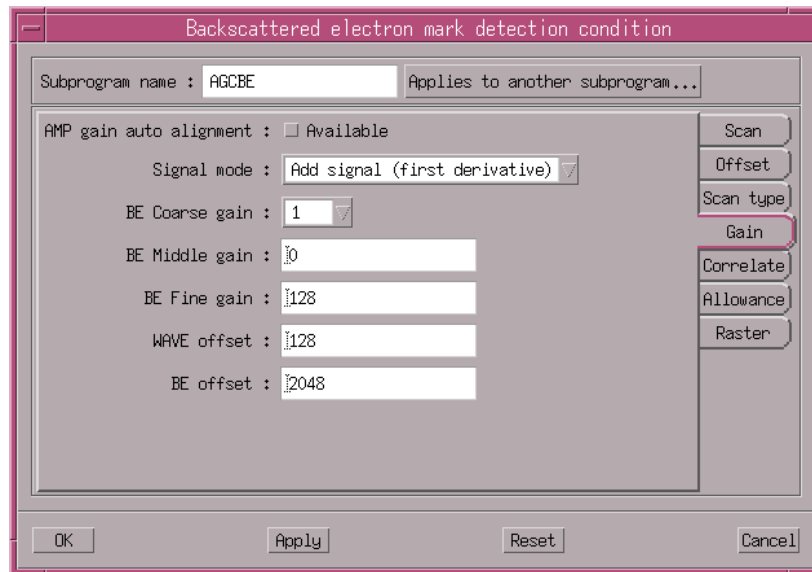
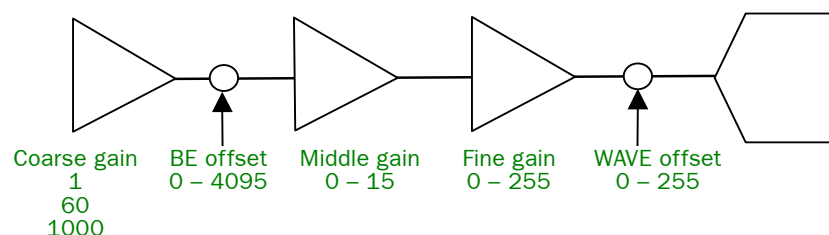


Fig. 89 Backscattered electron mark detection condition (Gain) window

- **AMP gain auto alignment**
The Coarse/Middle/Fine gain and the Fine offset are automatically calculated and set with consideration given to current fluctuation.
- **Signal mode**
You can select Add signal (first derivative), Add signal (raw), or Difference signal (raw) as the signal-processing mode.
- **BE Coarse gain**
This field lets you select the value of the BE coarse gain of the BE amplifier.
- **BE Middle gain**
This field lets you specify the value of the BE middle gain of the BE amplifier.
- **BE Fine gain**
This field lets you specify the value of the BE fine gain of the BE amplifier.
- **WAVE offset**
This field lets you specify the fine-offset value of the BE amplifier.
- **BE offset**
This field lets you specify the rough-offset value of the BE amplifier.

✂ The figure below illustrates the gain settings. You can set the parameters in the following ranges.



✂ In normal use, the results of running AGCBE are used as the BE-mark gain parameter settings, so it is rare to change the gain parameters manually.

■ Window for setting correlation calculation conditions

Here you can set the correlation-calculation parameters.

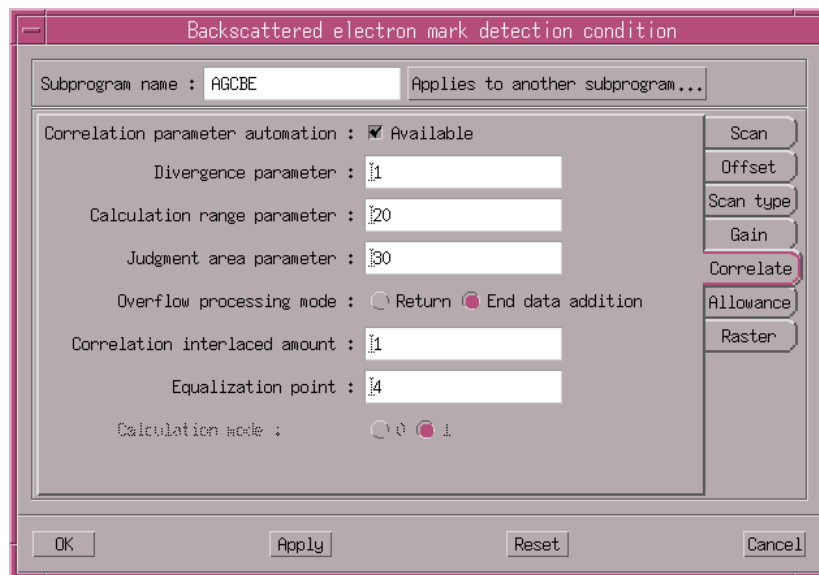


Fig. 90 Backscattered electron mark detection condition (Correlate) window

- Correlative auto calculation mode

You can specify whether to enable or disable automatic setting of correlation-calculation parameters. Enabling the automatic setting will automatically calculate the following Divergence, Calculation range, and Judgment area parameters.

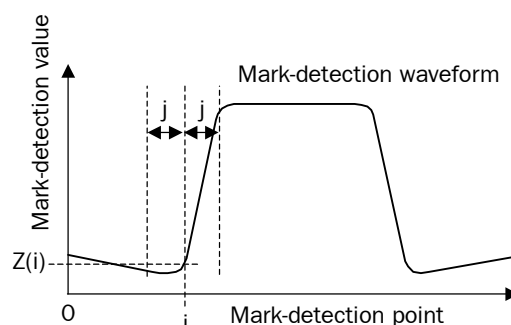
 The definitions of the correlation calculation parameters are shown below.

- Divergence parameter (j)

In order to obtain the mark-detection value $Z(i)$ at one mark-detection point i , this parameter specifies the number of points before and after i needed for correlation calculation of their detection values. It is to be expressed using an integer from 0 to 4000.

- Supplementary explanation

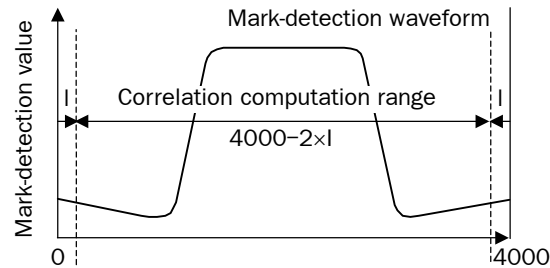
The definition of the divergence parameter (j) is shown below.



- Calculation range parameter (l)

Specifies the range over which correlation calculation is to be performed on the mark-detection waveform. It is to be expressed using an integer from 0 to 4000.

- The definition of parameter l is shown below.

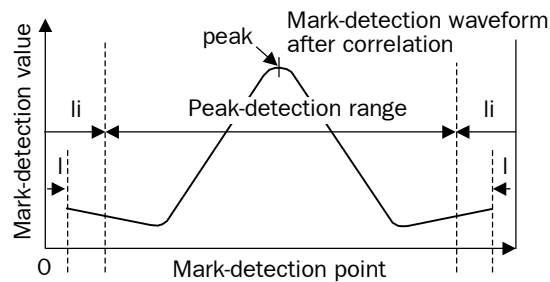


- Judgment area parameter (li)

Specifies the range over which the peak value of the mark-detection waveform after correlation calculation is to be detected. It is to be expressed using an integer from 0 to 4000. If you omit this parameter, the system sets li to the same value that you specified for the parameter l.

- Supplementary explanation

The definition of parameter li is shown below.



- Overflow processing mode (ex)

In order to perform a correlation calculation, it may be necessary to use mark-detection values that are to be obtained both before the start of the mark-detection waveform and after the end of that waveform. In that correlation calculation, this parameter specifies the way in which the necessary mark-detection values are to be used. It is to be expressed using the integer 0 or 1.

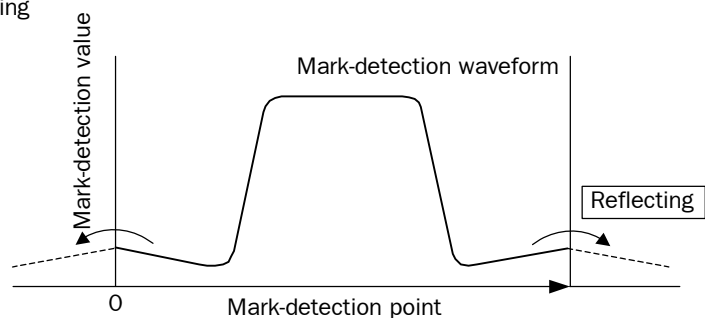
The meanings of the numbers are as follows.

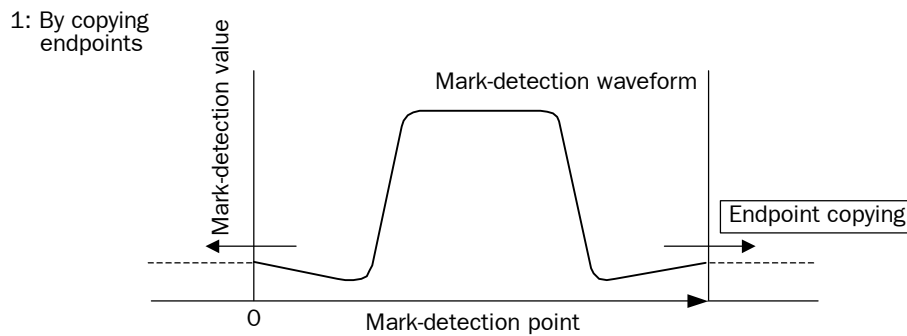
ex	Meaning
0	The values are to be obtained by reflecting the waveform.
1	The values are to be obtained by copying the end points of the waveform.

- Supplementary explanation

The definition of parameter ex is shown below.

0: By reflecting





- Bit shift amount (bs)
This parameter reduces the mark-detection waveform resulting from correlation computation. It converts the mark-detection value into a binary value, then specifies the number of bits to shift digits. It is to be expressed using an integer from 0 to 4.
- Averaging point (av)
Specifies the number of data points to be used when smoothing the mark-detection waveform resulting from correlation computation. It is to be expressed using an integer from 1 to 2000.
- Calculation mode

■ Window for setting allowable values

Here you can set the allowable values for the mark-detection calculation.

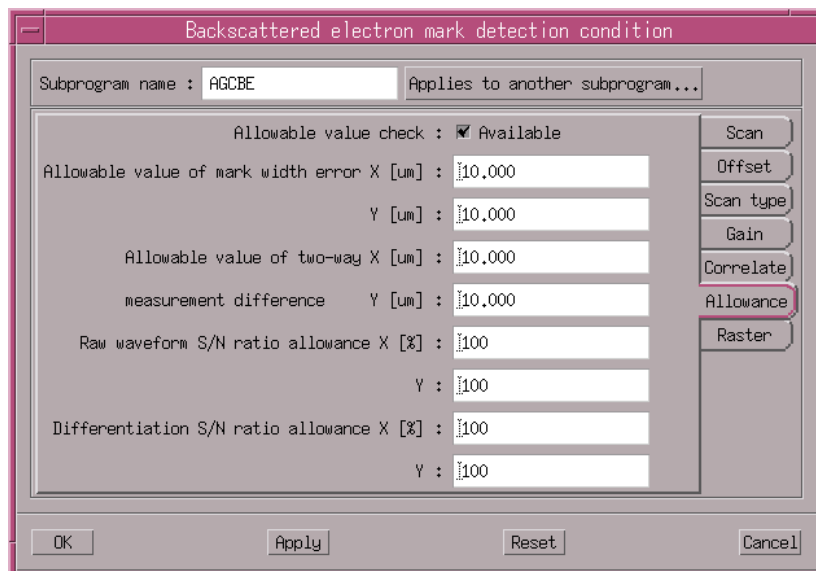


Fig. 91 Backscattered electron mark detection condition (Allowance) window

- Allowable value check
If you want to put the following parameters into effect, put a check in the **Available** box.
- Allowable value of mark width error X, Y [μm]
This field lets you specify the tolerance for the difference between the detected and expected mark widths.

- Allowable value of two-way measurement difference X, Y [μm]
This field lets you specify the tolerance for the difference between the detected and expected mark positions.
- Raw waveform S/N ratio allowance X, Y
The S/N ratio is defined below. Normally, specify a value two or three times the S/N ratio in percent (%). In MD waveform diagram 1 (Section 4.42.4, Fig. 105), the S/N ratio decreases to 100 or less. On the other hand, in MD waveform diagram 2 (Section 4.42.4, Fig. 106), the S/N ratio might exceed 500.

$$\text{S/N ratio} = (\text{noise amplitude} / \text{signal amplitude}) \times 100 (\%)$$

$$\text{Signal amplitude} = \text{average signal of mark part} - \text{average signal of noise part}$$

$$\text{Noise amplitude} = \text{maximum signal near scan start point} - \text{minimum signal near scan start point}$$
- Differentiation S/N ratio allowance X, Y
These values affect the differentiation waveform. The description of the S/N allowance is the same as with the raw waveform.

■ Window for setting raster scan conditions

Here you can set the parameters related to the raster scan.

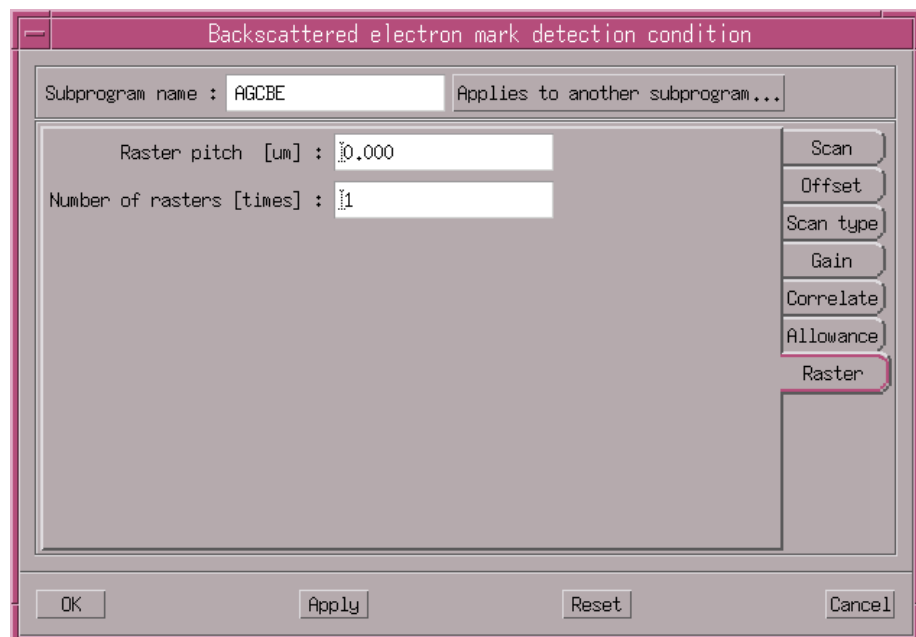


Fig. 92 Backscattered electron mark detection condition (Raster) window

- Raster pitch [μm]
This field lets you specify the raster-scan pitch.
- Number of rasters [times]
This field lets you specify the number of raster scans.

4.42.3 Revising registration mark parameter

The Registration mark parameter compilation window lets you set the RG (registration) mark-detection conditions. It consists of these seven windows:

- Scan
- Offset
- Scan type
- Gain setting
- Correlation
- Allowable values
- Retry

■ Applies to another subprogram

Use this feature if you want to apply the mark-detection conditions to the subprograms that use the RG mark.

1. In the item that is Set item box, click on the item to which you want to apply the subprograms.
2. In the Subprogram that is Mark type box, click on the subprogram to which you want to apply the mark-detection condition.

Selecting some subprograms, click on the subprograms while you press **ctrl** key.
Selecting all subprograms click on the All select button.

3. After selecting the subprogram, press the **Subprogram that can be set** button to apply the scan condition: then to close the window, click on the **Close** button.

Completing the above procedure applies the mark-detection conditions to the subprograms that use the RG mark.

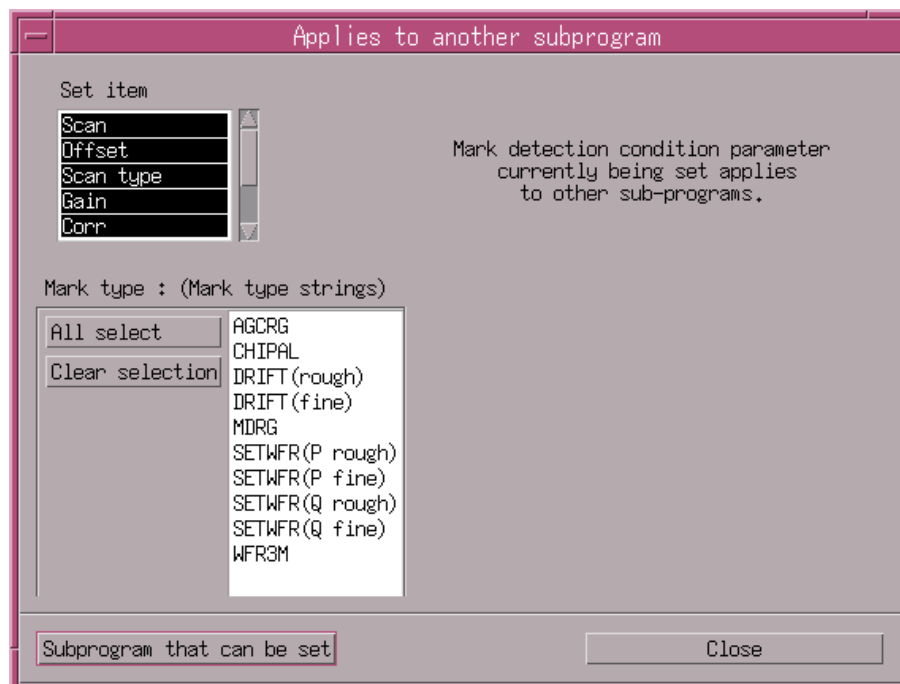
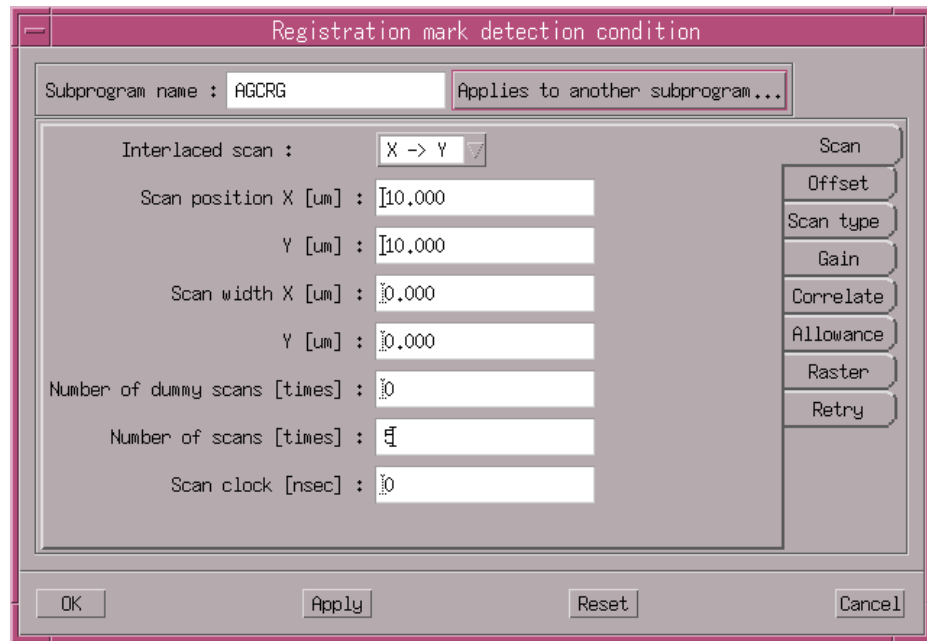


Fig. 93 Applies to another subprogram

■ Window for setting scan conditions

Here you can set the parameters related to the mark scan.



The image shows a software window titled "Registration mark detection condition". Inside, there's a section for "Subprogram name" with a text box containing "AGCRG" and a button "Applies to another subprogram...". Below this is a large area for setting scan parameters. On the left, there are labels and input fields: "Interlaced scan" with a dropdown menu showing "X -> Y", "Scan position X [μm]" with a text box "10.000", "Y [μm]" with a text box "10.000", "Scan width X [μm]" with a text box "0.000", "Y [μm]" with a text box "0.000", "Number of dummy scans [times]" with a text box "0", "Number of scans [times]" with a text box "4", and "Scan clock [nsec]" with a text box "0". On the right side of this area, there are several buttons: "Scan", "Offset", "Scan type", "Gain", "Correlate", "Allowance", "Raster", and "Retry". At the bottom of the window, there are four buttons: "OK", "Apply", "Reset", and "Cancel".

Fig. 94 Registration mark detection condition (Scan) window

- Sweep sequence

The selectable sweep sequences are as follows.

Sweep sequence	Description
X -> Y	The X scan is performed and then the Y scan is performed.
Y -> X	The Y scan is performed and then the X scan is performed.
X only	Only the X scan is performed.
Y only	Only the Y scan is performed.

- Scan position X, Y [μm]

This field lets you set the scan position from the mark center.

 For the BOX scan, you cannot specify zero.

- Scan width X, Y [μm]

This field lets you specify the scan width.

 Setting the sweep width to 4 μm or less results in measuring 4000 or fewer points, reducing the mark-detection time.

 Setting the sweep width to 4 μm or longer results in measuring 4000 points, making the scan pitch coarse.

- Number of dummy sweeps [times]

This field lets you specify the number of dummy scans at one scan position.

 The dummy sweeps are performed only for the first scan position.

- Number of scans [times]

This field lets you specify the number of actual scans at one scan position.

- Scan clock [nsec]

This field lets you specify the travel time per step for the beam to scan.

■ Offset

The Offset tab lets you set the parameters related to the mark-scan type.

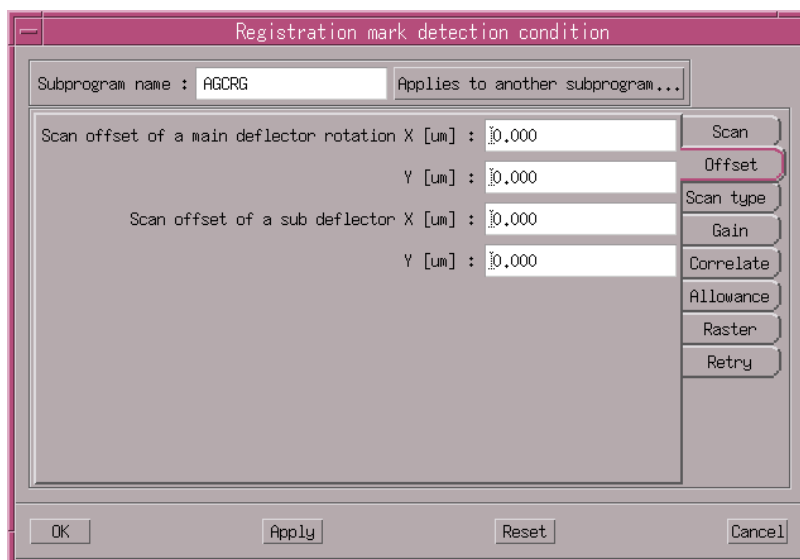


Fig. 95 Registration mark detection condition (Offset) window

- Scan offset X, Y [μm]

Setting the scan offset to zero will perform a scan with the deflection center of the main/sub deflector at the center. On the other hand, setting the scan offset to a certain position will perform the scan with the position at the center.

■ Window for setting scan type

Here you can set the parameters related to the type of mark scan.

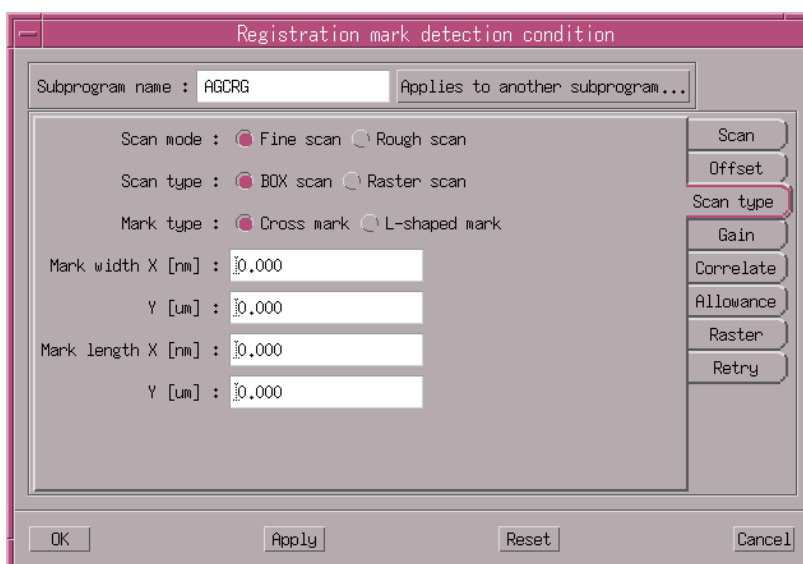


Fig. 96 Registration mark detection condition (Scan type) window

- Scan mode

You can select Fine scan or Rough scan.

- Scan type
You can select BOX scan or Raster scan.
- Mark type
You can select Cross mark or L-shaped mark.
- Mark width X, Y [μm]
This field lets you specify the mark width.
- Mark length X, Y [μm]
This field lets you specify the mark length.

■ Window for setting gain

Here you can set the gain and offset of the amplifier in the signal-processing system.

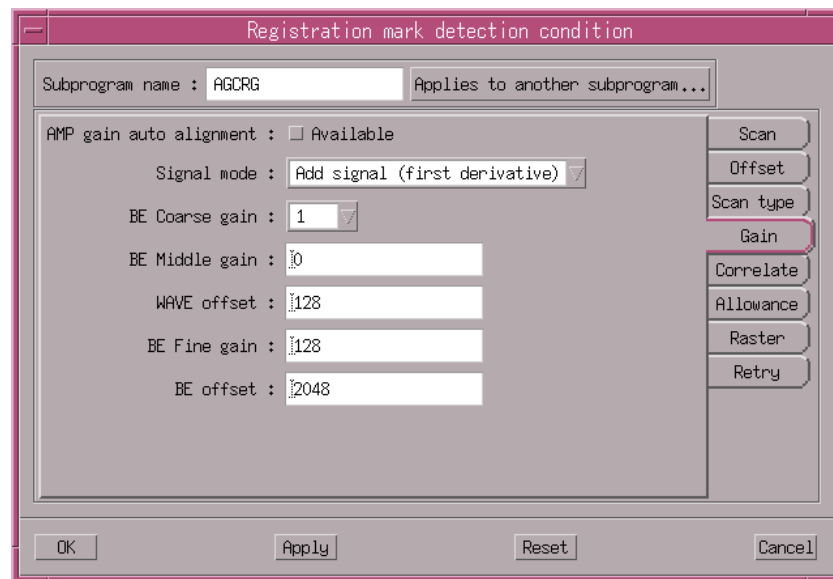
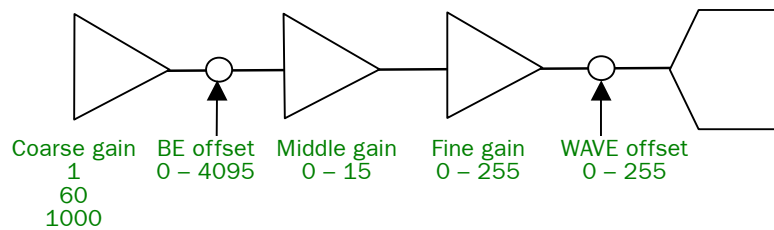


Fig. 97 Registration mark detection condition (Gain) window

- AMP gain auto alignment
The Coarse/Middle/Fine gain and the Fine offset are automatically calculated and set with consideration given to current fluctuation.
 - Signal mode
You can select Add signal (first derivative), Add signal (raw), or Difference signal (raw) as the signal-processing mode.
 - BE Coarse gain
This field lets you select the value of the BE coarse gain of the BE amplifier.
 - BE Middle gain
This field lets you specify the value of the BE middle gain of the BE amplifier.
 - WAVE offset
This field lets you specify the value of the BE fine gain of the BE amplifier.
 - BE Fine gain
This field lets you specify the fine-offset value of the BE amplifier.
 - BE Rough offset
This field lets you specify the rough-offset value of the BE amplifier.
- ✍ The figure below illustrates the gain settings. You can set the parameters in the following ranges.



 In normal use, the results of running AGCRG are used as the RG-mark gain parameter settings, so it is rare to change the gain parameters manually.

■ Window for setting correlation calculation conditions

Here you can set the correlation-calculation parameters.

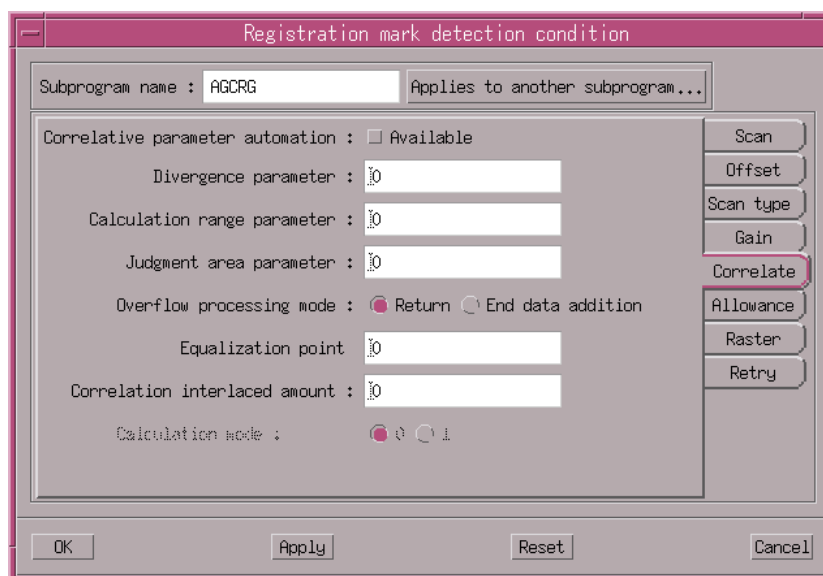


Fig. 98 Registration mark detection condition (Correlate) window

- Correlative parameter automation mode

You can specify whether to enable or disable automatic setting of correlation-calculation parameters. Enabling automatic setting will automatically calculate the following Divergence, Calculation range, and Judgment area parameters. The description of the parameters is the same as with the BE mark.

■ Window for setting allowable values

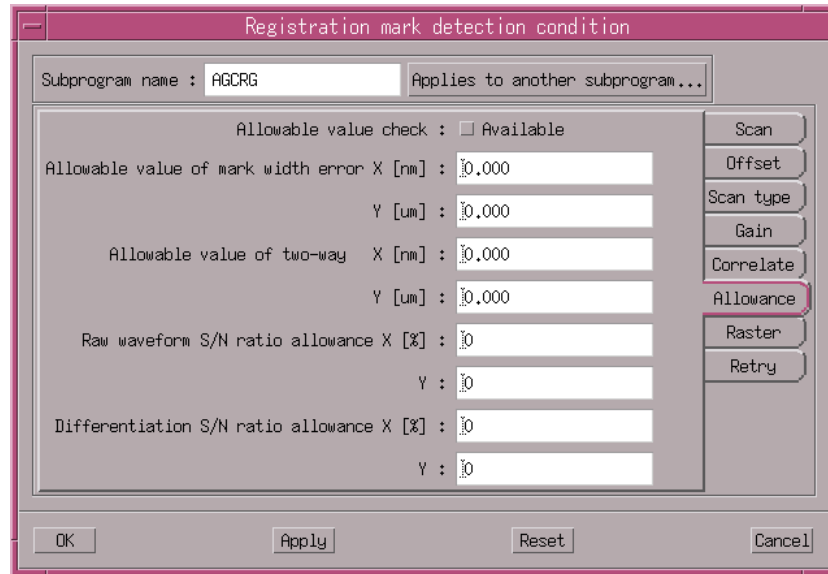


Fig. 99 Registration mark detection condition (Allowance) window

- Allowable value check
If you want to put the following parameters into effect, put a check in the **Available** box.
- Allowable value of mark width error X, Y [μm]
This field lets you specify the tolerance for the difference between the detected and expected mark widths.
- Allowable value of two-way X, Y [μm]
This field lets you specify the tolerance for the difference between the detected and expected mark positions.
- Raw waveform S/N ratio allowance X, Y
The S/N ratio is defined below. Normally, specify a value two or three times the S/N ratio in percent (%). In MD waveform diagram 105 (Section 4.42.4, Common MD parameters), the S/N ratio decreases to 100 or less. On the other hand, in MD waveform diagram 106 (Section 4.42.4, Common MD parameters), the S/N ratio might exceed 500.

$$\text{S/N ratio} = (\text{noise amplitude} / \text{signal amplitude}) \times 100 (\%)$$

$$\text{Signal amplitude} = \text{average signal of mark part} - \text{average signal of noise part}$$

$$\text{Noise amplitude} = \text{maximum noise near scan start point} - \text{minimum noise near scan start point}$$
- Differentiation S/N ratio allowance X, Y
These values affect the differentiation waveform. The description of the S/N allowance is the same as with the raw waveform.

■ Window for setting raster scan conditions

Here you can set the parameters related to the raster scan.

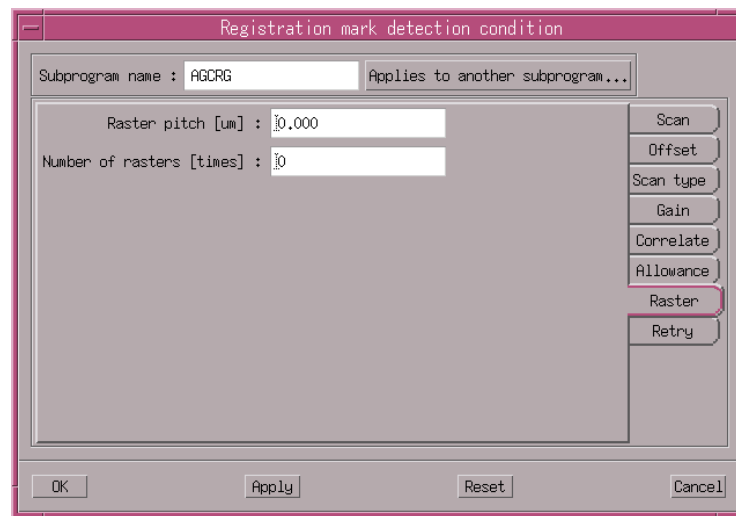


Fig. 100 Registration mark detection condition (Raster) window

- Raster pitch [μm]
This field lets you specify the raster-scan pitch.
- Number of rasters [times]
This field lets you specify the number of raster scans.

■ Window for setting retry conditions

Here you can set the parameters for the retries to perform if the mark detection fails.

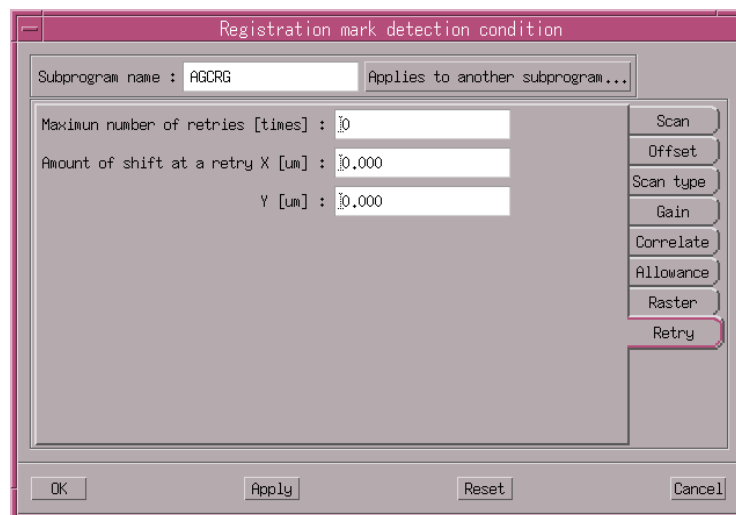


Fig. 101 Registration mark detection condition (Retry) window

- Maximum number of retries [times]
This field lets you specify the number of retries to perform if the mark detection fails.
- Amount of shift at a retry X, Y [μm]
This field lets you specify the step size to the next scan position after mark detection.

4.42.4 Common MD parameters

■ Sweep position and Sweep width

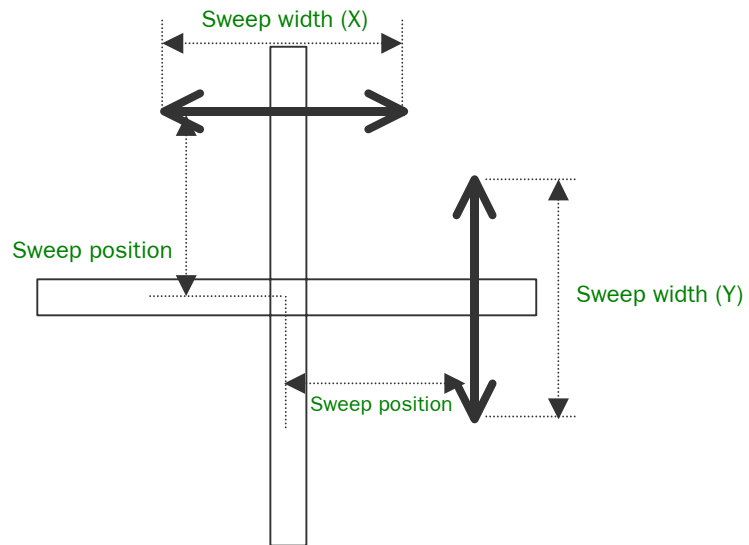


Fig. 102

■ Mark width and Mark length

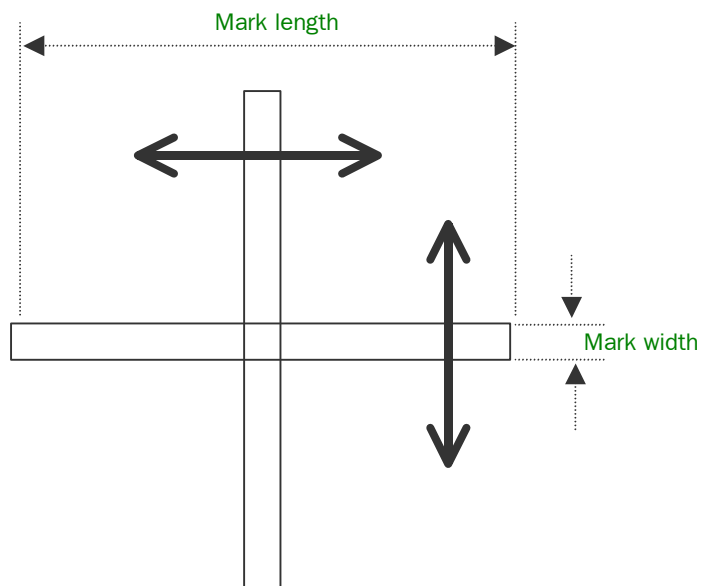
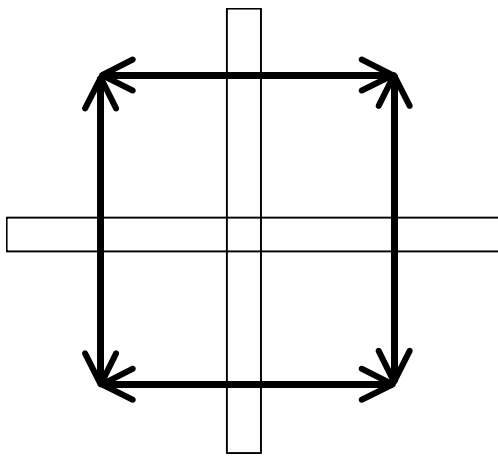
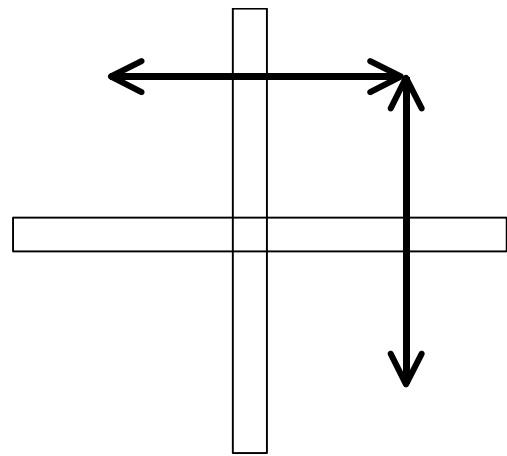


Fig. 103

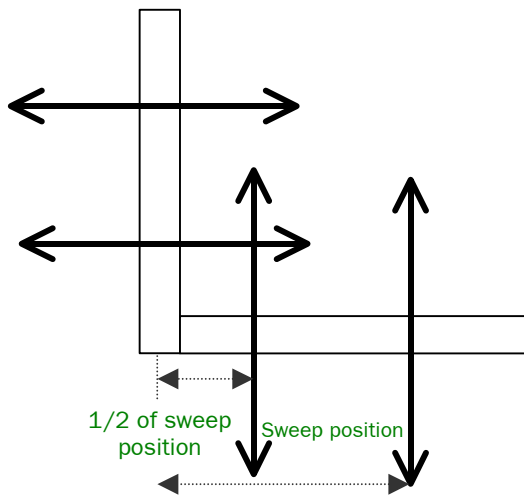
■ Scan types and mark types



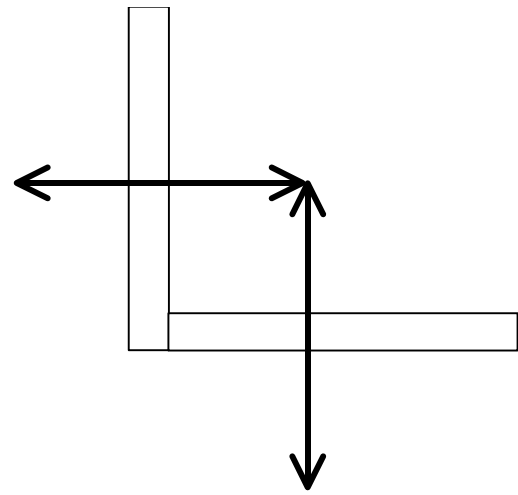
BOX scan & cross mark



Raster scan & cross mark



BOX scan & L-shaped mark



Raster scan & L-shaped mark

Fig. 104

■ MD waveform diagram 1

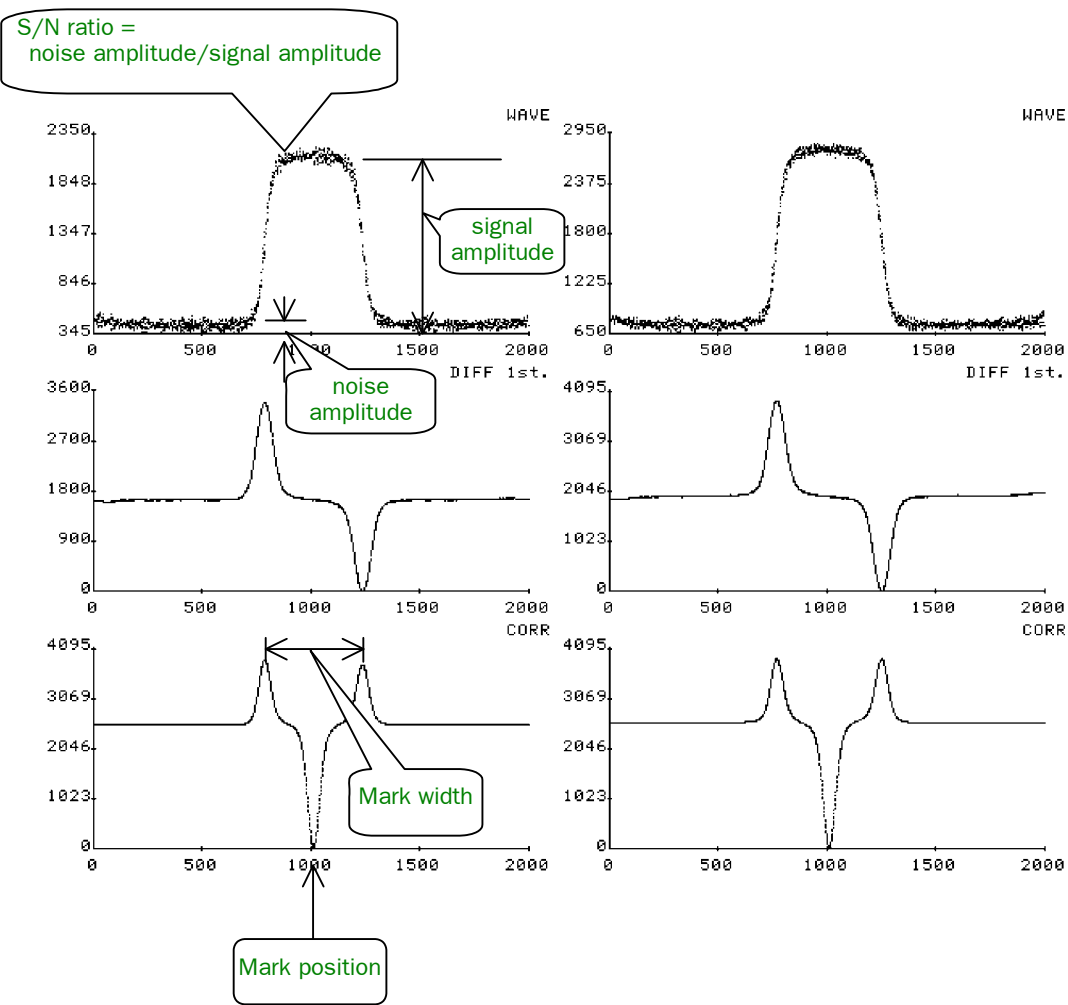


Fig. 105

■ MD waveform diagram 2 (with a lot of noise)

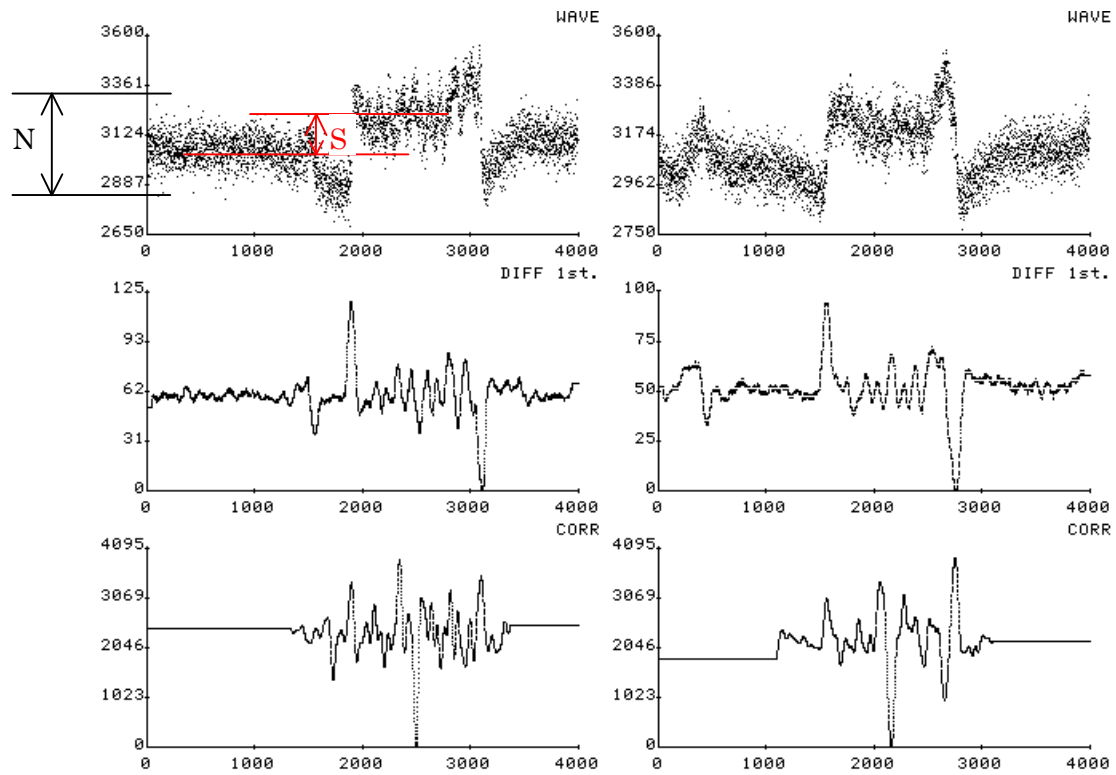


Fig. 106

4.43 SAVE

Under the conditions for starting the calibration program, SAVE saves the system status information. If you want to perform writing using the saved information, run the RESTOR subprogram.

The SAVE subprogram saves the system information for each calibration condition name.

During writing, if you want to change the system conditions, run RESTOR using the information saved by SAVE.

Clicking **Acquisition of latest status** displays the system information in the Result box.

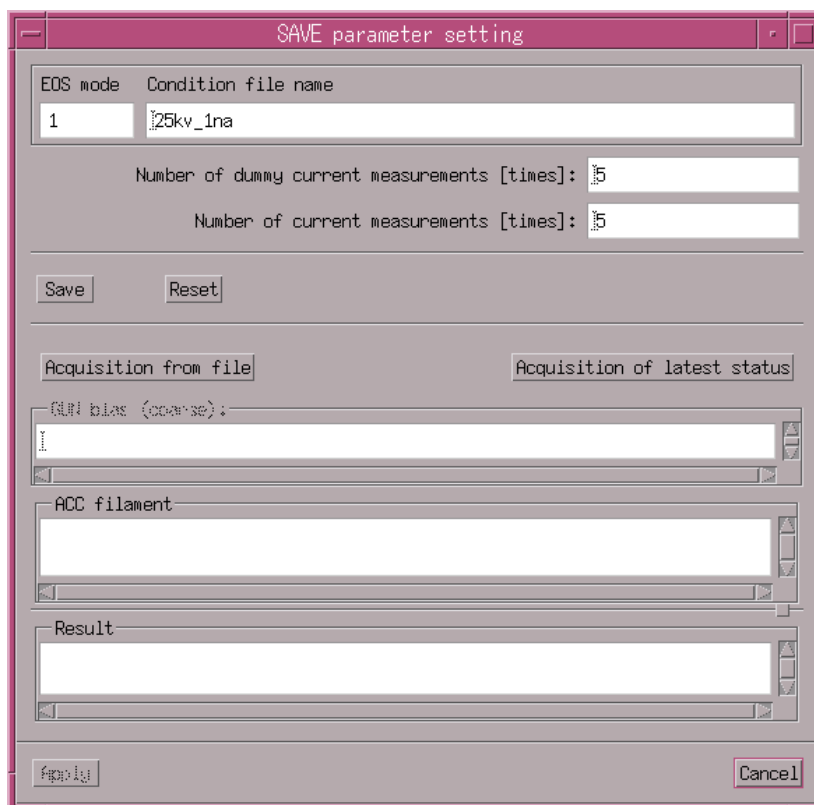
The image shows a software window titled "SAVE parameter setting". It contains several input fields and buttons. At the top, there are two labels: "EOS mode" and "Condition file name". Below "EOS mode" is a text box containing the number "1". Below "Condition file name" is a text box containing "25kv_1na". Below these are two labels: "Number of dummy current measurements [times]:" and "Number of current measurements [times]:". Each label is followed by a text box containing the number "5". Below these are two buttons: "Save" and "Reset". Below the buttons are two more buttons: "Acquisition from file" and "Acquisition of latest status". Below these are three text boxes: "GUN bias (coarse):", "ACC filament", and "Result". Each text box has a vertical scrollbar on its right side. At the bottom of the window are two buttons: "Apply" and "Cancel".

Fig. 107 SAVE parameter setting window

- Number of dummy current measurements [times], Number of current measurements [times]
During SAVE execution, the dummy current is measured and then the current is measured. You can specify how many times to measure each current.
- Acquisition from file
This button reads the present parameter settings from a file.
- Acquisition of latest status
This button reads the present system parameter settings.
- Comment
This box lets the user enter any comments.
- ACC filament
This box displays the ACC filament information read by clicking **Acquisition of latest status**.

- Result

This box displays the parameters read by clicking **Acquisition of latest status**.

Button	Function
Save	Saves the parameter settings presently shown in the window in a condition file.
Reset	Sets the parameters to the values read from a condition file, discarding changes to the parameters.
Apply	Runs the program, using the present parameter settings in the window.
Cancel	Closes the window without saving changes to the parameters.

4.44 RESTOR

Use RESTOR when you want to restore the system information saved by the SAVE subprogram.

The RESTOR subprogram restores the system information for each calibration condition name.

The RESTOR specification in writing (or in job) performs the same operation.

If you want to demagnetize the objective lens, put a check in the DEMAG execution box in the RESTOR parameter section. Demagnetization is recommended before RESTOR.

RESTOR parameter setting

Before change After change

EOS mode: 1 => 1

Condition file name: 25kv_1na => 25kv_1na

Comment:

: ☐ DEMAG execution

Allowable current fluctuation value [%]: 0.0

```

[SUBDEF DAC value calculated from lens4 + height]
ROT X,Y = -4 3
SUBDEF SHIFT X,Y = 0 0
GAIN X,Y = -1 1
ROT X,Y = 3 1
Material center position X Y [nm] = 105000000 85000000
Average shift amount X Y [nm] = 0.0 0.0
M1 position (stage coordinate) X Y [nm] = 185033787 -914239
Correction of gain and rotation for workpiece.
PDEF GAIN X,Y = 0 0
ROT X,Y = 0 0
SUBDEF SHIFT X,Y = 0 0
GAIN X,Y = 0 0
ROT X,Y = 0 0

=====
Current change allowance value [%] = 0.0
SAVE measurement current value [pA] = -1087.000

```

Save Execute Reset Cancel

Fig. 108 RESTOR parameter setting window

- Before change/After change

The EOS modes and the calibration condition names before and after RESTOR are displayed. You cannot edit them.

- Comment

The comments saved by SAVE are displayed. You cannot edit them.

- DEMAG execution

If you want to demagnetize the lens for setting the EOS parameters, put a check here.

- Allowable current fluctuation value [%]

This field lets you specify the fluctuation tolerance for changing the current.

Button	Function
Save	Saves the parameter settings presently shown in the window in a condition file.
Reset	Sets the parameters to the values read from a condition file, discarding changes to the parameters.
Execute	Runs the program, using the present parameter settings in the window.
Cancel	Closes the window without saving changes to the parameters.